

Physics Extra Edge

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BASIC MATHEMATICS USED IN PHYSICS & VECTORS

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Current has magnitude as well as direction but still not considered as vector.

Reason (R) : Current do not follow vector algebra.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

2. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Definite integral of a function is defined as area under the curve.

Reason (R) : Definite integral of a function is always positive.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

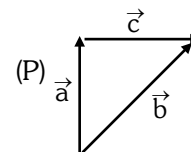
3. Two vectors $\vec{A}$ & $\vec{B}$ have equal magnitude equal to Z . If angle between $\vec{A}$ & $\vec{B}$ is 60° then match the following :

- | | |
|--------------------------------|------------------------------|
| (A) $ \vec{A} + \vec{B} $ | (P) $\frac{\sqrt{3}}{2} Z^2$ |
| (B) $ \vec{A} - \vec{B} $ | (Q) Z |
| (C) $\vec{A} \cdot \vec{B}$ | (R) $\sqrt{3} Z$ |
| (D) $ \vec{A} \times \vec{B} $ | (S) None |

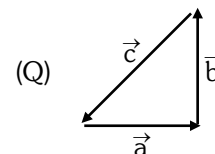
- (1) (A)–(R), (B)–(S), (C)–(Q), (D)–(P)
- (2) (A)–(R), (B)–(Q), (C)–(S), (D)–(P)
- (3) (A)–(P), (B)–(Q), (C)–(R), (D)–(S)
- (4) (A)–(Q), (B)–(P), (C)–(S), (D)–(P)

4. Match the following

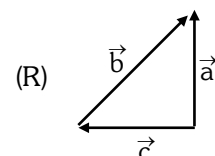
(A) $\vec{a} + \vec{b} = \vec{c}$



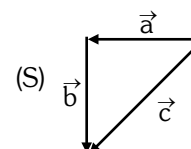
(B) $\vec{a} - \vec{c} = \vec{b}$



(C) $\vec{b} - \vec{a} = \vec{c}$



(D) $\vec{a} + \vec{b} + \vec{c} = \vec{0}$



- (1) (A)–(S), (B)–(R), (C)–(P), (D)–(Q)
- (2) (A)–(R), (B)–(S), (C)–(Q), (D)–(P)
- (3) (A)–(P), (B)–(Q), (C)–(R), (D)–(S)
- (4) (A)–(S), (B)–(R), (C)–(Q), (D)–(P)

5. Given below are two statements :

Statement I : Resultant of 2 forces of magnitude 4N and 5N can be 2N in magnitude.

Statement II : $||\vec{a}| - |\vec{b}|| \leq |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

TG: @Chalnaayaaar

6. Given below are two statements :

Statement I : Two null vector have same direction.

Statement II : $\vec{A} \times \vec{B}$ lies in the plane of $\vec{A} + \vec{B}$
In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

7. Which of the following is correct :

- (i) $\vec{A} \cdot \vec{B}$ is a vector quantity
- (ii) $\vec{A} \times \vec{B}$ is perpendicular to plane of $\vec{A} + \vec{B}$
- (iii) For two orthogonal vectors $\vec{A} \cdot \vec{B} = 0$
- (iv) If vectors are parallel or antiparallel, then $\vec{A} \times \vec{B} = \vec{0}$

- (1) (i) only
- (2) (i) & (ii)
- (3) (iii) & (iv) only
- (4) (ii), (iii) & (iv)

8. Which of the following is incorrect :

- (i) In third quadrant $\sin \theta$ of angle is positive
 - (ii) For an increasing function $\frac{dy}{dx} > 0$
 - (iii) Definite integral of a function gives magnitude of area between given limits
- (1) (i) only
 - (2) (ii) only
 - (3) (i) & (iii)
 - (4) (iii) only

9. Given below are two statements :

Statement I : For every small angle θ , we may use approximation $\sin \theta \approx \theta \approx \tan \theta$.

and

Statement II : For very small angle θ , the hypotenuse and the base become approximately of the same length.

- (1) Statement-I is true, Statement-II is true; Statement-II is a correct explanation for Statement-I.
- (2) Statement-I is true, Statement-II is true; Statement-II is not a correct explanation for Statement-I.
- (3) Statement-I is true, Statement-II is false.
- (4) Statement-I is false, Statement-II is true.

10. Suggest suitable match between function given in the first column and its description given in the second column.

Column-I

- (A) $\sin(390^\circ)$
- (B) $\sin(-30^\circ)$
- (C) $\cos 120^\circ$
- (D) $\tan(-120^\circ)$

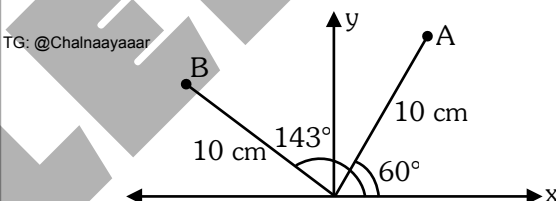
Column-II

- (P) Positive
- (Q) Negative
- (R) Zero
- (S) Modulus is greater than one
- (T) Modulus is less than one

- (1) $A \rightarrow PT, B \rightarrow QT, C \rightarrow QT, D \rightarrow PS$
- (2) $A \rightarrow PT, B \rightarrow QS, C \rightarrow QT, D \rightarrow PS$
- (3) $A \rightarrow QT, B \rightarrow QS, C \rightarrow PT, D \rightarrow PS$
- (4) $A \rightarrow QS, B \rightarrow PT, C \rightarrow QT, D \rightarrow PS$

11. Refer the given figure and identify correct statement(s)

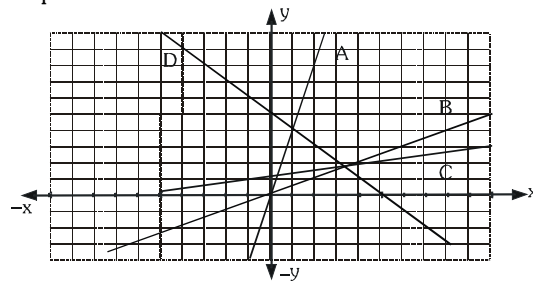
- (A) Distance of A from x-axis is $5\sqrt{3}$ cm.
- (B) Distance of B from x-axis is 6 cm.
- (C) Distance of A from y-axis is 5 cm.
- (D) Distance of B from y-axis is 8 cm.



Options :-

- (1) A, C
- (2) A, D
- (3) A, B, C
- (4) A, B, C, D

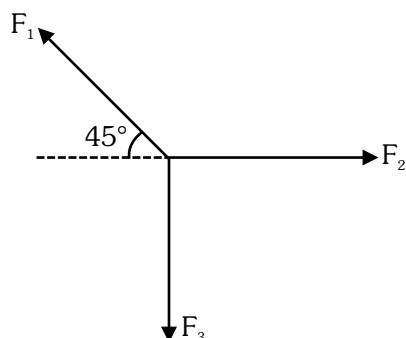
12. In the following graph, several straight lines are shown. Arrange them in order of increasing slope modulus



Options :-

- (1) C, B, A, D
- (2) C, B, D, A
- (3) A, D, B, C
- (4) D, A, B, C

13. Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.


Column I

(Combination)

(A) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$

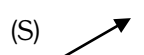
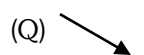
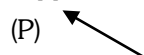
(B) $\vec{F}_1 - \vec{F}_2 + \vec{F}_3$

(C) $\vec{F}_1 - \vec{F}_2 - \vec{F}_3$

(D) $\vec{F}_2 - \vec{F}_1 - \vec{F}_3$

Column II

(Approximate Direction)


Options :-

- (1) A $\rightarrow$ R, B $\rightarrow$ Q, C $\rightarrow$ P, D $\rightarrow$ S
- (2) A $\rightarrow$ Q, B $\rightarrow$ P, C $\rightarrow$ R, D $\rightarrow$ S
- (3) A $\rightarrow$ Q, B $\rightarrow$ R, C $\rightarrow$ P, D $\rightarrow$ S
- (4) A $\rightarrow$ S, B $\rightarrow$ P, C $\rightarrow$ R, D $\rightarrow$ Q

14. Which of the following statement is/are true ?

- (a) Two vectors of unequal magnitude can add up to zero.
- (b) Three vectors of unequal magnitude can add up to zero, if they lie in a plane.
- (c) Three vectors of unequal magnitude can add up to zero, if they do not lie in same plane.

Options :-

- (1) Only a
- (2) a, b and c
- (3) only b
- (4) only c

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	1	3	2	1	1	4	4	3	1	1	4	2	3	3	

UNIT & DIMENSION

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Absolute error in a physical quantity may be +ve, -ve or zero.

Reason (R) : Absolute error is the difference of measured value and true value of a physical quantity.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

2. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Maximum possible error in a reading is taken as least count of the measuring instrument.

Reason (R) : Error in a measurement can not be greater than least count of the measuring instrument.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

3. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Higher is the accuracy of measurement, if instrument has smaller least count.

Reason (R) : Smaller the percentage error, higher is the accuracy of measurement.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

4. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a measurement, two reading obtained are 20.004 and 20.0004. The second measurement is more precise.

Reason (R) : Measurement having more decimal places is more precise.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

5. Given below are two statements :

Statement I : Absolute error is unitless and dimensionless.

Statement II : All types of errors are unitless and dimensionless.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

6. Given below are two statements :

Statement I : An exact number has infinite number of significant digits.

Statement II : A number which is not a measured value, has infinite number of significant digits.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

7. Given below are two statements :

Statement I : Two physical quantities having same dimensions, may have different units.

Statement II : Shake and light year, both measure time.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

8. Given below are two statements :

Statement I : Least count of a screw gauge is directly proportional to the number of divisions on circular scale.

Statement II : A screw gauge having a smaller value of pitch has greater accuracy.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

9. Match the quantities having same dimensions in column-1 and column-2 :

	Column-1		Column-2
A	Stress	p	Pressure
B	Strain	q	Energy density
C	Modulus of elasticity	r	Angle
D	Torque	s	Energy

- (1) $A \rightarrow r$; $B \rightarrow p$; $C \rightarrow s$; $D \rightarrow q$
- (2) $A \rightarrow q$; $B \rightarrow r$; $C \rightarrow p$; $D \rightarrow s$
- (3) $A \rightarrow p$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$
- (4) $A \rightarrow q$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$

10. Match the column :

	Column-1		Column-2
A	Electrical resistance	p	$M^{-1}L^{-2}T^4A^2$
B	Capacitance	q	$ML^2T^{-2}A^{-2}$
C	Magnetic field	r	$ML^2T^{-3}A^{-2}$
D	Inductance	s	$MT^{-2}A^{-1}$

- (1) $A \rightarrow r$; $B \rightarrow p$; $C \rightarrow s$; $D \rightarrow q$
- (2) $A \rightarrow s$; $B \rightarrow p$; $C \rightarrow r$; $D \rightarrow q$
- (3) $A \rightarrow p$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$
- (4) $A \rightarrow q$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$

- 11.** Match the quantities in column 1 to dimensions in column 2. R is resistance, L is inductance, C is capacitance, H is latent heat and S is specific heat.

	Column-1		Column-2
A	LC	p	L^2T^{-2}
B	LR	q	$L^2T^{-2}K^{-1}$
C	H	r	T^2
D	S	s	$M^2L^4T^{-5}A^{-4}$

- (1) $A \rightarrow r$; $B \rightarrow s$; $C \rightarrow p$; $D \rightarrow q$
 (2) $A \rightarrow s$; $B \rightarrow p$; $C \rightarrow r$; $D \rightarrow q$
 (3) $A \rightarrow p$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$
 (4) $A \rightarrow q$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$

- 12.** Match the column :

	Column-1		Column-2
A	Boltzmann constant	p	ML^2T^{-1}
B	Coefficient of viscosity	q	$ML^{-1}T^{-1}$
C	Planck constant	r	$MLT^{-3}K^{-1}$
D	Thermal conductivity	s	$ML^2T^{-2}K^{-1}$

- (1) $A \rightarrow r$; $B \rightarrow s$; $C \rightarrow p$; $D \rightarrow q$
 (2) $A \rightarrow p$; $B \rightarrow s$; $C \rightarrow r$; $D \rightarrow q$
 (3) $A \rightarrow p$; $B \rightarrow r$; $C \rightarrow s$; $D \rightarrow q$
 (4) $A \rightarrow s$; $B \rightarrow q$; $C \rightarrow p$; $D \rightarrow r$

- 13.** Which two of the following five physical quantities have the same dimensions?

- (a) Energy density (b) Refractive index
 (c) Dielectric constant (d) Young's modulus
 (e) Magnetic field
 (1) (a) and (e) (2) (b) and (d)
 (3) (c) and (e) (4) (a) and (d)

- 14.** Which of the following are not a unit of time?

- (a) Second (b) Parsec
 (c) Year (d) Light year
 (1) (b) and (d) (2) (b) and (c)
 (3) (a) and (b) (4) (c) and (d)

- 15.** Photon is quantum of radiation with energy $E = hv$ where v is frequency and h is Planck's constant. The dimensions of h are the same as that of

- (a) Linear impulse (b) Angular impulse
 (c) Linear momentum (d) Angular momentum
 (1) (a) and (c) (2) (b) and (d)
 (3) (b) and (c) (4) (a) and (d)

- 16.** Which of the following ratios express pressure?

- (a) Force / area (b) Energy / volume
 (c) Energy / area (d) Force / volume
 (1) (a) and (c) (2) (b) and (d)
 (3) (b) and (c) (4) (a) and (b)

- 17. Statement-1 :** The displacement cannot be added with distance travelled.

and

Statement-2 : Addition of scalar and vector of same dimensions is not a meaningful algebraic operation.

- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
 (3) Statement-1 is True, Statement-2 is False.
 (4) Statement-1 is False, Statement-2 is True.

- 18.** Match the column - I with column - II.

	Column-I (Number)		Column-II (Number of significant digits)
(A)	1001	(P)	5
(B)	0100.1	(Q)	4
(C)	100.100	(R)	6
(D)	0.001001	(S)	3

Options :-

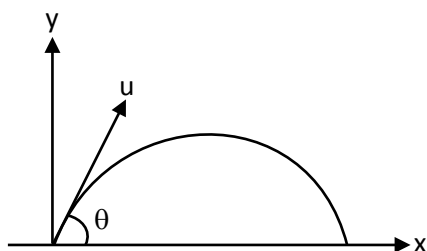
- (1) $A \rightarrow Q$, $B \rightarrow Q$, $C \rightarrow R$, $D \rightarrow Q$
 (2) $A \rightarrow Q$, $B \rightarrow P$, $C \rightarrow R$, $D \rightarrow R$
 (3) $A \rightarrow Q$, $B \rightarrow S$, $C \rightarrow S$, $D \rightarrow Q$
 (4) $A \rightarrow Q$, $B \rightarrow Q$, $C \rightarrow P$, $D \rightarrow Q$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	3	1	1	4	1	2	3	2	1	1	4	4	1	2
Que.	16	17	18												
Ans.	4	1	1												

KINEMATICS

1. For ground to ground projectile motion with initial velocity u at angle θ from horizontal.



(A)	Change in speed from initial to final point	(i)	$u - u \cos \theta$
(B)	Magnitude of change in velocity from initial to final point	(ii)	$u \sin \theta$
(C)	Change in speed from initial to top	(iii)	$2u \sin \theta$
(D)	Magnitude of change in velocity from initial to top	(iv)	zero

Options :-

- (1) (A)–(i), (B)–(ii), (C)–(iii), (D)–(iv)
 (2) (A)–(iv), (B)–(ii), (C)–(i), (D)–(iii)
 (3) (A)–(iv), (B)–(iii), (C)–(i), (D)–(ii)
 (4) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)
2. For initial velocity $(\vec{v}_i) = 3\hat{i} - 4\hat{j}$ and final velocity $(\vec{v}_f) = 3\hat{i} + 4\hat{j}$:-

- (A) $|\Delta \vec{v}|$ (i) $8\hat{j}$
 (B) $\Delta |\vec{v}|$ (ii) $6\hat{i}$
 (C) $\vec{v}_f - \vec{v}_i$ (iii) 0
 (D) $\vec{v}_f + \vec{v}_i$ (iv) 8

- (1) (A)–(iii), (B)–(iv), (C)–(ii), (D)–(i)
 (2) (A)–(iii), (B)–(iv), (C)–(i), (D)–(ii)
 (3) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)
 (4) (A)–(iv), (B)–(iii), (C)–(i), (D)–(ii)

3. A particle is projected vertically upward from ground with initial velocity u such that it clears the top of a pole of height h after time t_1 in its path. It takes further time t_2 to reach the ground.

- (A) u (i) $\frac{(t_1 + t_2)^2}{8}g$
 (B) h (ii) $(t_1 + t_2)$
 (C) Time of flight (iii) $\frac{1}{2}g(t_1 + t_2)$
 (D) Maximum height (iv) $\frac{1}{2}gt_1t_2$

- (1) (A)–(ii), (B)–(i), (C)–(iii), (D)–(iv)
 (2) (A)–(i), (B)–(ii), (C)–(iii), (D)–(iv)
 (3) (A)–(iii), (B)–(iv), (C)–(ii), (D)–(i)
 (4) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)

4. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The distance between two floating objects on a river flowing uniformly does not change with time.

Reason (R) : The floating object has the same velocity as that of the river.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

5. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Horizontal component of velocity is constant in projectile motion under gravity.

Reason (R) : Two projectiles having same horizontal range must have the same time of flight.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

TG: @Chalnaayaaar

6. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A particle with constant acceleration always moves along a straight line.

Reason (R) : A particle with constant acceleration will not change direction of motion.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) Both **(A)** and **(R)** are false.
7. An object may have-
- (A) varying speed without having varying velocity
 - (B) varying velocity without having varying speed
 - (C) non-zero acceleration without having varying velocity
 - (D) non-zero acceleration without having varying speed
- (1) Only B is correct
 - (2) Only D is correct
 - (3) Both B and D are correct
 - (4) All are correct
8. The velocity of a particle is zero at $t = 0$, then -
- (A) the acceleration at $t = 0$ must be zero
 - (B) the acceleration at $t = 0$ may be zero
 - (C) if the acceleration is zero from $t = 0$ to $t = 10$ s, speed is also zero in this interval
 - (D) if the speed is zero from $t = 0$ to $t = 10$ sec, the acceleration is also zero in the interval
- (1) (A), (C) and (D) are correct
 - (2) (B), (C) and (D) are correct
 - (3) (A) and (D) are correct
 - (4) (B) and (C) are correct
9. Consider the motion of the tip of the minute hand of a clock for one hour motion which of the following is/are correct.
- (A) the displacement is zero
 - (B) the distance covered is zero
 - (C) the average speed is zero
 - (D) the average velocity is zero
- Option

- (1) (A) & (D)
- (2) (B) & (C)
- (3) (A) & (C)
- (4) (B) & (D)

10. Instantaneous velocity of a particle -
- (A) depends on instantaneous position
 - (B) depends on instantaneous speed
 - (C) independent of instantaneous position
 - (D) independent of instantaneous speed
- (1) Both (A) & (B) are correct
 - (2) Both (C) & (D) are correct
 - (3) Both (A) & (D) are correct
 - (4) Both (B) & (C) are correct
11. For a body moving on a straight line if x is position coordinate and t is time then acceleration of body is constant when -
- (A) x and velocity is linear
 - (B) x and square of velocity is linear
 - (C) t and velocity is linear
 - (D) t and square of velocity is linear.
- (1) Both (A) & (B) are correct
 - (2) Both (C) & (D) are correct
 - (3) Both (A) & (D) are correct
 - (4) Both (B) & (C) are correct
12. If the velocity of a body is constant -
- (A) $|\text{Velocity}| = \text{speed}$
 - (B) $|\text{Average velocity}| = \text{speed}$
 - (C) $\text{Velocity} = \text{average velocity}$
 - (D) $\text{Speed} = \text{average speed}$
- (1) Only (A) is correct
 - (2) (A) and (B) are correct
 - (3) (C) and (D) are correct
 - (4) All are correct
13. A particle can move only along x -axis. Three pairs of initial and final positions of particle at two successive times are given
- | Pair | Initial position | Final position |
|------|------------------|----------------|
| 1 | - 3m | + 5m |
| 2 | - 3m | - 7m |
| 3 | + 7m | - 3m |
- Find the sum of magnitudes of displacement in the pairs which give negative displacement in m.
- (1) 14
 - (2) 12
 - (3) 20
 - (4) 22

14. In first and second columns of the following table, unit vectors $\hat{a}$ and $\hat{b}$ of some geographical directions are mentioned. Complete the third column.

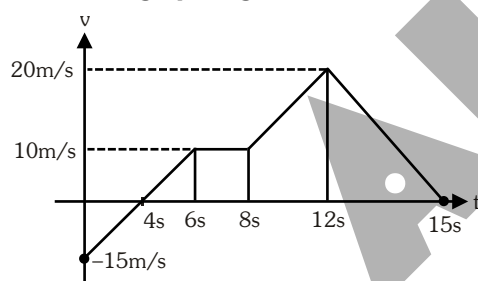
	Unit vector $\hat{a}$	Unit Vector $\hat{b}$	Scalar Product $\hat{a} \cdot \hat{b}$
A	West	Up	
B	South	East	
C	30° North of west	Down	
D	30° Down of south	Up	

Options :-

- (1) $A \rightarrow 0, B \rightarrow 0, C \rightarrow 0, D \rightarrow -\frac{1}{2}$
 (2) $A \rightarrow 0, B \rightarrow 0, C \rightarrow \frac{1}{2}, D \rightarrow -\frac{1}{2}$
 (3) $A \rightarrow 1, B \rightarrow 1, C \rightarrow \frac{1}{2}, D \rightarrow \frac{1}{2}$
 (4) $A \rightarrow 1, B \rightarrow 1, C \rightarrow -\frac{1}{2}, D \rightarrow \frac{1}{2}$

Paragraph for Question Nos. 15 to 17

A moving particle is acted upon by three forces at different times to bring it to rest. Its velocity versus time graph is given below



15. The average speed for the first 6 s is

- (1) zero
 (2) $\frac{5}{3} \text{ ms}^{-1}$
 (3) $\frac{10}{3} \text{ ms}^{-1}$
 (4) $\frac{20}{3} \text{ ms}^{-1}$

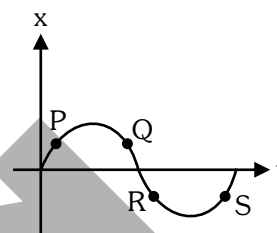
16. The average velocity for the first 12s is

- (1) zero
 (2) 5 m/s
 (3) $\frac{10}{3} \text{ m/s}$
 (4) 10 m/s

17. The average acceleration from $t=5\text{s}$ to $t=15\text{s}$ is

- (1) zero
 (2) -0.5 m/s^2
 (3) $+0.5 \text{ m/s}^2$
 (4) 1 m/s^2

18. A car is moving along a straight line. It's displacement (x) - time(t) graph is shown in column I. Match the entries in column I with points on graph.



Column I

- (A) $x \rightarrow \text{negative}, v \rightarrow \text{positive}, a \rightarrow \text{positive}$
 (B) $x \rightarrow \text{positive}, v \rightarrow \text{negative}, a \rightarrow \text{negative}$
 (C) $x \rightarrow \text{negative}, v \rightarrow \text{negative}, a \rightarrow \text{positive}$
 (D) $x \rightarrow \text{positive}, v \rightarrow \text{positive}, a \rightarrow \text{negative}$

Column II

- (P) P
 (Q) Q
 (R) R
 (S) S

Options :-

- (1) $A \rightarrow S, B \rightarrow Q, C \rightarrow R, D \rightarrow P$
 (2) $A \rightarrow P, B \rightarrow R, C \rightarrow Q, D \rightarrow S$
 (3) $A \rightarrow S, B \rightarrow R, C \rightarrow Q, D \rightarrow P$
 (4) $A \rightarrow S, B \rightarrow Q, C \rightarrow R, D \rightarrow S$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	3	1	3	4	3	2	1	4	4	4	1	1	4
Que.	16	17	18												
Ans.	2	2	1												

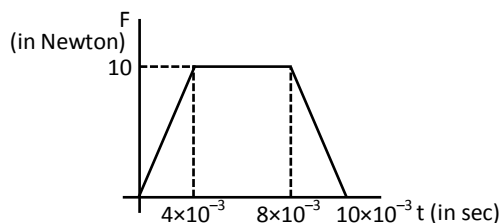
NEWTON'S LAW'S OF MOTION AND FRICTION

1. The force exerted by the lift on the foot of a person standing in it, is more than his weight than the lift is -

- (a) going up and slowing down
(b) going up and speeding up
(c) going down and slowing down
(d) going down and speeding up

- (1) (a, b) (2) (b, c) (3) (a, c) (4) (b, d)

2. A particle of mass 70 g moving at 50 cm/s, is acted upon by a variable force opposite to its initial direction of its motion. Force F is shown as a function of time ' t ' -



- (a) It speed will be 50 cm/sec after the force stop acting
(b) Its direction of motion gets reversed
(c) It average acceleration will be 1 m/s^2 , during the interval force acts
(d) Its average acc. will be 10 m/s^2 , during the interval in which force act

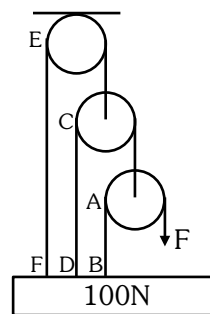
- (1) (a, b) (2) (a, c)
(3) (b, c) (4) (b, d)

3. A reference frame attached to the earth -

- (a) is an inertial frame by definition
(b) cannot be an inertial frame because the earth is revolving round the sun
(c) is an inertial frame because Newton's laws are applicable in this frame
(d) cannot be an inertial frame because the earth is rotating about its own axis

- (1) (a, b) (2) (b, c) (3) (a, c) (4) (b, d)

4. In the arrangement shown below force F is just sufficient to keep equilibrium of 100 N block, T_1 , T_2 and T_3 are tension, in strings AB, CD and EF and T_4 is total force of all tensions on block 100 N



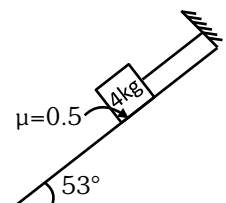
Match the following :

Column I

Column II

- (a) T_1 (P) $\frac{400}{7} \text{ N}$
(b) T_2 (Q) $\frac{100}{7} \text{ N}$
(c) T_3 (R) $\frac{200}{7} \text{ N}$
(d) T_4 (S) 100 N
(1) (a) - Q (b) - R (c) - P (d) - S
(2) (a) - P (b) - R (c) - Q (d) - S
(3) (a) - P (b) - Q (c) - S (d) - R
(4) (a) - P (b) - S (c) - Q (d) - S

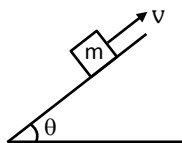
5. For the given figure, match column-I to column-II



Column-I		Column-II	
(a)	Tension in the string	(P)	24 N
(b)	Normal on the block	(Q)	20 N
(c)	Contact force between block and inclined plane	(R)	12 N
(d)	Friction on the block	(S)	27 N

- (1) (a) - Q, (b) - P, (c) - S, (d) - R
(2) (a) - P, (b) - Q, (c) - S, (d) - R
(3) (a) - P, (b) - Q, (c) - R, (d) - S
(4) (a) - Q, (b) - P, (c) - R, (d) - S

6. A block of mass m is thrown upwards with some initial velocity as shown. On the block



Column-I		Column-II	
(a)	Net force in horizontal direction	(P)	zero
(b)	Net force in vertical direction	(Q)	$m(g\sin\theta + \mu g\cos\theta)$
(c)	Net force along the plane	(R)	$m(g\sin\theta\cos\theta + \mu g\cos^2\theta)$
(d)	Net force perpendicular to plane	(S)	$m(g\sin^2\theta + \mu g\sin\theta\cos\theta)$
		(T)	None

(1) (a) – P (b) – R (c) – Q (d) – P

(2) (a) – R (b) – S (c) – Q (d) – P

(3) (a) – P (b) – S (c) – Q (d) – P

(4) (a) – Q (b) – R (c) – P (d) – R

7. If the net force acting on a system is represented by $\vec{F}$ and its momentum is $\vec{P}$. Then match the entries of column I with the entries of Column II.

Column-I		Column-II	
(a)	If $\vec{F}$ is constant	(P)	$\vec{P}$ may change its direction
(b)	If $\vec{F}$ is changing in magnitude only	(Q)	$\vec{P}$ must change its magnitude
(c)	If $\vec{F}$ is changing in direction only	(R)	$\vec{P}$ may not change its direction
(d)	If $\vec{F}$ is zero	(S)	$\vec{P}$ may be constant in magnitude

(1) (a) – P, Q, R (b) – P, Q, R (c) – P, S (d) – S

(2) (a) – P, R (b) – P, Q (c) – P, R (d) – S

(3) (a) – P, Q (b) – Q, R (c) – P, R (d) – S

(4) (a) – P, R (b) – P, Q, R (c) – P (d) – S

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When a body is at rest on a horizontal surface then the contact force on the body by the surface must be equal to the weight of body.

Reason (R) : The contact force applied by a surface on a body placed on it is the total force applied by surface on the body.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
- (A) is true but (R) is false.
- (A) is false but (R) is true.

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When a block is placed in a lift which is accelerating upwards then the body experiences the following three forces :

- Weight (mg)
- Normal reaction (N) and
- Pseudo force (ma) if 'a' is acceleration of lift

Reason (R) : Pseudo force is applied on a body only when body is seen from an accelerated observer.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
- (A) is true but (R) is false.
- (A) is false but (R) is true.

10. Equilibrium means zero acceleration while rest means zero velocity.

- True
- False
- Partially true
- None of the above

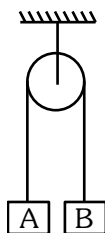
11. Newton's first law is a special case of second law.

- True
- False
- Partially true
- None of the above

12. A body released from a balloon rising up continues to move up along with the balloon due to inertia of motion.

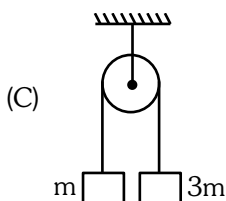
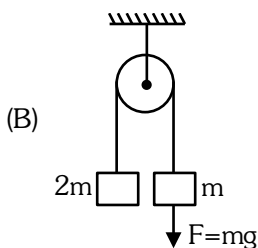
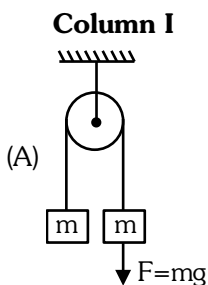
(1) True (2) False
(3) Partially true (4) None of the above

13. Two blocks of mass 2 kg and 3 kg are arranged as shown and released from rest. Assuming string and pulley to be ideal. Find the separation between them after 1 sec. [$g = 10 \text{ m/s}^2$]



(1) 1 m (2) 2 m (3) 4 m (4) None

14. Match the situations in column I to the accelerations of blocks in the column II (acceleration due to gravity is g and F is an additional force applied to one of the blocks)

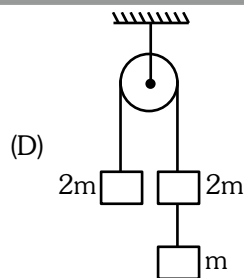


Column II

(P) $\frac{g}{5}$

(Q) $\frac{g}{3}$

(R) $\frac{g}{2}$



(S) 0

Options :-

- (1) $A \rightarrow R, B \rightarrow S, C \rightarrow R, D \rightarrow P$
(2) $A \rightarrow R, B \rightarrow S, C \rightarrow Q, D \rightarrow P$
(3) $A \rightarrow Q, B \rightarrow P, C \rightarrow R, D \rightarrow S$
(4) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$

15. If a body is placed on a rough inclined plane, the nature of forces acting on the body is(are)

(A) gravitational (B) electromagnetic
(C) nuclear (D) weak

- (1) Only A (2) A and D
(3) A, B, C, D (4) A and B

16. **Statement-I** : When a cloth with a lot of dirt in it, is jerked the dirt gets away from it.

and

Statement-II : When an object applies force on another object the other object also applies an equal force but opposite in direction.

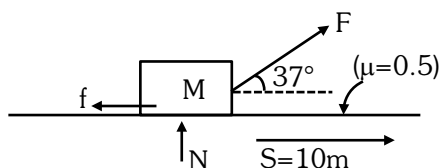
- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is True.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	4	1	1	2	1	4	4	1	2	2	2	1	4
Que.	16	TG: @Chalnaayaaar													
Ans.	2														

WORK ENERGY & POWER

1. A block of mass $M = 10\text{kg}$ is dragged 10m by force $F = 100\text{N}$ on rough horizontal surface ($\mu = 0.5$)

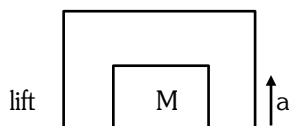


Force on block		Work done by force on block	
A	Applied (F)	P	Zero
B	Normal (N)	Q	600 J
C	Friction (f)	R	800 J
D	Net resultant (F_{net})	S	- 200 J

Match the column

- (1) [(A - Q), (B - P), (C - S), (D - R)]
 (2) [(A - R), (B - P), (C - S), (D - Q)]
 (3) [(A - P), (B - Q), (C - R), (D - S)]
 (4) [(A - S), (B - R), (C - Q), (D - P)]

2. Block of mass $M = 10\text{kg}$ is kept inside a lift, which starts motion from rest in vertical upward direction with uniform acceleration ($a = 2\text{m/sec}^2$)



Power delivered to block by force		Power (in watt)	
A	By normal at 5 sec	P	- 500
B	By normal in 5 sec	Q	600
C	By Mg (gravity) at 5 sec	R	- 1000
D	By Mg (gravity) in 5 sec	S	1200

Match the column

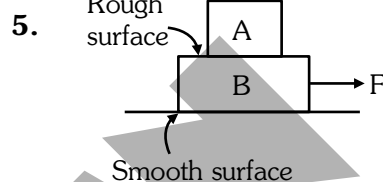
- (1) [(A - P), (B - Q), (C - R), (D - S)]
 (2) [(A - Q), (B - S), (C - R), (D - P)]
 (3) [(A - S), (B - Q), (C - P), (D - R)]
 (4) [(A - S), (B - Q), (C - R), (D - P)]

3. Select the correct statements -
- Work done by conservative force along closed path is zero.
 - A body with negative energy can't have linear momentum
 - Net work done by all the internal force of a system is independent of choice of reference frame
 - Work done by a non conservative force is always negative
- (1) a, b and c (2) a and c
 (3) a, b and d (4) All

4. Select the correct statements -

- Potential energy is defined only for gravitational force.
- Potential energy can be associated with frictional force.
- Work done by spring force is equal to change in potential energy of the spring.

- (1) a and b
 (2) a and c
 (3) all
 (4) none of these



If both blocks A and B are moving together, then select correct statements -

- Work done by the friction on block 'A' is zero
- Work done by the friction on block 'A' is positive
- Work done by the friction on block 'B' is negative
- Work done by friction forces on system (A and B) are zero

- (1) a, c and d
 (2) b, c and d
 (3) a and d
 (4) none of these

6. For a system of particles -

- The work done by all external forces is equal to change in kinetic energy
- The work done by all forces is equal to change in mechanical energy

Select the correct statements :

- (1) only a
 (2) only b
 (3) both a and b
 (4) none of these

7. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Kinetic energy of a system can be increased without applying external force on the system.

Reason (R) : If external forces are absent then work done by internal forces is equal to change in kinetic energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

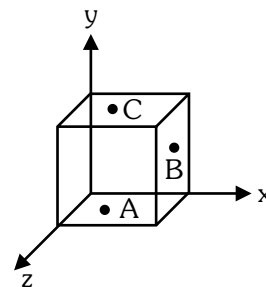
Assertion (A) : Work done is positive when force acts in the direction of displacement.

Reason (R) : Work done by frictional force cannot be positive.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

9. A particle is shifted from A to B and then from B to C where A, B and C are the midpoints of the corresponding faces of a cube of side 2m. If a force $\vec{F} = (3\hat{i} + 4\hat{j} - 5\hat{k})\text{N}$ is continuously acting on the particle, then select incorrect alternative



- (1) work done from A to B is 7 J
- (2) work done from B to C is 1 J
- (3) work done A to C is 8 J
- (4) force $\vec{F}$, is nonconservative force

10. **Statement : I** : Two balls of different masses are thrown vertically upwards with same speed. They will pass through their point of projection in the downward direction with the same speed in absence of air resistance.

and

Statement II : In absence of air resistance, the mechanical energy of a projectile is conserved.

- (1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.
- (2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.
- (3) Statement-I is true, statement-II is false.
- (4) Statement-I is false, statement-II is true.

11. Assume that in space potential energy U depends on x-coordinate only by :

$$U = (x - 3)^2 (2 - x)$$

A mass m can be kept on x-axis. In each position indicated in column-I, comment on the situation from column-II.

Column I

Column II

- | | |
|---------------|--------------------------|
| (A) $x = 7/3$ | (P) Stable equilibrium |
| (B) $x = 2$ | (Q) Unstable equilibrium |
| (C) $x = 3$ | (R) Neutral equilibrium |
| | (S) Not in equilibrium |

Options :-

- (1) $A \rightarrow P, B \rightarrow S, C \rightarrow R$
- (2) $A \rightarrow S, B \rightarrow P, C \rightarrow Q$
- (3) $A \rightarrow P, B \rightarrow S, C \rightarrow Q$
- (4) $A \rightarrow Q, B \rightarrow S, C \rightarrow P$

- 12.** The potential energy (in joules) function of a particle in a region of space is given as :

$$U = (2x^2 + 3y^3 + 2z)$$

Here x , y and z are in metres. Find the magnitude of x component of force (in newton) acting on the particle at point P (1m, 2m, 3m).

- (1) 4 (2) 2
(3) 6 (4) None

TG: @Chalnaayaaar

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans.	2	4	2	4	2	4	1	3	4	1	3	1	

CIRCULAR MOTION

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In circular motion, work done by the centripetal force is always zero.

Reason (R) : Centripetal force is always perpendicular to the instantaneous displacement. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

2. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In uniform circular motion speed of body is constant but its velocity is variable.

Reason (R) : In uniform circular motion tangential acceleration of body is zero, but body has normal acceleration

In the light of the above statements, choose the most appropriate answer from the options given below:

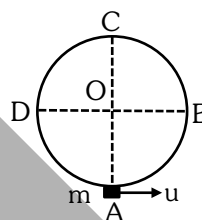
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

3. For a body moving in a circular path of radius $R = 1\text{m}$ in anticlockwise direction with constant speed V .

	Column-I		Column-II
(A)	Angle between instantaneous velocity and instantaneous acceleration	(P)	Non-zero
(B)	Total acceleration of the body	(Q)	Zero
(C)	Tangential acceleration of the body	(R)	$\pi/2$
		(S)	$\pi/3$

- (1) $A \rightarrow R, B \rightarrow P, C \rightarrow Q$
- (2) $A \rightarrow P, B \rightarrow Q, C \rightarrow R$
- (3) $A \rightarrow Q, B \rightarrow R, C \rightarrow P$
- (4) $A \rightarrow R, B \rightarrow Q, C \rightarrow P$

4. A particle suspended from a string of length ℓ is given a horizontal speed $u = 3\sqrt{g\ell}$ at the bottom. Then for the particle match the following column –



	Column-I		Column-II
(A)	Speed at B	(P)	7 mg
(B)	Speed at C	(Q)	$\sqrt{5g\ell}$
(C)	Tension in string at B	(R)	$\sqrt{7g\ell}$
(D)	Tension in string at C	(S)	4 mg

- (1) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$
- (2) $A \rightarrow R, B \rightarrow P, C \rightarrow Q, D \rightarrow S$
- (3) $A \rightarrow P, B \rightarrow Q, C \rightarrow S, D \rightarrow R$
- (4) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow R$

5. A body is connected to a string of length r and revolved in vertical circle with one end of string as the centre of circle. Its velocity at bottom most point v_L is twice that of its value v_H at the top most point. Then

Choose correct statement –

- (A) Velocity at highest point is $\sqrt{4gr/3}$
- (B) $v_L^2 - v_H^2$ is equal to $4gr$
- (C) Tension in string at bottom most point is 6 mg
- (D) Ratio of maximum tension to minimum tension in string is $10/9$

- (1) A & C
- (2) A & B
- (3) B & D
- (4) B & C

6. Speed of a body moving in a circular path changes with time as $v = 2t$, then –
 (A) Magnitude of acceleration remains constant
 (B) Magnitude of acceleration increases
 (C) Angle between velocity and acceleration remains constant
 (D) Angle between velocity and acceleration increases
 (1) A & C (2) A & D
 (3) B & C (4) B & D
7. A car of mass M is moving on a horizontal circular path of radius r . At an instant its speed is v and is increasing at a rate ' a '. Choose correct statements
 (A) the acceleration of the car is towards the centre of the path
 (B) the magnitude of the frictional force on the car is greater than mv^2/r
 (C) magnitude of its net acceleration will remain constant
 (D) magnitude of its net acceleration will be varying with time
 (1) A, B & C (2) A, B & D
 (3) B & C (4) B & D
8. A person applies a constant force $\vec{F}$ on a particle of mass m and finds that the particle moves in a circle of radius r with a uniform speed v . Choose correct statements
 (A) this is not possible
 (B) there are other forces also on the particle
 (C) the resultant of other forces is mv^2/r towards centre
 (D) the resultant of the other forces varies in magnitude as well as direction
 (1) B, C (2) only A
 (3) B, D (4) All are correct

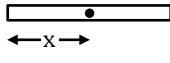
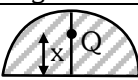
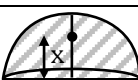
ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	1	1	1	1	2	4	4	3

CENTRE OF MASS

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Two blocks of masses m_1 & m_2 are at rest. They are now moving towards each other under a mutual internal force. The velocity of centre of mass is zero.
Reason (R) : If no external force acts on the system, then velocity of centre of mass remains unchanged but can never be zero.
In the light of the above statements, choose the most appropriate answer from the options given below:
 - (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) **(A)** is false but **(R)** is true.
2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Quick collisions are more violent than slow collisions.
Reason (R) : Quick collision are inelastic in nature.
In the light of the above statements, choose the most appropriate answer from the options given below:
 - (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) **(A)** is false but **(R)** is true.
3. Which of following statement is true.
 - (1) COM of a rigid body is always inside the body.
 - (2) Mass must be present physically at COM.
 - (3) COM of two particle system is closer to particle with greater mass.
 - (4) All above statements are true.
4. Which of following statement is false.
 - (1) KE in an elastic collision between two bodies remains same before and after collision.
 - (2) There may or may not be physical contact between two bodies for collision to occur.
 - (3) KE of two object system remains conserved during collision in elastic collision.
 - (4) None of the above
5. A projectile is projected at a speed u at an angle θ with horizontal. At the highest point projectile splits into two fragments of mass ratio 1 : 2. The smaller fragment coming to rest. Then consider following statements :-
 - (A) In a given time both parts will cover equal vertical displacement.
 - (B) COM of two fragments will follow the same path as the object would have followed if not exploded.
 - (C) Linear momentum is conserved in vertical direction only.
 Which of the following is correct ?
 - (1) A, B, C
 - (2) A, B
 - (3) B, C
 - (4) A, C
6. A particle of mass m moving with a velocity $(3\hat{i} + 2\hat{j})$ m/s collides with stationary body of mass M and finally moves with velocity $(-2\hat{i} + \hat{j})$ m/s then
 - (A) Impulse received by $m = m(5\hat{i} - 5\hat{j})$
 - (B) Impulse received by $m = m(-5\hat{i} - \hat{j})$
 - (C) Impulse received by $M = m(5\hat{i} + \hat{j})$
 - (D) Impulse received by $M = m(5\hat{i} - \hat{j})$
 then correct statements are
 - (1) B, C
 - (2) A, C
 - (3) B, D
 - (4) A, D

7. Match the following columns with respective positions of centre of mass -

Column - I		Column - II	
(A)	 uniform rod length = R	(P)	$x = \frac{3R}{8}$
(B)	 uniform thin semicircular ring of radius R	(Q)	$x = \frac{R}{2}$
(C)	 uniform semicircular disc of radius R	(R)	$x = \frac{4R}{3\pi}$
(D)	 uniform solid hemisphere of radius R	(S)	$x = \frac{2R}{\pi}$

- (1) A : P, B : R, C : Q, D : S
 (2) A : Q, B : P, C : R, D : S
 (3) A : Q, B : S, C : R, D : P
 (4) A : S, B : P, C : Q, D : R

8. Particle of mass m , kinetic energy K and momentum P collides head on elastically with another particle of mass $2m$ at rest. After collision:

Column I		Column II	
(A)	Momentum of I st particle	(P)	$\frac{4}{3}P$
(B)	Momentum of II nd particle	(Q)	$\frac{K}{9}$
(C)	KE of I st particle	(R)	$\frac{8K}{9}$
(D)	KE of II nd particle	(S)	$-\frac{P}{3}$

- (1) A → S; B → P, C → Q, D → R
 (2) A → P; B → S, C → Q, D → R
 (3) A → P; B → S, C → R, D → Q
 (4) A → S; B → P, C → R, D → Q

ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	3	3	3	3	2	1	3	1

ROTATIONAL MOTION

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Moment of inertia of a particle is a scalar quantity.

Reason (R) : Moment of inertia of a particle about any axis is given by $m(\vec{r} \cdot \vec{r})$ where m is mass of particle and $\vec{r}$ is position of particle w.r.t. axis.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Moment of inertia of a disc about an axis perpendicular to its plane passing through its centre is $\frac{mR^2}{2}$

Reason (R) : Moment of inertia of a disc about any axis passing through its plane remains equal.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

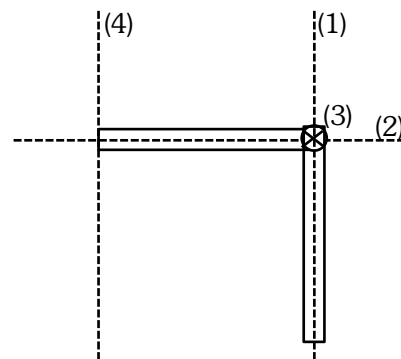
Assertion (A) : In a conical pendulum angular momentum about its vertical axis remains constant

Reason (R) : Net torque about vertical axis of conical pendulum is not zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

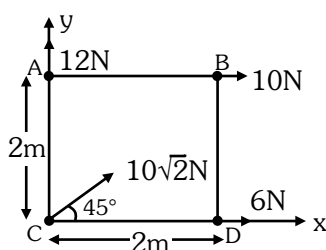
4. In diagram axis (3) is perpendicular to plane. Both rods are mass m and length ' ℓ ' then moment of inertial of system



i	About axis (1)	P	$\frac{m\ell^2}{3}$
ii	About axis (2)	Q	$\frac{4m\ell^2}{3}$
iii	About axis (3)	R	$\frac{2m\ell^2}{3}$
iv	About axis (4)		

- (1) i – P, ii – P, iii – R, iv – Q
- (2) i – Q, ii – Q, iii – P, iv – R
- (3) i – R, ii – R, iii – P, iv – Q
- (4) i – P, ii – R, iii – Q, iv – P

5. Value of net torque about given points.



i	Torque about A	P	$-64\hat{k}$
ii	Torque about B	Q	$32\hat{k}$
iii	Torque about C	R	$-12\hat{k}$
iv	Torque about D	S	$-20\hat{k}$

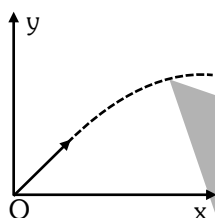
(1) i - P, ii - Q, iii - R, iv - S

(2) i - Q, ii - P, iii - P, iv - S

(3) i - Q, ii - R, iii - S, iv - P

(4) i - R, ii - S, iii - P, iv - S

6. A particle is projected from point 'O' having mass 'm' as shown in figure, moving on projectile path



i	Torque about 'O'	P	Zero
ii	Torque about x-axis	Q	Constant non zero
iii	Angular momentum about O	R	Increasing
iv	Angular momentum about x-axis	S	Decreasing

(1) i - P, ii - Q, iii - R, iv - S

(2) i - R, ii - P, iii - R, iv - P

(3) i - Q, ii - R, iii - R, iv - P

(4) i - P, ii - R, iii - R, iv - S

7. Four identical rods each of mass m and length ' ℓ ' are joined to form a rigid square, centre is at origin and sides are parallel to x and y axis then moment of inertia about

i	An axis parallel to z -axis and passing through a corner	P	$\frac{5}{3}m\ell^2$
ii	One side	Q	$\frac{2}{3}m\ell^2$
iii	The x -axis	R	$\frac{4}{3}m\ell^2$
iv	The z -axis passing through centre	S	$\frac{10}{3}m\ell^2$

(1) i - S, ii - P, iii - Q, iv - R

(2) i - P, ii - Q, iii - R, iv - S

(3) i - Q, ii - R, iii - P, iv - S

(4) i - S, ii - P, iii - R, iv - Q

8. A sphere is released on a smooth inclined plane.

Its angular momentum is

- (A) Conserved about every point
(B) Conserved about point of contact only
(C) Conserved about centre of sphere only
(D) Conserved about any point on a line parallel to inclined plane and passing through centre of ball.

From above, true statement/s is/are

- (1) A, B, C (2) Only D
(3) C and D (4) Only C

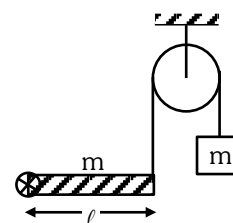
9. If lamina is in x - y plane then from theorem of perpendicular axis

- (A) $I_x - I_y = I_z$ (B) $I_x - I_z = I_y$
(C) $I_x + I_y = I_z$ (D) $I_y + I_z = I_x$

True statements is/are -

- (1) B and C (2) Only D
(3) Only C (4) C and D

10. Hinge is smooth and system is released from rest. Select correct statement -

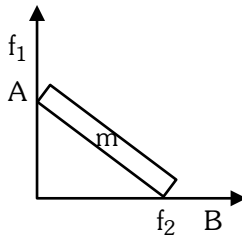


- (A) Acceleration of block is $g/2$ down ward
(B) Acceleration of block is $3g/8$ down ward
(C) Tension in string is $mg/2$
(D) Tension in string is $5mg/8$

True statements is/are

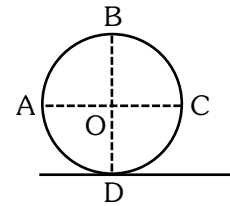
- (1) A, C (2) A, D
(3) B, C (4) B, D

- 11.** Wall and floor is rough. Rod is about to slide. Then find correct statement from the following (Given f_1 and f_2 are friction forces)



- (A) f_1 is in upward direction and f_2 towards the wall
 (B) normal on B is less than mg
 (C) normal on B is greater than mg
 (D) normal on B is equal to mg
 (1) A and B are true (2) A and C are true
 (3) A and D are true (4) Only D is correct

- 12.** A disc is rolling without friction then select correct statement from following –



- (A) w.r.t. earth kinetic energy of upper half is more than lower half
 (B) w.r.t. centre of disc K.E. of upper and lower half is same
 (C) w.r.t. centre of disc K.E. of part AOD and BOC is same
 (D) K.E. of parts of disc is independent of frame of reference
 (1) A, B, C (2) A, B only
 (3) D only (4) B, C only

TG: @Chalnaayaaar

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans.	1	3	3	1	3	2	1	2	3	4	1	1	

GRAVITATION

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : An artificial satellite is moving in a circular orbit of the earth. If the gravitational pull suddenly disappears, then it moves with the same speed tangential to the original orbit.
Reason (R) : The orbital speed of a satellite decreases with the increase in radius of the orbit.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 - (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) **(A)** is false but **(R)** is true.
2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Total energy is conserved in moving a satellite to higher orbit.
Reason (R) : For a satellite change in potential energy and kinetic energy is same in magnitude and opposite in nature.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 - (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) Both **(A)** and **(R)** are false.
3. As observed from earth, the sun appears to move in approximate circular orbit. For the motion of another planets like mercury as observed from the earth. This would :-
 - (1) be approximately
 - (2) not be true because the force between earth & mercury is not inverse square law
 - (3) not be true because the major gravitational force on mercury is due to sun
 - (4) not be true because mercury is influenced by forces other than gravitational
4. Choose the correct statement(s) from the following:-
 - (A) The magnitude of the gravitational force between two bodies of mass 1 kg each and separated by a distance of 1 m is 9.8 N.
 - (B) Higher the value of escape velocity for a planet, higher is the abundance of lighter gases in its atmosphere.
 - (C) Force of friction arises due to gravitational attraction.
 - (D) The gravitational force of attraction between two bodies of ordinary mass is not noticeable because the value of the gravitational constant is extremely small.
 (1) A, B (2) B, C, D (3) D (4) B, D
5. A satellite is moving round the earth in a circular orbit. The following statements are given :-
 - (i) It is moving with a constant velocity.
 - (ii) It has no acceleration.
 - (iii) It's angular momentum w.r.t. centre of the earth remains conserved.
 - (iv) Its distance from center must be equal to $\sqrt{2}$ times of earth's radius.
 The correct option is :-
 - (1) i & ii are true
 - (2) i, iii & iv are true
 - (3) only iii is true
 - (4) i & iv are true

6. Consider the following statement for a satellite S moving in an elliptical orbit around the earth (mass of satellite is negligibly small compared to earth)

I : The acc. of S is always towards earth.

II : The total mechanical energy of S varies periodically with time.

III : The linear momentum of 'S' remain constant in magnitude.

The correct sequence of true (T) and false (F) for the above statement is :-

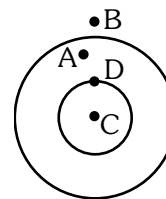
- (1) TTF (2) TFT (3) TFF (4) FFF

7. A satellite is in a circular equatorial orbit of radius 7000 km around the earth. If it is transferred to a circular orbit of double of radius :-

Column - I		Column - II	
(A)	Angular momentum	(P)	Increases
(B)	Area of earth covered by satellite signal	(Q)	Decreases
(C)	Potential energy	(R)	Becomes double
(D)	Kinetic energy	(S)	Becomes half

- (1) (A) → Q; (B) → P; (C) → P, R; (D) → Q, S
 (2) (A) → P; (B) → P; (C) → P; (D) → Q, S
 (3) (A) → Q; (B) → Q; (C) → P, S; (D) → P, R
 (4) (A) → P; (B) → Q; (C) → P; (D) → P, R

8. Two concentric spherical shells of non-zero masses are as shown in the figure :-



Column - I		Column - II	
(A)	Potential at A	(P)	Greater than B
(B)	Gravitational field at A	(Q)	Less than B
(C)	As one moves from C to D	(R)	Potential remains constant
(D)	As one moves from D to A	(S)	Gravitational field decreases
		(T)	Data insufficient

- (1) (A) → P; (B) → Q; (C) → R; (D) → S
 (2) (A) → Q; (B) → T; (C) → R; (D) → S
 (3) (A) → P; (B) → Q; (C) → S; (D) → R
 (4) (A) → Q; (B) → P; (C) → S; (D) → R

ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	2	4	1	4	3	1	2	2

PROPERTIES OF MATTER AND FLUID MECHANICS

1. Check the correctness of given statements :

Statement-I : Water wets glass capillary while mercury does not wet.

Statement-II : Excess pressure insides the soap bubble is inversely proportion to the radius of the bubble.

- (1) Both statements are correct.
- (2) Only statement-I is correct.
- (3) Only statement-II is correct.
- (4) Both statements are wrong.

2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The shape of small liquid drop is spherical.

Reason (R) : Due to surface tension, liquid tends to occupy minimum surface area and for given volume a sphere has minimum surface area.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When an ice cube, floating in a glass of water melts, the water level remains unchanged.

Reason (R) : The volume of ice on melting decreases.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

4. Match the column for followings

Column-A		Column-B	
i.	Bernoulli's theorem	A	$v = \sqrt{2gh}$
ii.	Law of continuity	B	$P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$
iii.	Torricelli's theorem	C	$Q = \frac{\pi P r^4}{8 \eta \ell}$
iv.	Poiseuilli's theorem	D	$Av = \text{constant}$

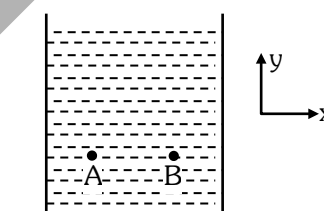
(1) $i \rightarrow B$, $ii \rightarrow D$, $iii \rightarrow A$, $iv \rightarrow C$

(2) $i \rightarrow B$, $ii \rightarrow A$, $iii \rightarrow D$, $iv \rightarrow C$

(3) $i \rightarrow C$, $ii \rightarrow D$, $iii \rightarrow A$, $iv \rightarrow B$

(4) $i \rightarrow A$, $ii \rightarrow B$, $iii \rightarrow C$, $iv \rightarrow D$

5. A container filled with liquid is placed on a smooth surface. Two points A and B are in same level as shown, state about pressures at these points for following cases, in which container accelerates along given direction



Column-A		Column-B	
i.	$a_x = 0$, $a_y \neq 0$	A	$P_A = P_B$
ii.	$a_x = 0$, $a_y = 0$	B	$P_A > P_B$
iii.	$a_x > 0$, $a_y = 0$	C	$P_A < P_B$
iv.	$a_x > 0$, $a_y > 0$		

(1) $i \rightarrow B$, $ii \rightarrow A$, $iii \rightarrow A$, $iv \rightarrow B$

(2) $i \rightarrow A$, $ii \rightarrow A$, $iii \rightarrow B$, $iv \rightarrow B$

(3) $i \rightarrow A$, $ii \rightarrow B$, $iii \rightarrow A$, $iv \rightarrow B$

(4) $i \rightarrow B$, $ii \rightarrow A$, $iii \rightarrow B$, $iv \rightarrow A$

6. There is a small hole near the bottom of an open tank filled with a liquid. The speed of the water ejected does not depend on (one or more options may be correct)

- (1) Area of the hole
- (2) Density of the liquid
- (3) Height of the liquid from the hole
- (4) Acceleration due to gravity

7. A solid floats in a liquid in a partially dipped position. (Select incorrect option)
- (1) The solid exerts a force, equal to its weight on the liquid.
 - (2) The liquid exerts a force of buoyancy on the solid which is equal to the weight of the solid.
 - (3) The weight of the displaced liquid equals the weight of the solid.
 - (4) The weight of the dipped part of the solid is equal to the weight of the displaced liquid.
8. A copper wire ($Y = 10^{11} \text{ Nm}^{-2}$) of length 8m and steel wire ($Y = 2 \times 10^{11} \text{ Nm}^{-2}$) of length 4m and each of cross-sectional area 0.5 cm^2 are fastened end to end and stretched with a tension of 500 N. Match the given columns and select the correct option from the codes given below.

Column-I

- Elongation in copper wire (in mm)
- Elongation in steel wire (in mm)
- Total elongation (in mm)

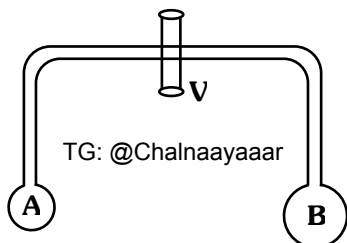
Column-II

- 0.2
- 1.0
- 0.8

Codes

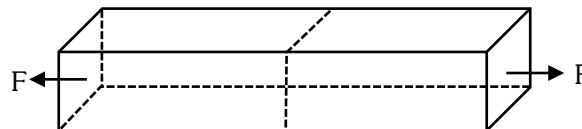
- (1) i - r, ii - p, iii - q
- (2) i - p, ii - q, iii - r
- (3) i - q, ii - r, iii - p
- (4) i - p, ii - r, iii - p

9. The valve V in the bent tubes initially kept closed. Two soap bubbles A (smaller) and B (larger) are formed at the two open ends of the tube. Valve V is now opened and air can flow freely between the bubbles.



- (1) There will be no change in the size of the bubbles
- (2) The bubbles will become of equal size
- (3) A will become smaller and B will become larger
- (4) The sizes of A and B will be interchanged

10. In the figure shown, forces of equal magnitude are applied to the two ends of a uniform rod. Consider A as the cross-sectional area of the rod. For this situation, mark out the correct statement.



- (1) The rod is in compressive stress
 - (2) The numerical value of stress developed in the rod is equal to F/A
 - (3) The stress is defined as internal restoring force per unit length
 - (4) None of the above
11. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
- Assertion (A) :** A needle placed carefully on the surface of water may float, whereas the ball of the same material will always sink.
- Reason (R) :** The buoyant force depends on the material and shape of body.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
 - (3) (A) is true but (R) is false.
 - (4) (A) is false but (R) is true.
12. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Spraying of water in form of small droplets causes cooling.

Reason (R) : For an isolated system, surface energy increase on the expense of internal energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

13. Given below are two statements :

Statement I : Viscosity of gases is greater than that of liquids.

Statement II : Surface tension of a liquid decreases due to the presence of insoluble impurities.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

14. Given below are two statements :

Statement I : For a substance the product of bulk modulus of elasticity and compressibility is 1.

Statement II : The shear modulus for an ideal rigid body is infinite.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

15. Match the column:-

Column 1		Column 2	
(i)	For Concave meniscus	(a)	angle of contact $> 90^\circ$
(ii)	For Convex meniscus	(b)	angle of contact $= 90^\circ$
(iii)	For Plane meniscus	(c)	angle of contact $< 90^\circ$

- (1) (i) $\rightarrow$ (c); (ii) $\rightarrow$ (a); (iii) $\rightarrow$ (b)
- (2) (i) $\rightarrow$ (a); (ii) $\rightarrow$ (b); (iii) $\rightarrow$ (c)
- (3) (i) $\rightarrow$ (a); (ii) $\rightarrow$ (c); (iii) $\rightarrow$ (b)
- (4) (i) $\rightarrow$ (b); (ii) $\rightarrow$ (a); (iii) $\rightarrow$ (c)

TG: @Chalnaayaaar

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	2	1	2	2	4	1	3	2	3	1	3	1	1

THERMAL PHYSICS

THERMAL EXPANSION

1. Match the column with scale of temperature and its range -

	Column-1		Column-2
(A)	Celsius scale	(P)	32° to 212°
(B)	Fahrenheit scale	(Q)	x° to y°
(C)	Kelvin scale	(R)	0° to 100°
(D)	Unknown scale	(S)	273.15 to 373.15

- (1) A → P, B → Q, C → R, D → S
 (2) A → R, B → P, C → S, D → Q
 (3) A → R, B → S, C → P, D → Q
 (4) A → R, B → Q, C → S, D → P

2. Choose the correct statement :-

- (a) When a bimetallic strip is heated it bends in the form of an arc.
 (b) It is used in thermostat
 (c) In bimetallic strip the material having lower temperature expansion co-efficient always goes on outer side.
 (d) A bimetallic strip on heating bends in the form of helix

- (1) (a, b) (2) (b, c) (3) (a, c) (4) (b, d)

3. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A brass disc is just fitted in a hole in steel plate. The system must be cooled to loosen the disc from the hole.

Reason (R) : The coefficient of linear expansion for brass is greater than the coefficient of linear expansion for steel.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

CALORIMETRY

4. If heat is supplied to a solid, its temperature :-

- (A) must increase
 (B) may increase
 (C) may remain constant
 (D) may decrease

Choice the correct statement :-

- (1) A (2) B,D
 (3) B,C,D (4) B,C

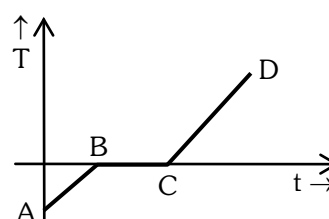
5. A good cooking utensil should have :-

- (A) low specific heat
 (B) high specific heat
 (C) low thermal capacity
 (D) high coefficient of thermal conductivity

Which statement are correct :-

- (1) A,C,D (2) B,C,D
 (3) A,B,D (4) B,C

6. Heat is supplied to a ice at a constant rate temperature variation with time is as shown in figure, then -



- (A) During AB volume of substance increases
 (B) During BC volume of substance decreases
 (C) Specific heat of substance in liquid phase is proportional to reciprocal of slope of portion AB of graph
 (D) Latent heat of fusion of substance is independent of portion AB of graph.

Then correct option is -

- (1) A,C (2) B,C
 (3) A,C,D (4) B,D

7. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In change of state from solid to liquid the temperature decrease until the entire amount of the solid substance melts.

Reason (R) : The phenomenon of refreezing is called melting.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) Both **(A)** and **(R)** are false.

8. Match the column :-

	Column-1		Column-2
(A)	Heat	(P)	Degree of hotness or coolness of a substance
(B)	Internal energy	(Q)	Amount of heat required to rise the temperature of unit mass of a substance by 1°C
(C)	Temperature	(R)	Energy in transit
(D)	Specific heat	(S)	The sum of energy due to attractive force between molecules and due to random motion

- (1) $A \rightarrow R, B \rightarrow S, C \rightarrow P, D \rightarrow Q$
- (2) $A \rightarrow R, B \rightarrow P, C \rightarrow S, D \rightarrow Q$
- (3) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$
- (4) $A \rightarrow R, B \rightarrow S, C \rightarrow Q, D \rightarrow P$

HEAT TRANSFER

9. Assume transmissivity $t = 0$ for all the cases – Which of the following statement is correct –

- (A) Bad absorber is bad emitter
- (B) Bad absorber is good reflector
- (C) Bad reflector is good emitter
- (D) Bad emitter is good absorber

- (1) A, B, C, D (2) A, C, D
- (3) A, B, D (4) A, B, C

10. Match the column in which column-1 gives the statement and column-2 its explanation by which law.

	Column-1		Column-2
(A)	When a person enters in an A.C. room then he feels cooler	(P)	Newtons law of cooling
(B)	Good absorber are good emitter of heat	(Q)	Wein's law
(C)	Colour of a star is an indication of temperature	(R)	Kirchoff's law
(D)	A hot cup of tea when placed in a room is cooled in some time	(S)	Prevost theory of mutual heat exchange

- (1) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
- (2) $A \rightarrow Q, B \rightarrow S, C \rightarrow P, D \rightarrow R$
- (3) $A \rightarrow S, B \rightarrow R, C \rightarrow Q, D \rightarrow P$
- (4) $A \rightarrow R, B \rightarrow P, C \rightarrow S, D \rightarrow Q$

11. A spherical black body of radius r radiates power P and its rate of cooling is R then –

- (a) $P \propto r$
- (b) $P \propto r^2$
- (c) $R \propto r^2$
- (d) $R \propto \frac{1}{r}$

Choice the correct one :-

- (1) a, b (2) a, c (3) b, d (4) a, b, d

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : For a given amount of an ideal gas, at constant temperature, the product of pressure and volume is constant.

Reason (R) : The root mean square velocity of the molecules is inversely proportional to molecular weight.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) Both **(A)** and **(R)** are false.

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : For higher temperature, the peak emission wavelength of a black body shifts to lower wavelengths.

Reason (R) : Peak emission wavelength of a black body is proportional to the fourth power of temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

KTG

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The rms speed of helium gas is equal to rms speed of hydrogen gas when absolute temperature of helium is twice that of hydrogen gas.

Reason (R) : RMS speed of gas depends only on the temperature of gas.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

15. In a mixture of nitrogen and helium kept at room temperature, as compared to a helium molecule nitrogen molecule strikes into the wall –

- (a) with greater average speed
- (b) with smaller average speed
- (c) with greater average kinetic energy
- (d) with smaller average kinetic energy

Choose the correct statements :-

- (1) a, c (2) b, d (3) a, d (4) b, c

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : An ideal gas is enclosed within a container fitted with a piston. When volume of this enclosed gas is increased at constant temperature. The pressure exerted by gas on piston decreases.

Reason (R) : In the above situation, the rate of molecules per unit area striking on the piston decreases.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Evaporation has a cooling effect.

Reason (R) : At the surface of liquid, some of the faster upward moving molecules have enough kinetic energy to overcome the attraction from other molecules and escape from the liquid. With these faster molecules the average kinetic energy left behind is reduced.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

18. For a monoatomic gas at temperature T , match the following column –

	Column-1		Column-2
(A)	Speed of sound	(P)	$\sqrt{\frac{2RT}{M_w}}$
(B)	RMS speed of gas molecule	(Q)	$\sqrt{\frac{8RT}{\pi M_w}}$
(C)	Average speed of gas molecule	(R)	$\sqrt{\frac{3RT}{M_w}}$
(D)	Most probable speed of gas molecule	(S)	$\sqrt{\frac{\gamma RT}{M_w}}$

- (1) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
- (2) $A \rightarrow Q, B \rightarrow S, C \rightarrow P, D \rightarrow R$
- (3) $A \rightarrow R, B \rightarrow P, C \rightarrow S, D \rightarrow Q$
- (4) $A \rightarrow S, B \rightarrow R, C \rightarrow Q, D \rightarrow P$

THERMODYNAMICS

19. Match the following column –

	Column-1		Column-2
(A)	Adiabatic bulk modulus	(P)	$-P/V$
(B)	Slope of PV curve in isothermal process	(Q)	$\frac{2}{\gamma - 1}$
(C)	Degree of freedom	(R)	γP
(D)	Molar heat capacity at constant pressure divide by R	(S)	$\frac{\gamma}{\gamma - 1}$

- (1) $A \rightarrow R, B \rightarrow P, C \rightarrow Q, D \rightarrow S$
- (2) $A \rightarrow P, B \rightarrow R, C \rightarrow S, D \rightarrow Q$
- (3) $A \rightarrow Q, B \rightarrow S, C \rightarrow P, D \rightarrow R$
- (4) $A \rightarrow S, B \rightarrow Q, C \rightarrow R, D \rightarrow P$

20. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

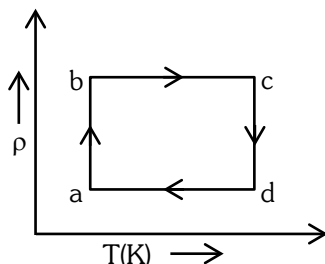
Assertion (A) : In an adiabatic process, change in internal energy of a gas is equal to workdone on the gas in the process.

Reason (R) : Temperature of gas remains constant in a adiabatic process.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

21. In the $\rho - T$ graph (for fixed mass of an ideal gas), shown in the figure match the following ($\rho \rightarrow$ density, $T \rightarrow$ temperature)



	Column-1		Column-2
(A)	Process ab	(p)	isochoric process
(B)	Process bc	(q)	$\Delta U = 0$
(C)	Process cd	(r)	P increasing
(D)	Process da	(s)	P decreasing

- (1) $A \rightarrow q, s$, $B \rightarrow p, q$, $C \rightarrow p, s$, $D \rightarrow r, q$
 (2) $A \rightarrow q, r$, $B \rightarrow p$, $C \rightarrow q, s$, $D \rightarrow p$
 (3) $A \rightarrow p, s$, $B \rightarrow q, r$, $C \rightarrow r, s$, $D \rightarrow s, r$
 (4) $A \rightarrow p, q$, $B \rightarrow q, r$, $C \rightarrow r, s$, $D \rightarrow s, p$

22. Match the column –

	Column-1		Column-2
(A)	$dU = nC_v dT$ is valid for	(P)	Adiabatic process
(B)	Temperature of the system can change in	(Q)	Isothermal process
(C)	$Q = dU + W$ is valid for	(R)	Polytropic process
(D)	The process in which heat exchange between the system and surrounding is zero.	(S)	Free expansion of ideal gas

- (1) $A \rightarrow P, Q, R, S$; $B \rightarrow P, R, S$; $C \rightarrow P, Q, R, S$; $D \rightarrow P, S$
 (2) $A \rightarrow P, Q, R, S$; $B \rightarrow P, Q$; $C \rightarrow P, Q, R, S$; $D \rightarrow Q, R$
 (3) $A \rightarrow P, Q, R, S$; $B \rightarrow P, S$; $C \rightarrow P, Q, R, S$; $D \rightarrow Q, R$
 (4) $A \rightarrow P, Q, R, S$; $B \rightarrow Q, S$; $C \rightarrow P, Q, R, S$; $D \rightarrow Q, R$

23. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : For an ideal gas $C_p > C_v$.

Reason (R) : Work is done in expansion of the gas at constant pressure.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

24. The internal energy of a system remains constant when it undergoes –

- (a) a cyclic process
 (b) an isothermal process
 (c) an adiabatic process
 (d) any process in which heat released by the system is equal to the work done on the system.

Choose the correct statements :-

- (1) a, b, c (2) b, c (3) a, b (4) a, b, d

25. The molar heat capacity for an ideal gas -

- (a) is zero for adiabatic process.
 (b) is infinite for an isothermal process
 (c) depends on the nature of gas for a process in which either volume or pressure is constant
 (d) is equal to the product of the molecular weight and gram specific heat capacity for any process.

Choose the correct statements :-

- (1) a, b (2) c, d
 (3) a, b, c & d (4) b, c, d

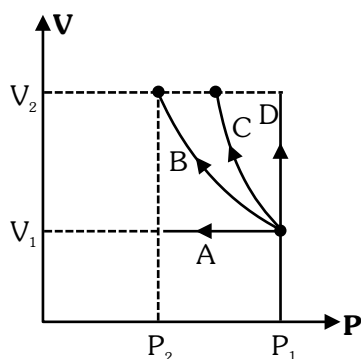
26. Match the following column –

	Column-1		Column-2
(A)	Adiabatic expansion	(P)	No work done
(B)	isobaric expansion	(Q)	Constant internal energy
(C)	Isothermal expansion	(R)	Increase in internal energy
(D)	Isochoric process	(S)	Decrease in internal energy

- (1) $A \rightarrow S$, $B \rightarrow R$, $C \rightarrow Q$, $D \rightarrow P$
 (2) $A \rightarrow P$, $B \rightarrow Q$, $C \rightarrow R$, $D \rightarrow S$
 (3) $A \rightarrow Q$, $B \rightarrow P$, $C \rightarrow S$, $D \rightarrow R$
 (4) $A \rightarrow R$, $B \rightarrow S$, $C \rightarrow P$, $D \rightarrow Q$

27. Volume versus pressure curves for one mole of an ideal gas are given for four processes as shown in figure.

(B $\rightarrow$ Adiabatic process, C $\rightarrow$ Isothermal process)



	Column-I		Column-II
(A)	For process A	(P)	Work done by the gas is positive.
(B)	For process B	(Q)	Temperature will increase.
(C)	For process C	(R)	Heat supplied is positive.
(D)	For process D	(S)	Change in internal energy is negative.

The correct option is :-

- (1) (A) – S, (B) – PS, (C) – PQ, (D) – PQR
 (2) (A) – PS, (B) – P, (C) – PR, (D) – PQQ
 (3) (A) – PS, (B) – S, (C) – PR, (D) – PQ
 (4) (A) – S, (B) – PS, (C) – PR, (D) – PQR

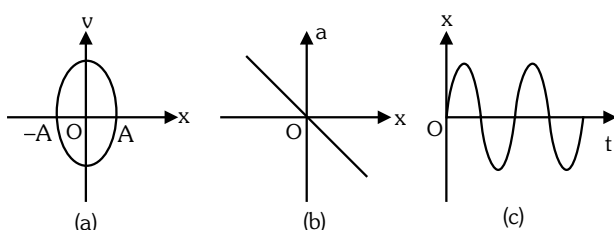
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	4	1	4	4	1	4	3	3	2	3	3	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans.	1	1	4	1	3	2	1	1	4	3	1	4			

OSCILLATIONS (S.H.M.)

1. The equation of motion for an oscillating particle is given by
 $x = 3\sin 4\pi t + 4\cos 4\pi t$
 where x is in cm and t is in sec. Choose the correct statements :
- (a) the motion is simple harmonic
 (b) amplitude of oscillation is 5 cm
 (c) motion is periodic
- (1) a, b, c (2) Only a, b
 (3) Only b, c (4) Only b

2. For a particle executing S.H.M.



Select the correct graph.

- (1) only a
 (2) only b
 (3) both (a) & (b)
 (4) all (a), (b) & (c)
3. Match the column

Column-I		Column-II	
(i)	$t=0$ M.P.	(a)	$x = A\cos\omega t$
(ii)	$t=0$ M.P. +A	(b)	$x = A\sin\omega t$
(iii)	$t=0$ M.P.	(c)	$x = -A\sin\omega t$
(iv)	$t=0$ -A M.P.	(d)	$x = -A\cos\omega t$

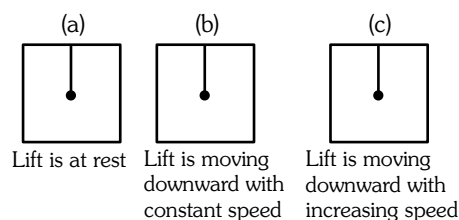
Choose the correct option :

- (1) i→b, ii→d, iii→a, iv→c
 (2) i→c, ii→d, iii→b, iv→a
 (3) i→b, ii→a, iii→c, iv→d
 (4) i→a, ii→c, iii→d, iv→b

4. A spring has natural length 40 cm and spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and other end of the spring is attached to the wall in horizontal position. The block release from the position, where the spring has length 45 cm then choose the correct statements :

- (A) The block will perform SHM of amplitude 5 cm
 (B) The block will have maximum velocity $50\sqrt{5}$ cm/sec.
 (C) The block will have maximum acceleration at released position
 (D) The minimum P.E. of the spring will be zero
- (1) only (A)
 (2) only (B)
 (3) both (A) & (B)
 (4) (A), (B), (C), (D)

5. A simple pendulum is suspended from the ceiling of a lift. Three cases of lift are given



In which case time period of pendulum is

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

- (1) only in (a)
 (2) only in (b)
 (3) only in (c)
 (4) in both (a) & (b)

6. A particle is executing linear SHM. For this match the following :-

i	If mass of the block is doubled (keeping other factor constant)	a	Energy of oscillation becomes 4 times
ii	If amplitude of oscillation is doubled (keeping other factor constant)	b	Max. speed of the particle becomes half times
iii	If force constant is doubled (keeping other factor constant)	c	Time period become $\sqrt{2}$ times
iv	If angular freq. is halved (keeping mass constant)	d	Max. potential energy becomes 2 times

- (1) i - c, ii - a, iii - d, iv - b
 (2) i - c, ii - a, iii - b, iv - d
 (3) i - a, ii - c, iii - b, iv - d
 (4) i - c, ii - b, iii - d, iv - a

7. A particle performing SHM with time period T according to the equation $x = A \sin \omega t$ then match the following :-

i	time taken to displace from $x = 0$ to $x = A/2$	a	$\frac{T}{24}$
ii	time taken to displace from $x = A/2$ to $x = A$	b	$\frac{T}{8}$
iii	time taken to go from $x = A/\sqrt{2}$ to $x = \sqrt{3} A/2$	c	$\frac{T}{12}$
iv	time taken to displace from $x = A/\sqrt{2}$ to $x = A$	d	$\frac{T}{6}$

- (1) i - c, ii - a, iii - d, iv - b
 (2) i - c, ii - a, iii - b, iv - d
 (3) i - c, ii - d, iii - a, iv - b
 (4) i - c, ii - b, iii - d, iv - a

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : General vibration of a polyatomic molecule about its equilibrium position is periodic but not SHM.

Reason (R) : A periodic motion can always be expressed as a sum of infinite number of harmonic motion with appropriate amplitude.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : All simple harmonic motion are periodic.

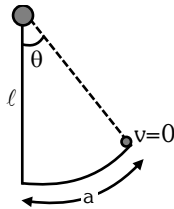
Reason (R) : Periodic motion is the motion in which the particle repeats its motion in a fixed time interval.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

10. Given below are two statements:

Statement I : The motion of a simple pendulum is simple harmonic only if $\theta \ll \ell$.



Statement II : The motion of a simple pendulum is simple harmonic for large angular displacement.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

11. Given below are two statements:

Statement I : SHM is not an example of uniformly accelerated motion.

Statement II : SHM is the projection of uniform circular motion along one of the diameter of the circle.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

12. Given below are two statements:

Statement I : In linear SHM net force always act in the direction of velocity.

Statement II : In linear SHM direction of velocity changes.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

13. Given below are two statements:

Statement I : In damped oscillation amplitude of the particle decreases exponentially with time.

Statement II : In forced oscillation amplitude of a particle can be maintained constant with time.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

14. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A small body suspended by a light spring performing SHM. When the entire system is immersed in a non-viscous liquid period of oscillation does not change.

Reason (R) : In simple harmonic motion the acceleration is always in a direction opposite to that of the displacement from mean position.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.
15. Motion of a ball bearing inside a smooth fixed bowl, when released from a point slightly above the lower point is
 (A) simple harmonic motion.
 (B) non-periodic motion.
 (C) periodic motion.
 (D) non-oscillatory motion
 (1) A,C (2) B,D (3) C,D (4) None
16. A particle moves such that its position from origin is given by $x = 2 + 3 \sin^2 \omega t$. Choose **CORRECT** statement(s) about its motion :-

(A) Particle performs S.H.M. of amplitude $\frac{3}{2}$ m

and frequency $\frac{\omega}{\pi}$

- (B) Particle performs oscillations but no S.H.M.
 (C) Particle performs S.H.M. about mean position at $x = \frac{7}{2}$ m
 (D) Particle performs oscillations of amplitude 3m
 (1) B,D (2) A,C (3) A,C,D (4) None

17. **STATEMENT - I :** In SHM let x be the maximum speed, y the frequency of oscillation and z the maximum acceleration, then $\left(\frac{xy}{z}\right)$ is a constant quantity.

and

STATEMENT - II : This is because $\left(\frac{xy}{z}\right)$ becomes a dimensionless quantity

- (1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.
 (2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.
 (3) Statement-I is true, statement-II is false.
 (4) Statement-I is false, statement-II is true.
18. **Statement-I :** Total energy in SHM will be always more than or equal to potential energy.
 and
Statement-II : Total energy in SHM will be always more than or equal to kinetic energy.
 (1) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I.
 (2) Statement-I is True, Statement-II is True ; Statement-II is not a correct explanation for Statement-I.
 (3) Statement-I is True, Statement-II is False.
 (4) Statement-I is False, Statement-II is True.
19. A vertical spring block system is made to oscillate.

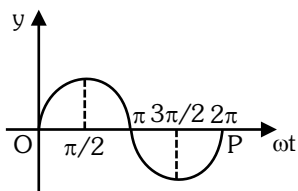
Statement I : Its time period on earth is more than that on the moon.

and

Statement II : Its extension on earth (in equilibrium) is more than that on the moon.

- (1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.
 (2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.
 (3) Statement-I is true, statement-II is false.
 (4) Statement-I is false, statement-II is true.

20. The graph plotted between phase angle (ϕ) and displacement of a particle from equilibrium position (y) is a sinusoidal curve as shown below. Then the best matching is



Column A		Column B	
(a)	K.E. versus phase angle curve	(i)	
(b)	P.E. versus phase angle curve	(ii)	
(c)	T.E. versus phase angle curve	(iii)	
(d)	Velocity versus phase angle curve	(iv)	

- (1) (a)-(i), (b)-(ii), (c)-(iii) & (d)-(iv)
 (2) (a)-(ii), (b)-(i), (c)-(iii) & (d)-(iv)
 (3) (a)-(ii), (b)-(i), (c)-(iv) & (d)-(iii)
 (4) (a)-(ii), (b)-(iii), (c)-(iv) & (d)-(i)

21. **Statement I** : Displacement-time equation of a particle moving along x-axis is $x=4 + 6 \sin \omega t$. Under this situation, motion of particle is not simple harmonic.

and

Statement II : for the given equation $\frac{d^2x}{dt^2}$ is not proportional to $-x$.

- (1) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I.
 (2) Statement-I is True, Statement-II is True ; Statement-II is not a correct explanation for Statement-I.
 (3) Statement-I is True, Statement-II is False.
 (4) Statement-I is False, Statement-II is True.

22. The angle made by the string of a simple pendulum with the vertical depends upon time as

$$\theta = \frac{\pi}{90} \sin \pi t. \text{ Find the length of the pendulum}$$

$$\text{if } g = \pi^2 \text{ ms}^{-2}$$

- (1) 1 m (2) 1 cm
 (3) 10 m (4) 10 cm

23. **Statement-I** : Cavalry troops are asked not to march in cadence across a suspension bridge otherwise bridge may collapse.

Statement-II : If the soldiers march in cadence, in every step they step down at exactly the same instant producing a very large impulse on the bridge, which may cause it to collapse.

- (1) Statement-I is true, statement-II is true; statement-II is a correct explanation for statement-I.
 (2) Statement-I is true, statement-II is true; statement-II is not a correct explanation for statement-I.
 (3) Statement-I is true, statement-II is false.
 (4) Statement-I is false, statement-II is true.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	4	3	4	4	1	3	2	1	2	1	3	1	2	1
Que.	16	17	18	19	20	21	22	23							
Ans.	2	2	3	4	2	4	1	3							

WAVE MOTION & DOPPLER'S EFFECT

1. Given below are two statements:

Statement I : For same length but different gas COP fundamental frequency are different.

Statement II : For same length but different gas COP fundamental wavelength remains same.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

2. Given below are two statements:

Statement I : Mechanical transverse waves cannot be generated in gaseous medium.

Statement II : Mechanical transverse waves can be produced only in such medium which have shearing property.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

3. Consider the following statements :

Statement I : The velocity of sound in the air increases due to presence of moisture in it.

Statement II : The presence of moisture in air lowers the density of air.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

4. The equation of travelling wave is given by (all quantities are in SI units) $y = 0.02 \sin 2\pi (10t - 5x)$

	Column-I		Column-II
i	Speed of wave	a	20π
ii	Angular frequency of wave	b	0.4π
iii	Wavelength of wave	c	2
iv	Maximum particle speed	d	0.2

(1) $i \rightarrow c$, $ii \rightarrow a$, $iii \rightarrow d$, $iv \rightarrow c$

(2) $i \rightarrow a$, $ii \rightarrow b$, $iii \rightarrow c$, $iv \rightarrow d$

(3) $i \rightarrow c$, $ii \rightarrow b$, $iii \rightarrow a$, $iv \rightarrow d$

(4) $i \rightarrow c$, $ii \rightarrow a$, $iii \rightarrow d$, $iv \rightarrow b$

5. In a resonance tube experiment, a close organ pipe of length 120 cm resonate when tuned with a tuning fork of frequency 340 Hz. If water is poured in the pipe then (given $v_{\text{air}} = 340\text{m/sec.}$):

(a) Minimum length of water column to have the resonance is 45 cm.

(b) The distance between two successive nodes is 50 cm.

(c) The maximum length of water column to create the resonance is 95 cm

(d) None of these

Choose the correct statements :

(1) a, b, c (2) b, c (3) a, c (4) None

6. Regarding speed of sound in gas match the following -

	Column-I		Column-II
i	Temperature of gas is made 4 times and pressure 2 times	a	Speed becomes $2\sqrt{2}$ times
ii	Only pressure is made 4 times without change in temperature	b	Speed becomes two times
iii	Only temperature changed to 4 times	c	Speed remain unchanged
iv	Molecular mass of the gas is made 4 times	d	Speed become half

(1) $i \rightarrow a$, $ii \rightarrow c$, $iii \rightarrow d$, $iv \rightarrow b$

(2) $i \rightarrow b$, $ii \rightarrow c$, $iii \rightarrow d$, $iv \rightarrow a$

(3) $i \rightarrow b$, $ii \rightarrow c$, $iii \rightarrow b$, $iv \rightarrow d$

(4) $i \rightarrow c$, $ii \rightarrow b$, $iii \rightarrow a$, $iv \rightarrow c$

7. For a closed organ pipe match the following :

	Column-I		Column-II
i	Third overtone frequency is x times the fundamental frequency. Here, x is equal to	a	3
ii	No. of nodes in second overtone	b	4
iii	No. of antinode in third overtone	c	5
		d	None

(1) $i \rightarrow a$, $ii \rightarrow d$, $iii \rightarrow b$

(2) $i \rightarrow d$, $ii \rightarrow a$, $iii \rightarrow b$

(3) $i \rightarrow b$, $ii \rightarrow a$, $iii \rightarrow d$

(4) $i \rightarrow c$, $ii \rightarrow d$, $iii \rightarrow b$

8. In a stationary wave pattern, all the particles -

- of the medium vibrate in the same phase
- in the region between two antinodes vibrate in the same phase
- in the region between two successive nodes vibrate in the same phase
- on either side of a node vibrate in opposite phase

Choose the correct statement :

- a, c, d
- a, b, c, d
- c, d
- a, b, c

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Sound travels faster on a rainy day than on a dry day.

Reason (R) : Moisture increases the pressure.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (A)** is true but **(R)** is false.
- (A)** is false but **(R)** is true.

10. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : For a closed pipe, the first resonance length is 60 cm. The second resonance position will be obtained at 120 cm.

Reason (R) : In a closed pipe $\ell_2 = 3\ell_1$.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (A)** is true but **(R)** is false.
- (A)** is false but **(R)** is true.

11. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In stationary waves, energy is confined within the wave region.

Reason (R) : Every particle is stationary in a stationary wave.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (A)** is true but **(R)** is false.
- (A)** is false but **(R)** is true.

12. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When two vibrating tuning forks having frequencies 256 and 512 are held near each other, beats cannot be heard.

Reason (R) : The principle of superposition is valid only if frequencies of oscillators are nearly equal.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (A)** is true but **(R)** is false.
- (A)** is false but **(R)** is true.

- 13.** The tension in a stretched string fixed at both ends is changed by 2%, the fundamental frequency is found to get changed by 15 Hz.
- Wavelength of the string of fundamental frequency does not change
 - Velocity of propagation of wave changes by 2%
 - Velocity of propagation of wave changes by 1%
 - Original frequency is 1500 Hz
- Choose the correct statement(s) –
- a, b, c, d
 - a, b, d
 - a, c, d
 - a, b, c
- 14.** Velocity of sound in air is 320 m/s. A pipe closed at one end has a length of 1 m. Neglecting end corrections, the air column in the pipe can resonate for sound of frequency.
- 80 Hz
 - 240 Hz
 - 320 Hz
 - 400 Hz
- Choose the correct statements:
- a, b, c
 - a, b, d
 - b, c, d
 - a, b, c, d
- 15.** The equation for the displacement of a stretched string is given by $y = 8 \sin^2 \pi \left[\frac{t}{0.02} - \frac{x}{100} \right]$ where y and x are in cm and t in second
- | Column I | Column II |
|--|--------------|
| (A) Amplitude of wave in cm | (P) 50 |
| (B) Frequency of wave in Hz | (Q) 4 |
| (C) Velocity of wave in ms^{-1} | (R) 4π |
| (D) Maximum particle velocity in cm/s | (S) 5000 |
| | (T) 400π |
- Options :-
- (A) → (Q); (B) → (P); (C) → (P); (D) → (T)
 - (A) → (R); (B) → (S); (C) → (P); (D) → (T)
 - (A) → (Q); (B) → (P); (C) → (S); (D) → (T)
 - (A) → (R); (B) → (S); (C) → (S); (D) → (T)
- 16.** Which of the following actions would make a pulse travel faster along a stretched string ?
- Move your hand up and down more quickly as you generate the pulse
 - Use a heavier string of the same length, under the same tension
 - Use a lighter string of the same length, under the same tension
 - Stretch the string tighter to increase the tension.
- (1) A,B,D (2) B,D (3) C,D (4) A,C
- 17.** Sinusoidal waves travel on five identical strings. Use the mathematical forms of the waves given below. In the expressions given below x is in m and y is in millimeters and t is in seconds.
- $y = (2\text{mm}) \sin(2x - 4t)$
 - $y = (3\text{mm}) \sin(4x - 10t)$
 - $y = (2.5\text{mm}) \sin(6x - 12t)$
 - $y = (1\text{mm}) \sin(8x - 16t)$
 - $y = (4\text{mm}) \sin(10x - 20t)$
- Four of the strings have the same tension, but any one has a different tension. Identify it.
- a
 - b
 - d
 - e
- 18.** A wave disturbance in a medium is described by $y(x, t) = 0.02 \cos(50\pi t + \pi/2) \cos(10\pi x)$, where x and y are in metres and t in seconds.
- A node occurs at $x = 0.15$ m
 - An antinode occurs at $x = 0.3$ m
 - The speed of the wave is 5.0 m/s
 - The wavelength is 0.2 m
- A,C,D
 - B,C,D
 - A,B,C,D
 - C,D

ANSWER KEY

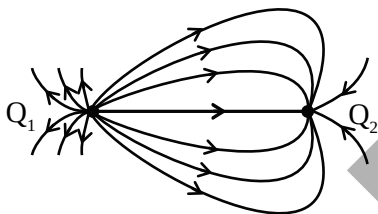
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	1	4	1	3	2	3	3	4	3	3	3	2	1
Que.	16	17	18	TG: @Chalnaayaaar											
Ans.	3	2	3												

ELECTROSTATICS

1. A ring with a uniform charge distribution with a total charge Q and radius R is placed in the yz plane with its centre at the origin then which of the following is incorrect about this situation:

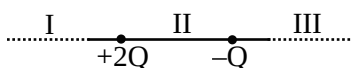
- (1) The electric field at the origin is zero
- (2) The electric potential at the origin is $\frac{1}{4\pi\epsilon_0} \frac{Q}{R}$
- (3) The electric field at the point $(x, 0, 0)$ is $\frac{1}{4\pi\epsilon_0} \frac{Q}{x^2}$
- (4) The electric field at the point $(x, 0, 0)$ is $\frac{1}{4\pi\epsilon_0} \frac{Qx}{(R^2 + x^2)^{3/2}}$

2. A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on their axis are shown in the figure. With the figure what information about charges and electric field is incorrect.



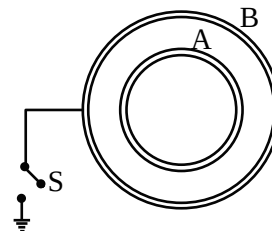
- (1) $|Q_1| > |Q_2|$
- (2) $Q_1 \rightarrow +ve, Q_2 \rightarrow -ve$
- (3) At a finite distance to the left of Q_1 the electric field is zero
- (4) At a finite distance to the right of Q_2 the electric field is zero

3. The figure shows, two point charges $q_1 = +2Q$ and $q_2 = -Q$. The charges divide the line joining them, in three parts I, II and III as shown in figure then which of the following statements is/are correct:

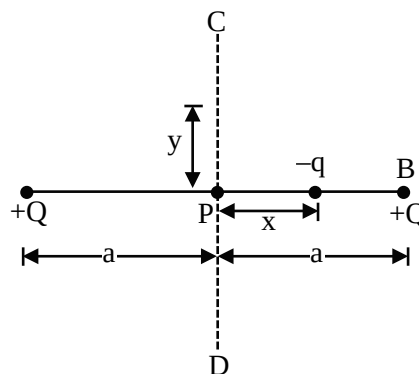


- (1) Electric field may be zero in region III
- (2) Electric field may be zero in region I
- (3) Electric field may be zero in region II
- (4) Electric field can not be zero anywhere on this line

4. Two concentric shells have radii R and $2R$ charges q_A and q_B and potentials $2V$ and $(3/2)V$ respectively. Now shell B is earthed by closing the switch S and let charges on the shells be changed to q_A' and q_B' . Then incorrect option is:



- (1) $|q_A|/|q_B| = 1$
 - (2) $|q_A'|/|q_B'| = 1$
 - (3) Potential of B after earthing becomes zero.
 - (4) Potential difference between the shells depends on charge of inner sphere.
5. For the arrangement shown in the Fig., the two positive charges, $+Q$ each, are fixed. Mark out the correct statement regarding a third charged particle $-q$ placed at mid point P that can be displaced along or perpendicular to the line connecting the charges,



- (1) The particle will perform SHM for $x \ll a$.
- (2) The particle will undergo oscillatory motion but not SHM about P for x comparable to a .
- (3) The particle will perform SHM for $y \ll a$.
- (4) The particle will undergo oscillatory motion but not SHM about P for $y \ll a$.

6. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis at $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y axis at a distance $b > 0$. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.

	List-I		List-II
P	Q_1, Q_2, Q_3, Q_4 all positive	1	+x
Q	Q_1, Q_2 positive, Q_3, Q_4 negative	2	-x
R	Q_1, Q_4 positive, Q_2, Q_3 negative	3	+y
S	Q_1, Q_3 positive, Q_2, Q_4 negative	4	-y

Codes:

- (1) P-3, Q-1, R-4, S-2
 (2) P-4, Q-2, R-3, S-1
 (3) P-3, Q-1, R-2, S-4
 (4) P-4, Q-2, R-1, S-3

7. Match Column-I with Column-II

	Column-I		Column-II
I.	Electric field due to Infinite plane sheet of charge	a.	0
II.	Electric field due to Infinite conducting plane sheet of uniform thickness	b.	$\frac{\sigma}{2\epsilon_0}$
III.	Electric field due to Non-conducting charged solid sphere at its surface	c.	$\frac{R\rho}{3\epsilon_0}$
IV.	Electric field due to Non-conducting charged solid sphere at its centre	d.	$\frac{\sigma}{\epsilon_0}$

- (1) I-c, II-d, III-a, IV-b
 (2) I-b, II-d, III-c, IV-a
 (3) I-b, II-d, III-a, IV-c
 (4) I-b, II-a, III-d, IV-c

8. Match the facts given in Column I with the Column II :

	Column-I		Column-II
I.	Charge cannot exist	a.	without charge
II.	Mass can exist	b.	without mass
III.	Charge is	c.	not conserved
IV.	Mass is	d.	conserved

- (1) I-b, II-a, III-d, IV-c (2) I-b, II-a, III-c, IV-d
 (3) I-b, II-d, III-a, IV-c (4) I-b, II-c, III-d, IV-a

9. Match Column-I with Column-II :-

	Column-I		Column-II
I.	ϵ_0	a.	$M^0 L^0 T^0 A^0$
II.	ϵ_r	b.	$[M^1 L^1 T^{-2}]$
III.	E	c.	$[M^{-1} L^{-3} T^4 A^2]$
IV.	F	d.	$[M^1 L^1 T^{-3} A^{-1}]$

- (1) I-c, II-a, III-b, IV-d
 (2) I-a, II-c, III-b, IV-d
 (3) I-b, II-c, III-a, IV-d
 (4) I-c, II-a, III-d, IV-b

10. Electric potential on the surface of a uniformly charged solid sphere is V. Radius of the sphere is 1m. Match the following two columns.

	Column-I		Column-II
(a)	Electric potential at $r = \frac{R}{2}$	(p)	$\frac{V}{4}$
(b)	Electric potential at $r = 2R$	(q)	$\frac{V}{2}$
(c)	Electric field at $r = \frac{R}{2}$	(r)	$\frac{3V}{4}$
(d)	Electric field at $r = 2R$	(s)	$\frac{11V}{8}$

- (1) a-p, b-s, c-r, d-q
 (2) a-p, b-r, c-q, d-s
 (3) a-s, b-q, c-q, d-p
 (4) a-q, b-q, c-r, d-s

11. Given below are two statements :

Statement I : Two concentric conducting spherical shells are charged. The charge on the outer shell is varied keeping the charge on inner shell constant, as a result the electric potential difference between the two shells does not change.

Statement II : In the above situation, the electric field in the region between both the shells depends only on the charge existing on the inner shell.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If a charge is released from rest in an electric field, it will always start to move along an electric field line.

Reason (R) : Force on a charged particle is always in the direction of electric field.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Electrostatic potential is a relative physical quantity.

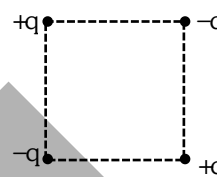
Reason (R) : Electric potential difference is an absolute physical quantity.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The electric potential and the electric field intensity at the centre of a square having four fixed point charges at their vertices as shown in figure are zero.



Reason (R) : If electric potential at a point is zero then the magnitude of electric field at that point must be zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

15. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The tyres of aircrafts are slightly conducting.

Reason (R) : If a conductor is connected to ground, the extra charge induced on conductor will flow to ground.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Any arbitrary displacement of charges inside a conducting shell does not introduce any change in the electrostatic field of the outer space.

Reason (R) : A closed conducting shell divides the entire space into the inner and outer parts which are completely independent of one another in respect of electric fields.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If a positive point charge is brought near an isolated neutral metal cube, the cube becomes negatively charged.

Reason (R) : If a positive point charge is brought near an isolated neutral metal cube, the cube remains neutral.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

18. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a given situation of arrangement of charges, an additional charge is placed outside the Gaussian surface. In this situation, in the Gauss theorem $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$, q_{in}

remains unchanged whereas electric field $\vec{E}$ is changed.

Reason (R) : Electric field $\vec{E}$ at any point on the Gaussian surface is due to inside charge only.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

19. If a body is charged by rubbing it with another body then weight of the system :-

- (A) may increase slightly
- (B) may decrease slightly
- (C) must increase slightly
- (D) remains precisely constant

- (1) Only A
- (2) AB
- (3) D
- (4) C

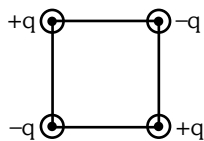
20. **Statement-I :** Coulombic force follows the principle of superposition.

and

Statement-II : Coulombic force is directly proportional to the charges on both particle.

- (1) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I.
- (2) Statement-I is True, Statement-II is True; Statement-II is not a correct explanation for Statement-I.
- (3) Statement-I is True, Statement-II is False.
- (4) Statement-I is False, Statement-II is True.

- 21. Statement-I :** The electric potential and the electric field intensity at the centre of a square having four point charges at their vertices (as shown) are zero.



Statement-II : Electric field is negative derivative of the potential.

- (1) Statement-I is True, Statement-II is True; Statement-II **is** a correct explanation for Statement-I
 (2) Statement-I is True, Statement-II is True; Statement-II **is NOT** a correct explanation for Statement-I
 (3) Statement-I is True, Statement-II is False
 (4) Statement-I is False, Statement-II is True.
- 22.** Two point charges $2 \mu\text{C}$ and $8 \mu\text{C}$ are placed at position $x = 0$ and $x = 4 \text{ m}$ respectively. The distance of the neutral point from the smaller charge is found to be at $x = \alpha/\beta$. Find the value of $(\alpha + \beta)$.
 (1) 7 (2) 5 (3) 4 (4) 3
- 23. Statement-I :** In a given situation of arrangement of charges, an extra charge is placed outside the Gaussian surface. In the Gauss Theorem

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{\text{in}}}{\epsilon_0}$$

Q_{in} remains unchanged where as electric field $\vec{E}$ at the site of the element is changed.

Statement-II : Electric field $\vec{E}$ at any point on the Gaussian surface is due to inside charge only.

- (1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.
 (2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.
 (3) Statement-I is true, statement-II is false.
 (4) Statement-I is false, statement-II is true.

- 24.** Two concentric spherical shells are given charges:-

- (A) Outer shell will be at higher potential if the charges given to both shells are negative
 (B) Inner shell will be at higher potential if the charges given to both shells are positive
 (C) Outer shell will always be at higher potential irrespective of the sign of charges given to two spheres
 (D) No prediction can be made using given data.

- (1) B (2) A,B (3) C (4) D

- 25.** In the following diagrams, all the charges have equal magnitude

Column-I		Column-II	
(A)	Equilateral triangle	(P)	The potential is zero at the centre
(B)	Square	(Q)	The electric field is zero at the centre
(C)	Square		
(D)	Rectangle		

- (1) $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow P, Q$; $D \rightarrow P$
 (2) $A \rightarrow P, Q$; $B \rightarrow P$; $C \rightarrow P, Q$; $D \rightarrow Q$
 (3) $A \rightarrow Q$; $B \rightarrow P, Q$; $C \rightarrow P$; $D \rightarrow P, Q$
 (4) $A \rightarrow P, Q$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow P$

26. Statement-I : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

and

Statement-II : Electric lines of forces are perpendicular to equipotential surface.

- (1) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I.
- (2) Statement-I is True, Statement-II is True ; Statement-II is not a correct explanation for Statement-I.
- (3) Statement-I is True, Statement-II is False.
- (4) Statement-I is False, Statement-II is True.

27. Statement I : Electric field outside the uniformly charged spherical shell (due to shell) is the same as that of a point charge at its center.

and

Statement II : The centre of mass of uniform shell is at its center.

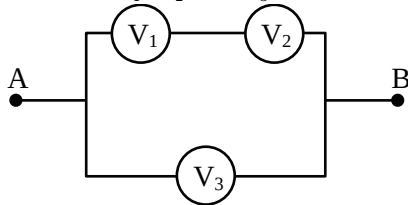
- (1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.
- (2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.
- (3) Statement-I is true, statement-II is false.
- (4) Statement-I is false, statement-II is true.

ANSWER KEY

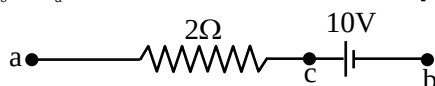
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	1	1	3	1	2	1	4	3	1	3	2	3	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans.	1	4	3	3	2	2	1	3	2	1	1	2			

CURRENT ELECTRICITY

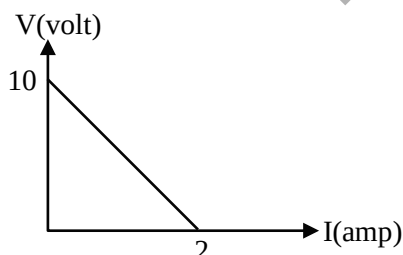
1. Three voltmeters all having different resistance, are joined as shown. When some potential difference is applied across A and B, their readings are V_1 , V_2 and V_3 . Then



- (a) $V_1 = V_2$ (b) $V_1 \neq V_2$
(c) $V_1 + V_2 = V_3$ (d) $V_1 + V_2 > V_3$
(1) only a is correct
(2) only b is correct
(3) only c is correct
(4) only b & c are correct
2. In the circuit shown in figure it is given that $V_b - V_a = 2$ volt. Choose the correct options.



- (a) Current in the wire is 6 A
(b) Direction of current is from a to b
(c) $V_a - V_c = -12$ volt
(d) $V_c - V_a = -12$ volt
(1) only a & b (2) only b & c
(3) only a & c (4) all
3. A battery of e.m.f E and internal resistance r is connected to a resistance R which can be adjusted to any value greater than or equal to zero. A graph is plotted between the current passing through the resistance (I) and potential difference (V) across it. Select the correct alternatives



- (a) Internal resistance of the battery is 5Ω
(b) emf of the battery is 10 V
(c) maximum current which can be taken from the battery is 2A
(d) V - I graph can never be a straight line as shown in the figure
(1) only a & b (2) a, b & c
(3) only b & c (4) all

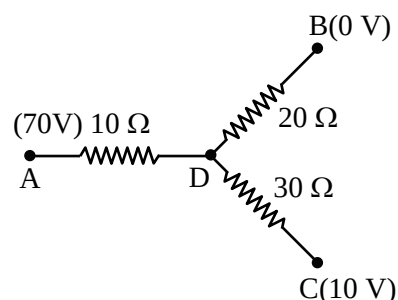
4. Which of the following statements is/are correct for potentiometer circuit?

- (a) Sensitivity varies proportionally with the length of the potentiometer wire
(b) Sensitivity is inversely proportional to the potential difference applied across the potentiometer wire
(c) Accuracy of a potentiometer can be increased only by increasing the length of wire
(d) Range depends upon the potential difference applied across the potentiometer wire
(1) only a & c (2) a, b & d
(3) a, b & c (4) only b & c

5. Two heaters designed for the same voltage V have different power ratings. When connected individually across a source of voltage V , they produce H amount of heat each in times t_1 and t_2 respectively. When used together across the same source, they produce H amount of heat in time t . Then correct option is

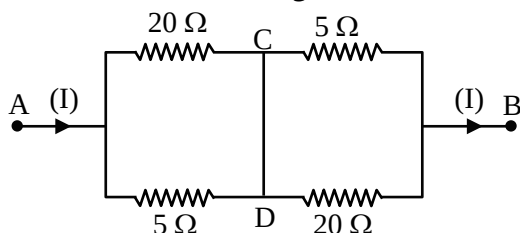
- (a) If they are in series, $t = t_1 + t_2$
(b) If they are in series, $t = 2(t_1 + t_2)$
(c) If they are in parallel, $t = \frac{t_1 t_2}{(t_1 + t_2)}$
(d) If they are in parallel, $t = \frac{t_1 t_2}{2(t_1 + t_2)}$
(1) only a & c (2) only a & b
(3) only c & d (4) all

6. In the network shown in figure, points A, B and C are at potentials of 70 V, 0 and 10 V respectively.



- (a) Point D is at a potential of 40V.
(b) The currents in the sections AD, DB, DC are in the ratio 3:2:1
(c) The currents in the sections AD, DB, DC are in the ratio 1:2:3.
(d) The network draws a total power of 200 W.
(1) only a is correct (2) a and c are correct
(3) a, b, d are correct (4) None

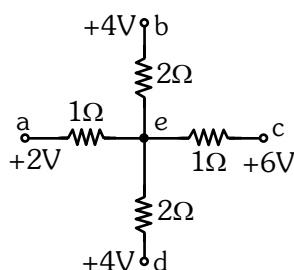
7. When some potential difference is maintained between A and B, current I enters the network at A and leaves at B see figure.



- (a) The equivalent resistance between A and B is 8Ω
 (b) C and D are at the same potential.
 (c) No current flows between C and D
 (d) Current $3I/5$ flows from D to C.

- (1) a, b, d are correct
 (2) a, d are correct
 (3) a, c, d are correct
 (4) all are correct

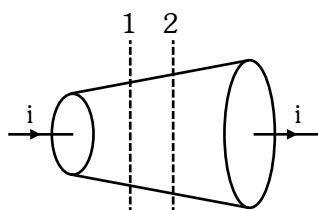
8. For the circuit shown in figure, match the two columns.



	Column-I		Column-II
(a)	Current in wire ae	(p)	1 A
(b)	Current in wire be	(q)	2 A
(c)	Current in wire ce	(r)	0.5 A
(d)	Current in wire de	(s)	Zero

- (1) a-q, b-s, c-q, d-s
 (2) a-q, b-q, c-s, d-p
 (3) a-p, b-r, c-s, d-q
 (4) a-q, b-r, c-p, d-s

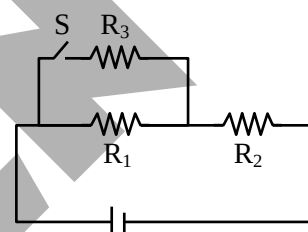
9. Current i is flowing through a wire of non-uniform cross section as shown. Match the following two columns.



	Column-I		Column-II
(a)	Current density	(p)	is more at 1
(b)	Electric field	(q)	is more at 2
(c)	Resistance per unit length	(r)	is same at both sections 1 and 2
(d)	Potential difference per unit length	(s)	data insufficient

- (1) a-p, b-p, c-p, d-p (2) a-q, b-r, c-s, d-p
 (3) a-q, b-q, c-p, d-p (4) a-p, b-q, c-r, d-s

10. In the circuit shown in figure, after closing the switch S, match the following two columns.



	Column-I		Column-II
(a)	Current through R_1	(p)	will increase
(b)	Current through R_2	(q)	will decrease
(c)	Potential difference across R_1	(r)	will remain same
(d)	Potential difference across R_2	(s)	data insufficient

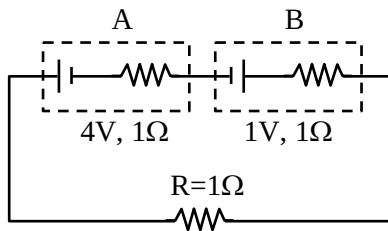
- (1) a-p, b-q, c-r, d-s (2) a-p, b-r, c-s, d-q
 (3) a-p, b-r, c-s, d-p (4) a-q, b-p, c-q, d-p

11. Match the following two columns.

	Column-I		Column-II
(a)	Electrical resistance	(p)	$[ML^3T^{-3}A^{-2}]$
(b)	Electrical potential	(q)	$[ML^2T^{-3}A^{-2}]$
(c)	Specific resistance	(r)	$[ML^2T^{-3}A^{-1}]$
(d)	Specific conductance	(s)	None of these

- (1) a-q, b-s, c-r, d-p (2) a-q, b-r, c-p, d-s
 (3) a-p, b-q, c-s, d-r (4) a-p, b-r, c-q, d-s

12. In the circuit shown in the figure, match the following two columns:



	Column-I		Column-II (In SI units)
(a)	Potential difference across battery A	(p)	Zero
(b)	Potential difference across battery B	(q)	1
(c)	Net power consumed by R	(r)	2
(d)	Potential difference across R	(s)	3

- (1) a-p, b-q, c-r, d-s
 (2) a-s, b-q, c-r, d-p
 (3) a-s, b-r, c-q, d-q
 (4) a-q, b-r, c-s, d-s

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Drift velocity of electron in a metallic wire will decrease if temperature of wire is increased

Reason (R) : On increasing temperature conductivity of metallic wire decreases.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Resistance of ammeter is less than resistance of milliammeter, (If made from same galvanometer)

Reason (R) : Value of shunt resistance in case of ammeter is more than a milliammeter.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

15. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Electric field is present inside the current carrying wire.

Reason (R) : The principle of conservation of charge is followed, when charges are in motion.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In balanced Wheatstone bridge, the current through cell depends on resistance of galvanometer.

Reason (R) : At balanced condition current through galvanometer is zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In series combination of electrical bulb, lower power rating bulb emits more light than that of higher power rating bulb.

Reason (R) : The lower power rating bulb in series gets more current than the higher power rating bulb.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

18. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When an external resistor of resistance R (connected across a cell of internal resistance r) is varied, power consumed by resistance R is maximum when $R = r$.

Reason (R) : Power consumed by a resistor of constant resistance R is maximum when current through it is maximum.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

19. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ohm's law holds only for small currents in metallic wire not for high currents.

Reason (R) : For metallic wire resistance increases with increase in temperature. TG: @Chalnaayaaar

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

20. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a Meter Bridge experiment, null point for an unknown resistance is measured. Now, the unknown resistance is put inside an enclosure maintained at a higher temperature. The null point can be obtained at the same point as before by decreasing the value of the standard resistance.

Reason (R) : Resistance of a metal increases with increase in temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

21. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Kirchoff's loop law represents conservation of energy

Reason (R) : If the sum of "Potential Differences" around a closed loop is zero, unlimited energy could be gained by repeatedly carrying a charge around a loop

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

22. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In potentiometer experiment, null point can not be obtained if emf of unknown cell is more than emf of source battery.

Reason (R) : By increasing the emf of source battery balancing length increases.

In the light of the above statements, choose the most appropriate answer from the options given below:

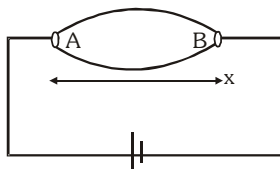
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

23. If the current in resistance of uniform cross section is doubled then, (assume temperature to be approximately constant).

- (A) the current density is doubled
- (B) the conduction electron density is doubled
- (C) the mean time between collision is constant
- (D) the electron drift speed is doubled

- (1) A, C, D
- (2) B, C, D
- (3) A, B, C
- (4) A, B, C, D

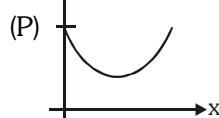
24. A resistor of varying cross-section is attached with a battery as shown. Variation of area from both ends are similar and areas are same at ends and maximum at centre. Match the approximate graphical representation of various physical properties in columns, where x axis represents position from end A of resistor and y-axis various physical properties.



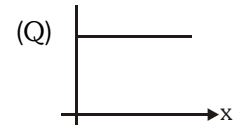
Column I

(A) i (current)

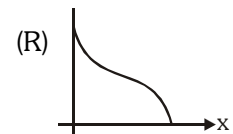
Column II



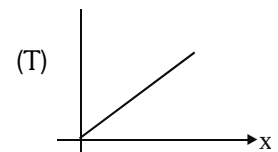
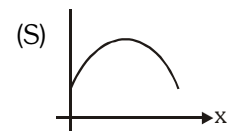
(B) j (current density)



(C) E (electric field)



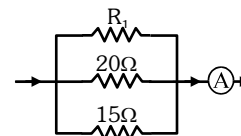
(D) V_d (potential)



Options :-

- (1) $A \rightarrow S, B \rightarrow P, C \rightarrow P, D \rightarrow R$
- (2) $A \rightarrow Q, B \rightarrow P, C \rightarrow P, D \rightarrow R$
- (3) $A \rightarrow T, B \rightarrow Q, C \rightarrow P, D \rightarrow S$
- (4) $A \rightarrow Q, B \rightarrow P, C \rightarrow P, D \rightarrow T$

25. In the given circuit the current flowing through the resistance $20\ \Omega$ is 0.3 ampere while the ammetre reads 0.9 ampere. What is the value of R_1 (in ohms).



- (1) $20\ \Omega$
- (2) $15\ \Omega$
- (3) $30\ \Omega$
- (4) $60\ \Omega$

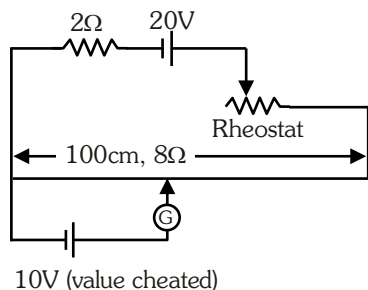
26. **Statement 1:** Voltage across a battery is always less than its emf.

and

Statement 2 : There is always some loss of energy due to internal resistance of the battery.

- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
- (3) Statement-1 is True, Statement-2 is False.
- (4) Statement-1 is False, Statement-2 is True.

27. A student was given the experiment to measure the emf of a unknown cell using potentiometer. He cheated result and wrote the observation arbitrarily and was caught. Which reading helped the teacher to arrive at the conclusion that he cheated?



S.N.	value of rheostat	Null point
I.	R_1	50 cm
II.	R_2	60 cm
III.	R_3	70 cm
IV.	R_4	80 cm

- (1) Only I (2) I, II
(3) II, IV (4) Only IV

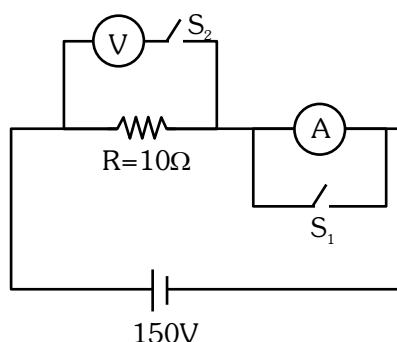
28. **Statement-1** : A potentiometer having 10 wires is more accurate than the potentiometer having 1 wire.

and

Statement-2 : Potential gradient of 10-wire potentiometer is much less.

- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 is False, Statement-2 is True.

29. Voltmeter of resistance 10Ω and ammeter of resistance 10Ω are attached in the following circuit with switches S_1 and S_2



Column I (switch position)	Column II (reading of voltmeter & ammeter)
-------------------------------	---

- (A) S_1 open, S_2 closed (P) 0, 7.5 A
(B) S_1 closed, S_2 open (Q) 50 V, 10A
(C) S_1 open, S_2 open (R) 0, 0
(D) S_1 closed, S_2 closed (S) 150 V, 0

Options :-

- (1) A → S, B → R, C → P, D → Q
(2) A → Q, B → R, C → P, D → S
(3) A → Q, B → P, C → R, D → S
(4) A → S, B → P, C → R, D → Q

30.

	Column-I (Situation)	Column-II (Values in MKS units)
(A)	Magnitude of electric field at (1,1,1) if potential is given by $V = -\frac{1}{x} - \frac{1}{y} - \frac{1}{z}$, is	(P) 0.4
(B)	The charge flowing through a resistance 6Ω varies with t as $Q = 4t - 2t^2$. The total heat produced in resistance till the current reach to zero.	(Q) $\sqrt{3}$
(C)	A milliammeter of range 10mA has a coil of resistance 0.99Ω . To use it as an ammeter of range 1A, the required shunt must have a resistance of	(R) 0.01
(D)	Two unlike point charges of same magnitude 10 mC are placed on x-axis at a separation of 8 cm. If both charges placed symmetrically about origin in external electric field $\vec{E} = 500\text{Vm}^{-1} \hat{i}$. Then potential energy of the system may be [Assume potential energy to be zero when dipole is perpendicular to electric field]	(S) 32

Options :-

- (1) A → Q, B → S, C → P, D → R
(2) A → S, B → Q, C → R, D → P
(3) A → Q, B → S, C → R, D → P
(4) A → S, B → Q, C → P, D → R

31. Statement-1 : In real battery total chemical energy lost in the battery can not be obtained at load resistance.

and

Statement-2 : Two heaters with different resistance are connected in parallel. Higher resistance heater will glow more.

- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
- (3) Statement-1 is True, Statement-2 is False.
- (4) Statement-1 is False, Statement-2 is True.

32. Match the column I with column II -

	Column-I (Instruments)		Column-II (Based on)
(A)	Hot wire ammeter	(P)	Heating effect of current
(B)	Moving coil galvanometer	(Q)	Magnetic effect of current
(C)	Potentiometer	(R)	Chemical effect of current
(D)	Voltameter	(S)	Peltier effect of current
		(T)	Zero deflection method

Options :-

- (1) $A \rightarrow P, B \rightarrow T, C \rightarrow Q, D \rightarrow R$
- (2) $A \rightarrow P, B \rightarrow T, C \rightarrow Q, D \rightarrow S$
- (3) $A \rightarrow P, B \rightarrow Q, C \rightarrow T, D \rightarrow R$
- (4) $A \rightarrow Q, B \rightarrow P, C \rightarrow T, D \rightarrow R$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	3	2	2	1	3	1	1	1	4	2	3	2	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	3	2	1	4	3	3	1	2	3	4	2	1	2	3
Que.	31	32	TG: @Chalnaayaaar												
Ans.	3	3													

CAPACITANCE

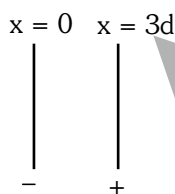
1. A parallel plate air capacitor is connected to a battery. The quantities charge, voltage, electric field and energy associated with the capacitor are given by Q_0 , V_0 , E_0 and U_0 respectively. A dielectric slab is now inserted to fill the space between the plates with battery still in connection. The corresponding quantities now given by Q , V , E and U are related to the previous one as :

(a) $Q > Q_0$ (b) $V > V_0$ (c) $E > E_0$ (d) $U > U_0$

- (1) only (a) is correct
(2) only a & c are correct
(3) only a & d are correct
(4) all of above are correct

2. A dielectric slab of thickness d is inserted in a parallel plate capacitor whose negative plate is at $x = 0$ and positive plate is at $x = 3d$. The slab is equidistant from the plates. The capacitor is given some charge. As one goes from 0 to $3d$:

- (a) The magnitude of the electric field remains the same
(b) The direction of the electric field remains the same
(c) The electric potential increases continuously
(d) The electric potential increases at first, then decreases and again increases



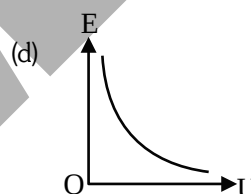
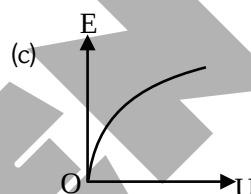
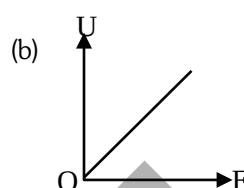
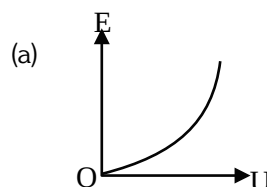
- (1) a and c are correct (2) b and c are correct
(3) a, b and c are correct (4) All are correct

3. The plates of a parallel plate capacitor are, not exactly parallel. The surface charge density therefore:

- (a) Is higher at the closer end
(b) The surface charge density will not be uniform
(c) Each plate will have the same potential at each point
(d) The electric field is smallest where the plates are closest

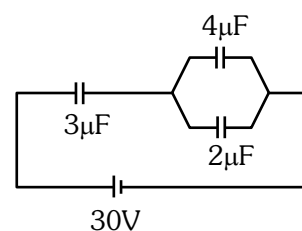
- (1) only a is correct
(2) only b & c are correct
(3) a & c are correct
(4) a, b & c are correct

4. If at distance r from a positively charged particle, electric field strength and energy density are E and U , respectively, which of the following graphs is correct?



- (1) only a (2) only b
(3) only c (4) a, b, d

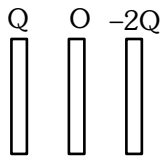
5. In the circuit shown in figure, match the following two columns.



	Column-I		Column-II
(I)	Charge given by battery	(a)	20 μC
(II)	Charge of $4\mu\text{F}$ capacitor	(b)	30 μC
(III)	Charge of $2\mu\text{F}$ capacitor	(c)	60 μC
(IV)	Charge of $3\mu\text{F}$ capacitor	(d)	40 μC

- (1) I-a, II-d, III-b, IV-c (2) I-c, II-d, III-a, IV-c
(3) I-b, II-a, III-c, IV-d (4) I-c, II-d, III-c, IV-a

6. Three identical metal plates with large surface areas are kept parallel to each other as shown in fig. The leftmost plate is given charge Q , the rightmost given charge $-2Q$ and the middle one remains neutral. Then



	Column-I		Column-II
i.	The charge appearing on outer surface of the rightmost plate	a.	$+\frac{Q}{2}$
ii.	The charge appearing on outer surface of the leftmost plate	b.	$-\frac{Q}{2}$
iii.	The charge appearing on left surface of the middle plate	c.	$-\frac{3Q}{2}$
iv.	The charge appearing on right surface of the middle plate	d.	$\frac{3Q}{2}$

- (1) i-b, ii-b, iii-d, iv-c (2) i-c, ii-b, iii-d, iv-c
(3) i-b, ii-a, iii-c, iv-d (4) i-b, ii-b, iii-c, iv-d

7. Match the following –

	List-I		List-II
(P)	Self-energy of a uniformly charged spherical shell	(1)	$\frac{1}{8\pi\epsilon_0} \frac{Q^2}{R}$
(Q)	Energy stored per unit volume of a charged air filled capacitor	(2)	$\frac{1}{4\pi\epsilon_0} \frac{2Q^2}{R^2}$
(R)	Force between two circular plates of a charged parallel plate capacitor, radius of the plates being R and separation $R/32$	(3)	$\frac{1}{4\pi\epsilon_0} \frac{Q^2}{R}$
(S)	Work done by electric field in moving a point charge Q from the surface of a uniformly charged sphere of radius R and moving it to infinity, total charge on the sphere = Q	(4)	$\frac{1}{2} \epsilon_0 E^2$

Codes :

	P	Q	R	S
(1)	3	1	4	2
(2)	2	1	3	4
(3)	1	4	2	3
(4)	4	1	2	3

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Mutual force of attraction between two plates of isolated capacitor is independent of presence of dielectric slab between plates.

Reason (R) : Force of interaction between two charged bodies is independent of medium.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : It is not possible to make a spherical conductor of capacitance one farad.

Reason (R) : It is possible for earth as its radius is 6400 km.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When two charged capacitors are connected in parallel, then total potential energy of system always decreases.

Reason (R) : During recombination charge flow from one capacitor to another capacitor so energy is lost in the form of heat and electromagnetic radiation.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

11. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Circuits containing high capacity capacitors, charged to high voltage should be handled with caution, even when the current in the circuit is switched off.

Reason (R) : When an isolated capacitor is touched by hand or any other part of the human body, there is an easy path to the ground available for the discharge of the capacitor.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If separation between plates of a parallel plate isolated charged capacitor is increased, its energy stored increases.

Reason (R) : Work done to more separate the plates get converted in electrostatic potential energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

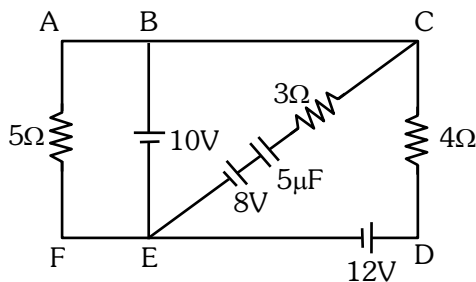
Assertion (A) : When a capacitor is charged by a battery, both the plates receive charge equal in magnitude, no matter size of plates are identical or not.

Reason (R) : The charge distribution on the plates of capacitor is in accordance with charge conservation principle.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

14. A network consisting of three resistors, three batteries, and a capacitor is shown in figure.



Column I

Column II

- | | |
|-----------------------------|----------------------|
| (A) Current in branch EB is | (P) $10 \mu\text{C}$ |
| (B) Current in branch CB is | (Q) 0.5 A |
| (C) Current in branch ED is | (R) 1.5 A |
| (D) Charge on capacitor is | (S) $5 \mu\text{C}$ |
| | (T) None of these |

Options :-

- (1) $A \rightarrow R, B \rightarrow Q, C \rightarrow Q, D \rightarrow P$
- (2) $A \rightarrow R, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
- (3) $A \rightarrow Q, B \rightarrow R, C \rightarrow R, D \rightarrow P$
- (4) $A \rightarrow R, B \rightarrow Q, C \rightarrow Q, D \rightarrow S$

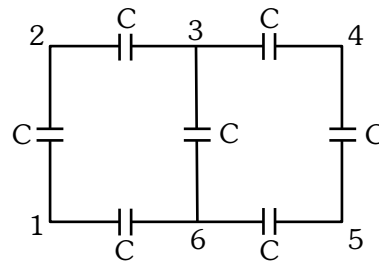
15. **Statement-1** : If capacitor is filled with, same thickness (t) of dielectric ($t < d$) and conducting sheet one after another, then capacitance are C_1 and C_2 respectively then $C_1 < C_2$.

and

Statement-2 : Capacitance is more in presence of metal sheet in compare to dielectric sheet as $K_{\text{Conduct}} > K_{\text{dielectric}}$

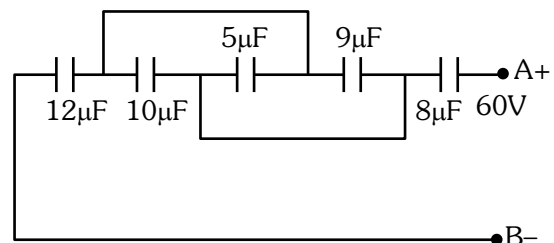
- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
- (3) Statement-1 is True, Statement-2 is False.
- (4) Statement-1 is False, Statement-2 is True.

16. For the given circuit, select the incorrect alternative

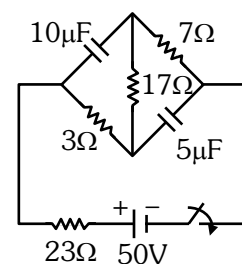


- (1) The equivalent capacitance between points 1 & 2 is $\frac{15C}{11}$
- (2) The equivalent capacitance between points 3 & 6 is $\frac{5C}{3}$
- (3) The equivalent capacitance between points 1 & 3 is $\frac{15C}{14}$
- (4) The equivalent capacitance between points 3 & 5 is $\frac{14C}{15}$

17. The equivalent capacitance between A and B is and charge on $5\mu\text{F}$ capacitor is

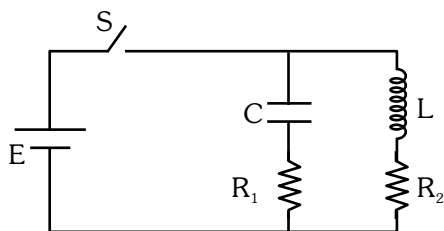


- (1) $4\mu\text{F}$ & $50 \mu\text{C}$
 - (2) $5\mu\text{F}$ & $40 \mu\text{C}$
 - (3) $8\mu\text{F}$ & $60 \mu\text{C}$
 - (4) None
18. Write down the charge in μC on $10\mu\text{F}$ capacitor in steady state.



- (1) $100 \mu\text{C}$
- (2) $150 \mu\text{C}$
- (3) $180 \mu\text{C}$
- (4) $200 \mu\text{C}$

19. In the circuit shown in figure $E=25V$, $L=2H$, $C=60\ \mu F$, $R_1 = 5\Omega$ and $R_2 = 10\Omega$. Switch S is closed at $t = 0$.



	Column-I		Column-II
(A)	Current through R_1 at (P) $t = 0$	(P)	0
(B)	Current through R_2 at (Q) $t = 0$	(Q)	5A
(C)	Current through R_1 at (R) $t = \infty$	(R)	2.5 A
(D)	Current through R_2 at (S) $t = \infty$	(S)	None of these

Options :-

- (1) $A \rightarrow P$, $B \rightarrow Q$, $C \rightarrow P$, $D \rightarrow R$
- (2) $A \rightarrow Q$, $B \rightarrow P$, $C \rightarrow P$, $D \rightarrow R$
- (3) $A \rightarrow Q$, $B \rightarrow P$, $C \rightarrow R$, $D \rightarrow P$
- (4) $A \rightarrow P$, $B \rightarrow R$, $C \rightarrow P$, $D \rightarrow Q$

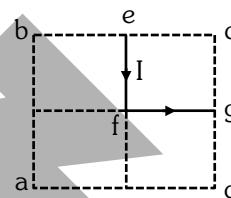
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	4	3	2	4	3	1	3	4	1	1	1	1	1
Que.	16	17	18	19	TG: @Chalnaayaaar										
Ans.	4	1	4	2											

MAGNETIC EFFECT OF CURRENT AND MAGNETISM

- Two circular coils of radii 5 cm and 10 cm carry currents of 2 A. The coils have 50 and 100 turns respectively and are placed in such a way that their planes as well as their centres coincide. Magnitude of magnetic field at the common centre of coils is
 - $8\pi \times 10^{-4}$ T if currents in the coils are in same sense
 - $4\pi \times 10^{-4}$ T if currents in the coils are in opposite sense
 - zero if currents in the coils are in opposite sense
 - $8\pi \times 10^{-4}$ T if currents in the coils are in opposite sense
 - b and d are correct
 - a and c are correct
 - only a are correct
 - only d are correct
- When a current carrying coil is placed in a uniform magnetic field with its magnetic moment anti-parallel to the field
 - Torque on it is maximum
 - Torque on it is zero
 - Potential energy is maximum
 - Dipole is in unstable equilibrium
 - b, c, d are correct
 - only b is correct
 - a and c are correct
 - none
- If a long cylindrical conductor carries a steady current parallel to its length
 - the electric field along the axis is zero
 - the magnetic field along the axis is zero
 - the magnetic field outside the conductor is zero
 - the electric field outside the conductor is zero
 - b and d are are correct
 - a and c are are correct
 - only d is correct
 - none

- A charged particle revolves in circular path in uniform transverse magnetic field after being accelerated by a potential difference of V volt. If V is doubled, then choose the correct options.
 - Kinetic energy of particle will become two times
 - Radius of circular path will become two times
 - Radius of circular path will become $\sqrt{2}$ times
 - angular velocity will remain unchanged
 - a and b are correct
 - a and c are correct
 - a, c, d are correct
 - none
- abcd is a square. There is a current I in wire efg as shown. Choose the correct options :



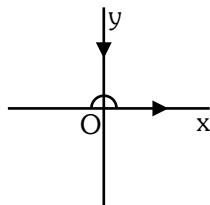
- Net magnetic field at a is inwards
 - Net magnetic field at f is zero
 - Net magnetic field at c is zero
 - Net magnetic field at g is zero
- a and b are correct
 - a, b, c are correct
 - b and c are correct
 - none

- In column I, a current carrying loop and a uniform magnetic field are shown. Match this with column II.

	Column-I		Column-II
(a)		(p)	$F \neq 0$
(b)		(q)	Maximum torque
(c)		(r)	Minimum potential energy
(d)		(s)	Positive potential energy

- a-s, b-p, c-r, d-q
- a-s, b-q, c-r, d-s
- a-s, b-r, c-p, d-q
- a-q, b-p, c-r, d-s

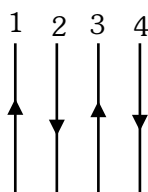
7. Equal currents are flowing in two infinitely long separate wires lying along x and y-axes in the directions shown in figure. Match the following two columns.



	Column-I		Column-II
(a)	Magnetic field at (a, a)	(p)	along positive y-axis
(b)	Magnetic field at (-a, -a)	(q)	along positive z-axis
(c)	Magnetic field at (a, -a)	(r)	along negative z-axis
(d)	Magnetic field at (-a, a)	(s)	zero

- (1) a-p, b-q, c-s, d-r
 (2) a-q, b-r, c-s, d-s
 (3) a-r, b-s, c-p, d-r
 (4) a-s, b-p, c-q, d-s

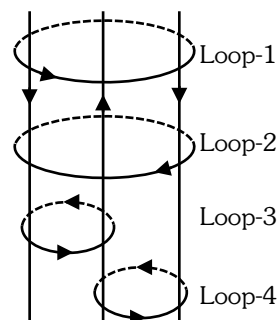
8. Equal currents are flowing in four infinitely long wires. Distance between two wires is same and directions of currents are shown in figure. Match the following two columns.



	Column-I		Column-II
(a)	Force on wire-1	(p)	Inwards
(b)	Force on wire-2	(q)	Leftwards
(c)	Force on wire-3	(r)	Rightwards
(d)	Force on wire-4	(s)	Zero

- (1) a-q, b-r, c-q, d-r (2) a-r, b-q, c-p, d-r
 (3) a-q, b-p, c-q, d-p (4) a-p, b-r, c-p, d-r

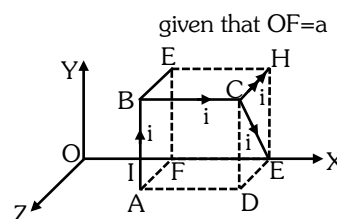
9. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown in figure. The direction of $d\vec{\ell}$ is shown in the figure.



	Column-I		Column-II
a	Along close Loop-1	p	$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$
b	Along close Loop-2	q	$\oint \vec{B} \cdot d\vec{\ell} = -\mu_0 i$
c	Along close Loop-3	r	$\oint \vec{B} \cdot d\vec{\ell} = 0$
d	Along close Loop-4	s	$\oint \vec{B} \cdot d\vec{\ell} = 2\mu_0 i$

- (1) a - q, b - p, c - r, d - r
 (2) a - p, b - r, c - s, d - s
 (3) a - r, b - p, c - s, d - p
 (4) a - r, b - q, c - q, d - q

10. Side of cube is 'a' and wire AB, BC, CH and CE are insulated with each other. If each having current i , then match the following. External magnetic field applied along positive x-axis as $\vec{B} = B_0 \hat{i}$



	List-I (Magnitude of force due to external magnetic field only)		List-II
(a)	On AB	(p)	Zero
(b)	On BC	(q)	$\sqrt{2}iB_0a$
(c)	On CH	(r)	iB_0a
(d)	On CE	(s)	$2iB_0a$

- (1) a - r, b - p, c - r, d - q
 (2) a - p, b - p, c - s, d - q
 (3) a - q, b - s, c - r, d - s
 (4) a - s, b - r, c - d, d - q

- 11.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : An electron cannot be accelerated by cyclotron.

Reason (R) : Mass of electron is too small.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 12.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Force on a current carrying wire of length $d\ell$ placed in magnetic field $\vec{B}$ is given by $d\vec{F} = id\vec{\ell} \times \vec{B}$

Reason (R) : Net force on a current carrying loop in a non-uniform magnetic field must be non-zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 13.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The ferromagnetic substances obey Curie's law.

Reason (R) : At Curie point ferromagnetic substances start behaving as paramagnetic substances.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 14.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The surface integral of magnetic field over any closed surface is always zero.

Reason (R) : Magnetic poles always exists in pair.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 15.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A moving charge produces magnetic field.

Reason (B) : A current element produces magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 16.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Parallel wires carrying currents in same direction attracts each other due to magnetic force.

Reason (R) : Two electron beams moving parallel to each other repel each other due to electric force.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 17.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ampere's circuital law is not suitable for a finite straight wire carrying current I .

Reason (R) : One cannot calculate magnetic field due to a finite straight current carrying wire by using Biot Savart Law.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 18.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The angle of dip at geomagnetic pole is zero

Reason (R) : In general, in a vertical plane perpendicular to the magnetic meridian the dip needle becomes exactly vertical

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 19.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The diamagnetic substances do not obey Curie's law.

Reason (R) : At Curie point ferromagnetic substances start behaving as paramagnetic substances.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 20.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Solenoid is used to generate magnetic field in television.

Reason (R) : Value of orbital magnetic moment, for first orbit of a single electron species is $0.927 \times 10^{-23} \text{ A-m}^2$.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

- 21. Statement-1 :** In an isolated conductor, free electrons keep on moving but no net magnetic force acts on a conductor in a magnetic field.

Statement-2 : In a conductor, the average velocity of thermal motion of electrons is zero. Hence no current flows through the conductor.

- (1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
- (3) Statement-1 is True, Statement-2 is False.
- (4) Statement-1 is False, Statement-2 is True.

- 22.** A particle with charge Q is moving in fields of combination given below. Match for the most correct possible path/s. Then match the following :-

	Column-I		Column-II
(A)	Uniform $\vec{E}$	(P)	Straight line
(B)	Uniform $\vec{B}$	(Q)	Parabola
		(R)	Circular

Options :-

- (1) $A \rightarrow PQ, B \rightarrow PR$
- (2) $A \rightarrow P, B \rightarrow R$
- (3) $A \rightarrow Q, B \rightarrow PR$
- (4) $A \rightarrow P, B \rightarrow PQ$

- 23. Match the column I with column II -**

	Column-I		Column-II
(A)	Paramagnetic substance	(P)	$\chi_m = -1$
(B)	Diamagnetic substance	(Q)	$\chi_m < 0$
(C)	Super conductor	(R)	$\chi_m \gg 1$
(D)	Ferromagnetic substance	(S)	$\mu_r > 1$

Options :-

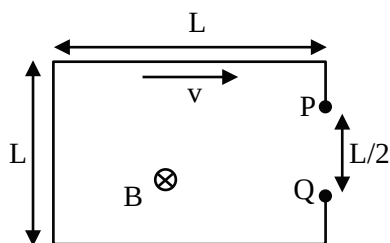
- (1) $A \rightarrow S, B \rightarrow Q, C \rightarrow P, D \rightarrow R$
- (2) $A \rightarrow R, B \rightarrow P, C \rightarrow Q, D \rightarrow S$
- (3) $A \rightarrow S, B \rightarrow P, C \rightarrow Q, D \rightarrow R$
- (4) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$

ANSWER KEY

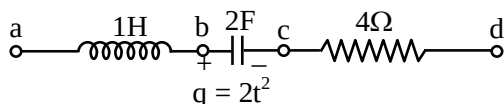
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	3	1	2	2	1	1	1	1	3	4	1	2
Que.	16	17	18	19	20	21	22	23	TG: @Chalnaayaaar						
Ans.	2	3	4	2	2	1	1	1							

ELECTROMAGNETIC INDUCTION (EMI)

1. The loop shown moves with a velocity v in a uniform magnetic field of magnitude B , directed into the paper as shown in the figure. The potential difference between points P and Q is e . Then

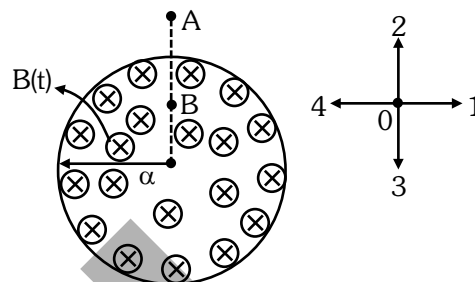


- (a) $e = \frac{1}{2}BLv$
 (b) $e = BLv$
 (c) P is positive with respect to Q
 (d) Q is positive with respect to P
 (1) only a is correct
 (2) only a & c are correct
 (3) only b is correct
 (4) only b & d are correct
2. Choose the correct options
 (a) SI unit of magnetic flux is henry-ampere
 (b) SI unit of coefficient of self-induction is J/A
 (c) SI unit of coefficient of self induction is $\frac{\text{volt} - \text{second}}{\text{ampere}}$
 (d) SI unit of magnetic induction is weber
 (1) only (a)
 (2) only (b)
 (3) (a) and (c)
 (4) (b) and (d)
3. In the figure shown, q is in coulomb and t in second.
 At time $t = 1s$



- (a) $V_a - V_b = 4V$
 (b) $V_b - V_c = 1V$
 (c) $V_c - V_d = 16V$
 (d) $V_a - V_d = 20V$
 (1) only (a) is correct
 (2) only (b) is correct
 (3) (a), (b) and (c) are correct
 (4) only (d) is correct

4. A uniform but time varying magnetic field $B(t)$ exists in a cylindrical region of radius α and is directed into the plane of the paper, as shown in fig. The magnetic field decreases at a constant rate inside the region. If r is the distance from the axis of the cylindrical region then match column I to column II.



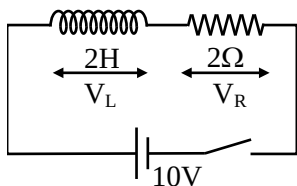
	Column-I		Column-II
I.	Induced electric field at point A	a.	0 to 1
II.	Induced electric field at point B	b.	Zero
III.	Force on an electron placed at point A	c.	0 to 4
IV.	Force on an electron placed at point B	d.	0 to 2

- (1) I - a, II - d, III - d, IV - a
 (2) I - d, II - a, III - c, IV - a
 (3) I - a, II - b, III - c, IV - d
 (4) I - a, II - a, III - c, IV - c
5. Match the column.

	Column-I		Column-II
(a)	Magnetic field	(p)	$[M^1T^{-2}A^{-1}]$
(b)	Coefficient of self induction	(q)	$[T^2]$
(c)	LC	(r)	$[ML^2T^{-2}A^{-2}]$
(d)	Magnetic flux	(s)	None of these

- (1) a - p, b - r, c - q, d - s
 (2) a - p, b - q, c - r, d - s
 (3) a - p, b - s, c - q, d - r
 (4) a - s, b - s, c - s, d - r

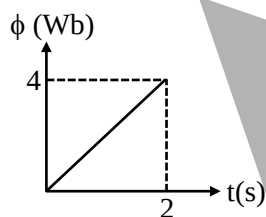
6. In the circuit shown in figure, switch is closed at time $t = 0$. Match the following two columns.



	Column-I		Column-II
(a)	V_L at $t = 0$	(p)	zero
(b)	V_R at $t = 0$	(q)	10V
(c)	V_L at $t = 1s$	(r)	$\frac{10}{e}V$
(d)	V_R at $t = 1s$	(s)	$\left(1 - \frac{1}{e}\right)10V$

- (1) a - q, b - p, c - r, d - s
 (2) a - q, b - r, c - p, d - s
 (3) a - r, b - q, c - r, d - s
 (4) a - s, b - r, c - p, d - q

7. Magnetic flux passing through a coil of resistance 2Ω is as shown in figure. Match the following two columns. In column II all physical quantities are in SI units.



	Column-I		Column-II
(a)	Induced emf produced	(p)	4
(b)	Induced current	(q)	1
(c)	Charge flow in 2s	(r)	8
(d)	Heat generation in 2s	(s)	2

- (1) a-s, b-q, c-s, d-p
 (2) a-q, b-r, c-p, d-s
 (3) a-p, b-q, c-r, d-s
 (4) a-r, b-q, c-s, d-p

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Transformer can't be used to step up or step down the steady dc voltage

Reason (R) : Transformers are based upon the principle of mutual induction for which a time varying voltage is necessary.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The direction of induced current is always such as to oppose the change that causes it.

Reason (R) : The direction of induced current is given by Lenz's law.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A time varying magnetic field induces an electric field.

Reason (R) :- An inductor always tends to keep the flux constant.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

11. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The mutual inductance of coil 1 with respect to coil 2 is the same as mutual inductance of coil 2 with respect to coil 1.

Reason (R) : Mutual inductance between coils is independent of current passing through the coil

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The electric field created by time-varying magnetic field is non-conservative.

Reason (R) : The line integral of induced electric field in a closed loop is always equal to zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Magnetic flux is a scalar quantity.

Reason (R) : Value of magnetic flux cannot be negative.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When a coil is rotated in a uniform magnetic field about an diametrical axis perpendicular to the field, emf is induced in it which is maximum for the orientation of coil in which magnetic flux through the coil is zero.

Reason (R) : In an electric generator, electrical energy is generated by rotating a coil in a magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below:

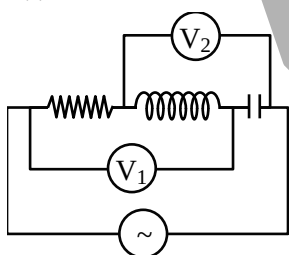
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	2	3	3	4	1	1	1	1	1	2	2	3	3	2	

ALTERNATING CURRENT

- Resonance occurs in a series LCR circuit when the frequency of the applied emf is 1000 Hz. Then
 - when frequency = 900 Hz, then the current through the voltage source will be ahead of e.m.f. of the source
 - the impedance of the circuit is minimum at $f = 1000$ Hz
 - at resonance the voltage across L and current in C differ in phase by 180°
 - If the value of C is doubled resonance occurs at $f = 2000$ Hz
 - only a & b are correct
 - a, b, c are correct
 - c & d are correct
 - all are correct
- If R, C, and L be the resistance, capacitance, and inductance in the circuit in which ac of frequency f is set up, then which of the following have the dimensions of R?
 - fC
 - fL
 - $1/fC$
 - L/f
 - only a & b
 - only b & c
 - only c & d
 - all
- In an RLC series circuit shown in figure, the readings of voltmeters V_1 and V_2 are 100 V and 120 V respectively. The source voltage is 130 V. For this situation mark out the correct statement(s).



- Voltage across resistor, inductor and capacitor are 50 V, 86.6 V and 206.6 V, respectively
 - Voltage across resistor, inductor and capacitor are 10V, 90 V and 30 V, respectively
 - Power factor of the circuit is $\frac{5}{13}$
 - The circuit is capacitive in nature
- a, c, d
 - b, c, d
 - c, d
 - None

- Match the column :

In column I, variation of current i with t is given in figures. In column II root mean square current i_{rms} , and average current are given. Match the column I with corresponding quantities given in Column II

	Column-I		Column-II
(A)		(P)	$i_{\text{rms}} = \frac{i_0}{\sqrt{2}}$
(B)		(Q)	Average current is i_0
(C)		(R)	$I_{\text{rms}} = \frac{i_0}{\sqrt{3}}$
(D)		(S)	None of above

- a - S, b - Q, c - Q, d - P
- a - P, b - Q, c - S, d - P
- a - Q, b - P, c - R, d - S
- a - P, b - P, Q, c - P, d - P

- Match the following :

	Column-I		Column-II
(A)	LCR circuits	(P)	Resonant curve will be flattened
(B)	Inductor	(Q)	Sharpness indicates sensitivity
(C)	More resistance	(R)	Have resonant frequency $\omega = \frac{1}{\sqrt{LC}}$ with maximum current
(D)	Radio Tuner's characteristic curve ($I - f$)	(S)	Analogous to mass

- A-R, B-S, C-Q, D-P
- A-Q, B-P, C-Q, D-R
- A-S, B-Q, C-R, D-P
- A-R, B-S, C-P, D-Q

6. Match the column

	Column-I		Column-II
(a)	In L-R series AC circuit	(p)	Current lags voltage by $\pi/2$
(b)	In R-C series AC circuit	(q)	Current lags voltage by an angle less than $\pi/2$
(c)	In L-C series AC circuit ($X_L > X_C$)	(r)	Current leads voltage by an angle less than $\pi/2$
(d)	In purely resistive AC circuit	(s)	Current and voltage are in phase

(1) a-q, b-r, c-p, d-s

(2) a-s, b-q, c-p, d-r

(3) a-s, b-r, c-q, d-p

(4) a-p, b-s, c-r, d-q

7. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Smaller the band width, sharper the resonance and easier it is to tune an LCR circuit.

Reason (R) : Resonant frequency is arithmetic mean of half power frequencies.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

TG: @Chalnaayaaar

8. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The impedance of series L-C-R circuit can be greater, equal or less than the resistance.

Reason (R) : The impedance of series LCR circuit depends over angular frequency of applied emf.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In a series LCR circuit at resonance, the voltage across the capacitor or inductor may be more than the applied voltage.

Reason (R) : At resonance in a series LCR circuit, the voltages across inductor and capacitor are equal & 180° out of phase.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

10. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Average power consumed in an ac circuit is equal to average power consumed by resistors in the circuit.

Reason (R) : Average power consumed by capacitor and inductor is zero

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

11. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

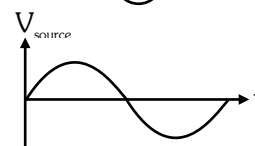
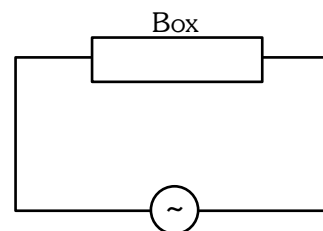
Assertion (A) : In an LC oscillation energy is not constant practically.

Reason (R) : Energy is lost in resistance of inductor and of connecting wires and also in form of electromagnetic radiations.

In the light of the above statements, choose the most appropriate answer from the options given below:

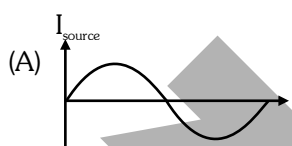
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

12. Box may have any series combination of L, C and R. Column-I represents source current and column-II represents possible statements.

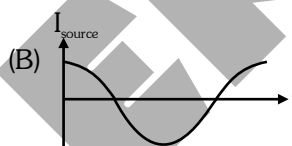


Column-I

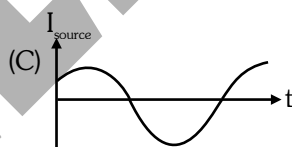
Column-I



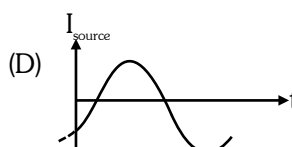
(P) Box may contain LCR.



(Q) Box may contain only LR.



(R) Power factor of box is zero.



(S) Box may contain only CR.

(T) State of resonance.

Options :-

- (1) $A \rightarrow P, B \rightarrow R, C \rightarrow PS, D \rightarrow P$
- (2) $A \rightarrow PT, B \rightarrow R, C \rightarrow PS, D \rightarrow PQ$
- (3) $A \rightarrow PT, B \rightarrow PR, C \rightarrow P, D \rightarrow PQ$
- (4) $A \rightarrow P, B \rightarrow PR, C \rightarrow P, D \rightarrow T$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans.	1	2	1	2	4	1	3	4	2	1	1	2	

ELECTROMAGNETIC WAVES (EMW)

1. For EM waves
- γ rays have higher penetrating power than x rays
 - x rays have higher penetrating power than γ rays
 - Infra red has higher wavelength than x-rays
 - Range of visible light is $4000 \text{ \AA}$ to $7000 \text{ \AA}$
- then correct options are
- a, b and d
 - a, c and d
 - a and c
 - a, b and d
2. Electric field associated with EM wave in free space is given by $\vec{E} = 30 \sin(6 \times 10^8 t + kz) \hat{i}$ then match the following columns.

	Column-I		Column-II
(a)	Direction of propagation of EM wave	(p)	-y axis
(b)	Wave length of EM wave	(q)	2 m^{-1}
(c)	Direction of magnetic field of EM wave	(r)	3.14 m
(d)	Propagation constant 'k' of EM wave	(s)	'-z' axis

- a-p, b-r, c-s, d-r
- a-s, b-r, c-p, d-q
- a-p, b-q, c-s, d-s
- a-s, b-r, c-p, d-p

3. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
- Assertion (A)** : MRI is less harmful medical diagnosis technique than CT-scan
- Reason (R)** : X-rays are used in CT-scan, while MRI uses radio waves.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (A)** is true but **(R)** is false.
 - (A)** is false but **(R)** is true.

4. Match the column I with column II -

	Column-I (Law)		Column-II (Mathematical formula)
(A)	Faraday law	(P)	$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$
(B)	Ampere Maxwell law	(Q)	$\oint \vec{B} \cdot d\vec{A} = 0$
(C)	Gauss law for magnetism	(R)	$\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\phi_B}{dt}$
(D)	Gauss law for electrostatics	(S)	$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i + \epsilon_0 \mu_0 \frac{d\phi_E}{dt}$

Options :-

- $A \rightarrow R, B \rightarrow Q, C \rightarrow S, D \rightarrow P$
- $A \rightarrow P, B \rightarrow Q, C \rightarrow S, D \rightarrow R$
- $A \rightarrow P, B \rightarrow S, C \rightarrow Q, D \rightarrow R$
- $A \rightarrow R, B \rightarrow S, C \rightarrow Q, D \rightarrow P$

ANSWER KEY

Que.	1	2	3	4
Ans.	2	2	1	4

RAY OPTICS

1. Choose correct answer :

	Column-I		Column-II
(A)	Human Eye	(P)	Enlarged and inverted image
(B)	Simple microscope	(Q)	Enlarged and erect image
(C)	Compound microscope	(R)	Diminished and inverted image
(D)	Astronomical telescope	(S)	Diminished and erect image

Optics

(1) $A \rightarrow P, B \rightarrow Q, C \rightarrow P, D \rightarrow P$

(2) $A \rightarrow S, B \rightarrow P, C \rightarrow Q, D \rightarrow R$

(3) $A \rightarrow R, B \rightarrow P, C \rightarrow R, D \rightarrow Q$

(4) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow R$

2. Given below are two statements:

Statement-1 : Magnifying power of a simple microscope is positive.

Statement-2 : Magnifying power of a compound microscope is negative.

In light of above statement choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both statements I and statements II are incorrect.

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Magnifying power of a simple microscope is inversely proportional to focal length of lens in normal adjustment.

Reason (R) : Magnifying power of simple microscope can be infinite.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

4. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : To correct the myopia a lens of negative power must be used. It may be concave or may be combination of convex and concave.

Reason (R) : In this defect image is formed before the retina. So to correct the eyes defect some divergence on incident rays is needed.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

5. A bird is flying at the height of 12 cm from the surface of a lake and a fish is swimming at a depth of 24 cm from the surface.

(Take is $\mu = 4/3$)

	Column-A		Column-B
(A)	Distance of fish from the surface as seen by bird	(P)	16 cm
(B)	Distance of bird from the surface as seen by fish	(Q)	40 cm
(C)	Distance between fish and bird as seen by bird	(R)	18 cm
(D)	Distance between fish and bird as seen by fish	(S)	30 cm

- (1) $A \rightarrow P, B \rightarrow R, C \rightarrow S, D \rightarrow Q$
 (2) $A \rightarrow R, B \rightarrow P, C \rightarrow Q, D \rightarrow S$
 (3) $A \rightarrow R, B \rightarrow P, C \rightarrow S, D \rightarrow Q$
 (4) $A \rightarrow P, B \rightarrow R, C \rightarrow Q, D \rightarrow S$

6. (1) When a light ray goes from denser to rarer medium it may suffer TIR
 (2) For TIR to take place angle of incidence should be greater than critical angle.
 (3) Laws of reflection are not valid for TIR
 (4) Intensity gets decreased during TIR

Incorrect statements are -

- (1) 1 and 2
 (2) 3 and 4
 (3) 1, 3 and 4
 (4) 1, 2, 3 and 4

7. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Keeping a point object fixed, if a plane mirror is moved, the image will also move always.

Reason (R) : In case of a plane mirror, distance of object and its image is equal from any point on the mirror.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A thin lens has two principal focal lengths which may be different in magnitude.

Reason (R) : The distance of both principal focus from optical centre of thin lens depend on the two radii of curvature of the lens. Distance of both principal focus from optical centre of a lens are same only if radii of curvature of both sides of lens are same.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) Both **(A)** and **(R)** are false.

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When the upper half of a converging lens is missing, a real image formed by the lens for a real object will lack its lower half.

Reason (R) : The real image formed by a thin lens for a real object will be always erected.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) Both **(A)** and **(R)** are false.

10. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A simple microscope may have different magnification for different persons.

Reason (R) : All persons must have the same near point distance of 25 cm.

In the light of the above statements, choose the most appropriate answer from the options given below:

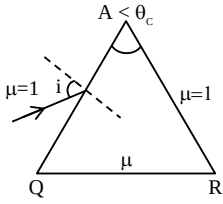
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

11. Column Matching :-

	Column-I		Column-II
(A)	An object is placed at a distance equal to focal length from pole before convex mirror	(P)	Magnification is ∞
(B)	An object is placed at focus before a concave mirror	(Q)	Magnification is 0.5
(C)	An object is placed at the center of curvature before a concave mirror	(R)	Magnification is $(1/3)$
(F)	As object is placed at a distance equal to radius of curvature before a convex mirror	(S)	Magnification is -1

- (1) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
- (2) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow R$
- (3) $A \rightarrow Q, B \rightarrow P, C \rightarrow R, D \rightarrow S$
- (4) $A \rightarrow P, B \rightarrow R, C \rightarrow Q, D \rightarrow S$

12. Match the column (θ_c is the critical angle b/w prism medium and surrounding medium)

(A)		(1)	Light ray always crosses the prism for any value of i
(B)	$A > 2\theta_c$	(2)	Light ray never crosses the prism for any value of i
(C)	$\theta_c < A < 2\theta_c$	(3)	Light ray may cross the prism depends on i

Option :

- (1) $A \rightarrow 1, B \rightarrow 1, C \rightarrow 3$
- (2) $A \rightarrow 3, B \rightarrow 2, C \rightarrow 1$
- (3) $A \rightarrow 2, B \rightarrow 1, C \rightarrow 2$
- (4) $A \rightarrow 1, B \rightarrow 2, C \rightarrow 3$

13. All the following statements are correct except :

(For real object)

- (1) The magnification produced by a convex mirror is always less than 1.
- (2) A virtual, erect, same sized image can be obtained by using only a plane mirror.
- (3) A virtual, erect, magnified image can be formed by using concave mirror.
- (4) A real inverted, same sized image can be formed by using a convex mirror.

14. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : There is no dispersion of light refracted through a rectangular slab.

Reason (R) : Dispersion of light is the phenomenon of splitting of a beam of light into its constituent colours.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

15. Match the following :-

(A)	$\mu = \tan i$	(P)	Snell's law
(B)	$\mu = \frac{1}{\sin \theta_c}$	(Q)	Brewsters angle
(C)	$\mu = \frac{\sin i}{\sin r}$	(R)	Prism
(D)	$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin \frac{A}{2}}$	(S)	Total internal reflection

- (1) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
- (2) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow R$
- (3) $A \rightarrow Q, B \rightarrow S, C \rightarrow P, D \rightarrow R$
- (4) $A \rightarrow P, B \rightarrow S, C \rightarrow S, D \rightarrow R$

16. The angle of a prism is 30° . The rays incident at 60° on one refracting face suffer a deviation of 30° . Then the angle of emergence is $3K^\circ$. Find the value of K.

- (1) 40°
- (2) 20°
- (3) 10°
- (4) 0°

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	3	1	3	2	4	4	4	3	2	4	4	4	3
Que.	16														
Ans.	4														

WAVE OPTICS
INTERFERENCE

1. (A) Phase difference between various particles on a wave front is zero
- (B) Normal to the wave front represents a ray of light.
- (C) A point source of light gives rise to spherical wavefronts.
- (D) A wave front always travels in forward direction in a medium.

Correct statements are –

- (1) A and B only
- (2) A, B and C only
- (3) C and D only
- (4) A, B, C and D
2. In YDSE, match the column-I & column-II

	Column-I		Column-II
(i)	Fringe width	(a)	$\frac{3\lambda D}{d}$
(ii)	Angular fringe width	(b)	$\frac{7\lambda D}{2d}$
(iii)	Position of 3 rd bright w.r.t. centre	(c)	$\frac{\lambda}{d}$
(iv)	Position of 4 th dark w.r.t. centre	(d)	$\frac{\lambda D}{d}$

- (1) (i – a), (ii – b), (iii – c), (iv – d)
- (2) (i – d), (ii – c), (iii – a), (iv – b)
- (3) (i – d), (ii – c), (iii – b), (iv – a)
- (4) (i – c), (ii – d), (iii – a), (iv – b)

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Wave nature can be proved by phenomena of interference and diffraction.

Reason (R) : Only transverse wave can show the phenomena of polarization.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

4. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When width of one of the slits in Young's double slit experiment is four times that of the other, then brighter fringes are nine times brighter than the dark fringes.

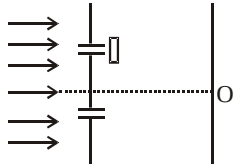
Reason (R) : The amplitude of the wave is proportional to the width of the slit.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

5. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In the figure shown zero order maxima will be above point "O".



Reason (R) : Zero order maxima means a point where the phase difference is zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

6. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If white light is used in place of monochromatic light in YDSE then central point is white. Although at other places coloured fringes will be obtained.

Reason (R) : At centre, path difference is zero for each wave length. Hence all wave will interfere constructively.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

DIFFRACTION

7. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In case of single slit diffraction intensity of higher order maxima decreases.

Reason (R) : Higher order maxima are at larger distance.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Out of radio waves and microwaves, the radio waves are more diffracted in daily life.

Reason (R) : Practically diffraction is more for waves which have longer wavelength.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Diffraction takes place for all types of waves mechanical or non-mechanical, transverse or longitudinal.

Reason (R) : Diffraction's effects are perceptible only if wavelength of wave is comparable to dimensions of diffracting device.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

10. Match the column :

	Column-A		Column-B
(A)	Fresnel distance equal to	(P)	$\theta = 90^\circ$
(B)	In a single slit experiment, when $\Delta = a$, then angular width	(Q)	$\frac{a^2}{\lambda}$
(C)	In single slit experiment, angular width for first dark fringe.	(R)	$n\lambda$
(D)	Path difference for dark fringe in single slit diffraction	(S)	$\frac{\lambda}{a}$

- (1) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow R$
- (2) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$
- (3) $A \rightarrow P, B \rightarrow R, C \rightarrow Q, D \rightarrow S$
- (4) $A \rightarrow S, B \rightarrow Q, C \rightarrow R, D \rightarrow P$

11. In a single slit diffraction, for first maxima:

(i) $\Delta x = \lambda$ (ii) $\Delta \phi = 2\pi$

(iii) $\Delta x = \frac{3\lambda}{2}$ (iv) $\Delta \phi = 3\pi$

- (1) (i) and (ii)
- (2) (i) and (iv)
- (3) (ii) and (iii)
- (4) (iii) and (iv)

POLARISATION

12. Match the column :

(R $\rightarrow$ Rarer medium, D $\rightarrow$ Denser medium)

	Column-I		Column-II
(A)	When light is going from R to D and angle of polarisation θ_p then	(P)	$\theta_p = \tan^{-1}(\operatorname{cosec} i_c)$
(B)	When light is going from D to R and angle of polarisation θ_p then	(Q)	$i_c = \operatorname{cosec}^{-1}(\cot \theta_p)$
(C)	When light is going from R to D and i_c is critical angle then	(R)	$\theta_p = \tan^{-1}(\sin i_c)$
(D)	When light is going from for D to R and i_c is critical angle then	(S)	$i_c = \sin^{-1}(\cot \theta_p)$

Which one of the following can be taken correct?

- (1) (A, S), (B, Q), (C, P), (D, R)
- (2) (A, Q), (B, S), (C, P), (D, C)
- (3) (A, S), (B, Q), (C, R), (D, P)
- (4) (A, Q), (B, S), (C, R), (D, A)

13. When incident light is polarized in a plane perpendicular to the plane of incidence and it is incident at the angle of polarization

- (A) Reflected light is perfectly polarised
- (B) Refracted light will be partially polarised
- (C) Reflected light is partially polarised
- (D) There will be no reflection

- (1) (A, B)
- (2) (B, C)
- (3) (B, D)
- (4) (D)

14. In Nicol prism we get polarized light using the phenomenon of :

- (A) Total internal reflection
- (B) Selective absorption
- (C) Scattering
- (D) Double refraction

- (1) A and B
- (2) B and D
- (3) A and D
- (4) C and D

15. When sun light is incident on water surface at an angle of incidence 53° , the reflected ray is found to be completely polarized then
- (A) The angle of refraction is 37°
- (B) The refractive index of water is $\frac{3}{2}$
- (C) The angle of refraction is 53°
- (D) The refractive index of water is $\frac{4}{3}$
- (1) A, B (2) B, C (3) C, D (4) A, D

16. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : If three polarisers are arranged such that the axis of any two successive polarisers make equal angle of 45° with each other. If unpolarised light of intensity I_0 incident on first polariser then intensity of emergent light after 3rd polariser is $\frac{I_0}{8}$.

Reason (R) : In given condition each time intensity becomes 50% by Malus law.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	2	2	3	1	1	2	1	2	1	4	1	4	3	4
Que.	16														
Ans.	1														

MODERN PHYSICS

RADIOACTIVITY

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Radioactive nuclei emits β^- particle.

Reason (R) : Electrons exist inside the nucleus
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

2. Match the following

	List-I		List-II
a	$n \rightarrow p + \dots$	(p)	α -decay
b	$p \rightarrow n + \dots$	(q)	Positron emission
c	x-ray emission	(r)	β^- emission
d	${}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + \dots$	(s)	K electron capture

- (1) $a \rightarrow r, b \rightarrow q, c \rightarrow q, d \rightarrow s$
(2) $a \rightarrow q, b \rightarrow r, c \rightarrow s, d \rightarrow p$
(3) $a \rightarrow r, b \rightarrow q, c \rightarrow s, d \rightarrow p$
(4) $a \rightarrow q, b \rightarrow s, c \rightarrow r, d \rightarrow p$

3. Given below are two statements:

Statement I : Two different radioactive substances have initially same number of nuclei. Their decay constants are λ_1 and λ_2 ($\lambda_2 < \lambda_1$). Then first radioactive substance decays with faster rate initially.

Statement II : Half life of first radioactive substance is lesser than second.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

4. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:

- (A) both Z and A will decrease if the process is α -decay
(B) Z will decrease but A will not change if process is β^+ decay
(C) Z will decrease but A will not change if process is β^- decay.
(D) Z and A will remain unchanged if process is γ -decay.

Choose the correct statement :

- (1) Only A (2) B, C, D (3) A, D (4) A, B, D

MATTER WAVE

5. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In Davisson-Germer experiment wavelength associated with the beam decreases with increases of speed of electrons.

Reason (R) : Wavelength associated with electron beam in Davisson-Germer experiment

is given by $\lambda = \frac{h}{\sqrt{2mK}}$

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

6. A proton and an electron both have energy 50 eV.

Statement I : Both have different wavelengths.

Statement II : Wavelength depends on energy, not on mass.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

7. Match the following ratio of De-Broglie wavelengths of a proton and α particle, when both

	List-I		List-II
A	Have same speed	(p)	2 : 1
B	Have same potential difference	(q)	1 : 1
C	Have same kinetic energy.	(r)	4 : 1
D	Have same linear momentum	(s)	$2\sqrt{2} : 1$

	A	B	C	D
1	p	q	r	s
2	r	s	p	q
3	p	r	q	s
4	r	q	s	p

8. The wavelength of a photon is twice the de Broglie wavelength of an electron. The speed

of the electron is $v_e = \frac{c}{100}$ then,

- (a) $\frac{E_e}{E_p} = 10^{-4}$ (b) $\frac{E_e}{E_p} = 10^{-2}$
 (c) $\frac{p_e}{m_e c} = 10^{-2}$ (d) $\frac{p_e}{m_e c} = 10^{-4}$
 (1) a, c, d (2) b, c
 (3) c, d (4) a, b, c

NUCLEAR PHYSICS

9. Choose correct statement/statements :

- (a) Nuclear force is independent of charge of nucleons.
 (b) Nuclear force has saturation property.
 (c) Nuclear force is always Attractive in nature.
 (d) Nuclear force exists in range of femtometre.
 (1) a,b,c (2) a,b,d (3) b,c,d (4) a, c, d

10. Match the following column:

For two nuclei having mass number 2 and 16

- (i) Ratio of nuclear radii (p) 1 : 8
 (ii) Ratio of densities (q) 1 : 2
 (iii) Ratio of volumes (r) 1 : 1
 (iv) Ratio of masses (s) 1 : 8

- (1) (i) $\rightarrow$ q, (ii) $\rightarrow$ r, (iii) $\rightarrow$ s, (iv) $\rightarrow$ p
 (2) (i) $\rightarrow$ r, (ii) $\rightarrow$ s, (iii) $\rightarrow$ p, (iv) $\rightarrow$ q
 (3) (i) $\rightarrow$ p, (ii) $\rightarrow$ q, (iii) $\rightarrow$ r, (iv) $\rightarrow$ s
 (4) (i) $\rightarrow$ r, (ii) $\rightarrow$ p, (iii) $\rightarrow$ s, (iv) $\rightarrow$ q

11. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Fission reactions are exothermic

Reason (R) : Total mass of fragments is lesser than the total mass of reactants.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.

12. Given below are two statements

Statement I : Binding energy per nucleon is more important than binding energy.

Statement II : Binding energy per nucleon is measurement of nuclear stability.

In light of above statement choose the most **appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct.
 (2) Statement I is correct and statement II is incorrect.
 (3) Statement I is incorrect and statement II is correct.
 (4) Both statements I and statements II are incorrect.

PHOTOELECTRIC EFFECT

13. The work function of caesium (Cs) metal is 2 eV. When monochromatic radiation beam of intensity 60 kWm^{-2} and frequency $6 \times 10^{14} \text{ Hz}$, is falling normally on Cs plate of 6.6 cm^2 in a photocell then

- (a) Energy of a photoelectron may be $8 \times 10^{-20} \text{ J}$
 (b) Stopping potential is 0.48 V
 (c) Saturation current is 16 A

Correct choices are

- (1) Only b (2) Only b & c
 (3) Only a & b (4) a, b, & c

14. Two statements are given below :

Statement I : Photosensitivity of a metal is high, if its work function is low.

Statement II : Work function for a metal is given by $h\nu_0$, where ν_0 is threshold frequency. Considering the above, choose the appropriate answer from the options given below :

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statements I and statements II are incorrect.

15. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Two photons each of energy 3 eV, can't eject electrons from a surface of work function 5 eV.

Reason (R) : Only a photon of energy greater than work function may emit photoelectron.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

16. On a photosensitive surface of threshold wavelength 5λ , two different monochromatic beams of wavelengths 3λ and λ are incident, one after another, then match the ratio of physical quantities specified in column – I to the ratio values given in column – II

	Column I		Column II
(a)	Maximum KE of photo electrons	(i)	1 : 3
(b)	Speed of incident photons	(ii)	1 : 6
(c)	Momentum of incident photons	(iii)	1 : $\sqrt{6}$
(d)	Fastest speeds of photo electrons	(iv)	1 : 1

Choose the correct answer from the option given below :

- (1) (a) – iii, (b) – ii, (c) – i, (d) – iv
- (2) (a) – ii, (b) – iv, (c) – iii, (d) – i
- (3) (a) – ii, (b) – iv, (c) – i, (d) – iii
- (4) (a) – i, (b) – ii, (c) – iv, (d) – iii

17. Match the column I with column II –

	Column-I (Radiation)		Column-II (Source)
(A)	X-rays	(P)	Nuclei of atoms and sudden deceleration of high energy free electrons
(B)	Gamma rays	(Q)	Inner electrons in atoms and sudden deceleration high energy free electrons
(C)	Radiowaves	(R)	Oscillating electric circuit
(D)	Infrared	(S)	Outer electrons in atoms and molecules
(E)	Ultraviolet	(T)	Outer electrons in atoms

Options :-

- (1) $A \rightarrow P$, $B \rightarrow Q$, $C \rightarrow T$, $D \rightarrow S$, $E \rightarrow R$
- (2) $A \rightarrow Q$, $B \rightarrow P$, $C \rightarrow R$, $D \rightarrow S$, $E \rightarrow T$
- (3) $A \rightarrow P$, $B \rightarrow Q$, $C \rightarrow S$, $D \rightarrow R$, $E \rightarrow T$
- (4) $A \rightarrow Q$, $B \rightarrow P$, $C \rightarrow S$, $D \rightarrow T$, $E \rightarrow R$

18. Match the column I with column II –

	Column-I (Radiation)		Column-II (Use)
(A)	γ -rays	(P)	Structure of crystal
(B)	X-rays	(Q)	Structure of nuclei
(C)	Microwaves	(R)	Radar and radio broadcasting
(D)	Infrared	(S)	Long distance wireless communication via satellite
(E)	Radio waves	(T)	Structure of molecules

Options :-

- (1) $A \rightarrow R, B \rightarrow P, C \rightarrow S, D \rightarrow T, E \rightarrow Q$
- (2) $A \rightarrow P, B \rightarrow Q, C \rightarrow S, D \rightarrow T, E \rightarrow R$
- (3) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow T, E \rightarrow R$
- (4) $A \rightarrow Q, B \rightarrow P, C \rightarrow T, D \rightarrow S, E \rightarrow R$

19. Match the column I with column II –

	Column-I		Column-II
(A)	Force between a proton and a proton inside nucleus	(P)	Gravitational force
(B)	Force between a proton and a neutron inside nucleus	(Q)	Electrostatic force
(C)	Force on an electron in an atom	(R)	Strong force
(D)	Conservative force	(S)	Weak force

Options :-

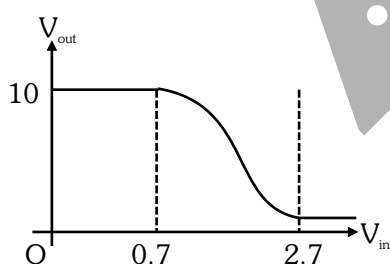
- (1) $A \rightarrow PQR, B \rightarrow PR, C \rightarrow PQ, D \rightarrow PQ$
- (2) $A \rightarrow R, B \rightarrow R, C \rightarrow Q, D \rightarrow RS$
- (3) $A \rightarrow PQR, B \rightarrow PQR, C \rightarrow PQ, D \rightarrow RS$
- (4) $A \rightarrow PQR, B \rightarrow P, C \rightarrow PQ, D \rightarrow RS$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	1	4	1	2	2	2	2	1	1	1	4	1	1
Que.	16	17	18	19											
Ans.	3	2	3	1											

SEMICONDUCTOR & DIGITAL ELECTRONICS

1. Given below are two statements:
Statement-I : Doping of indium in Si crystal leads to p-type semiconductor.
Statement-II : Indium is a pentavalent atom.
 In light of above statement choose the most **appropriate** answer from the options given below:
- (1) Both statement I and statement II are correct.
 - (2) Statement I is correct and statement II is incorrect.
 - (3) Statement I is incorrect and statement II is correct.
 - (4) Both statements I and statements II are incorrect.
2. Match the column-I of special diodes with their utilization given in column-II:
- | Column I | Column II |
|-----------------|-------------------------------|
| (a) Zener diode | (1) As a light sensor |
| (b) LED | (2) Voltage regulator circuit |
| (c) Photodiode | (3) As a bulb of light |
- (1) $a \rightarrow 3, b \rightarrow 2, c \rightarrow 1$
 - (2) $a \rightarrow 3, b \rightarrow 1, c \rightarrow 2$
 - (3) $a \rightarrow 1, b \rightarrow 3, c \rightarrow 2$
 - (4) $a \rightarrow 2, b \rightarrow 3, c \rightarrow 1$
3. For a given transfer characteristic curve of common emitter transistor



Which of the followings are correct –

- (a) At $V_{in} = 0.5$ V transistor is used as a 'OFF switch'
 - (b) At $V_{in} = 1.5$ V transistor is used as a 'ON switch'
 - (c) At $V_{in} = 1.5$ V transistor is used as an amplifier
 - (d) At $V_{in} = 3$ V transistor is used as a 'OFF switch'.
- (1) a, b, c & d
 - (2) a, b & d
 - (3) b & d
 - (4) a & c

4. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : The number of electrons in n-type silicon semiconductor is higher than the number of electrons in a pure silicon semiconductor.
Reason (R) : The law of mass action is applicable only to n-type semiconductors.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) **(A)** is false but **(R)** is true.
5. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Transistor as an amplifier must be used in active region.
Reason (R) : In active region output voltage approximately linearly change with input voltage.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) **(A)** is false but **(R)** is true.
6. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Temperature coefficient of resistance for semiconductor is negative
Reason (R) : Conductivity of semiconductor increases with increment in temperature.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 - (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 - (3) **(A)** is true but **(R)** is false.
 - (4) **(A)** is false but **(R)** is true.

7. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The OR gate function corresponds to switches in parallel.

Reason (R) : The AND gate function corresponds to switches in series.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In an unbiased p-n junction, holes diffuse from the p-region to n-region.

Reason (R) : Hole concentration in p-region is more as compared to n-region.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
- (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
- (3) **(A)** is true but **(R)** is false.
- (4) **(A)** is false but **(R)** is true.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	
Ans.	2	4	4	3	1	1	2	1	

BASIC MATHEMATICS USED IN PHYSICS & VECTORS

2. Definite integral of a function may be positive, negative or zero.
3. $|\vec{A}| = |\vec{B}| = Z$, $\theta = 60^\circ$
 $|\vec{A} + \vec{B}| = 2a \cos\left(\frac{\theta}{2}\right) = \sqrt{3}Z$
 $|\vec{A} - \vec{B}| = 2a \sin\left(\frac{\theta}{2}\right) = Z$
 $\vec{A} \cdot \vec{B} = AB \cos \theta = \frac{1}{2} \cdot Z^2$
 $|\vec{A} \times \vec{B}| = AB \sin \theta = \frac{\sqrt{3}}{2} \cdot Z^2$
4. According to triangle law.
6. Direction of null vector is not defined and direction of $\vec{A} \times \vec{B}$ is given by right hand thumb rule which lies in $\perp$ plane of $\vec{A}$, $\vec{B}$, $(\vec{A} + \vec{B})$ and $(\vec{A} - \vec{B})$.
7. $\vec{A} \cdot \vec{B}$ is a scalar quantity that's why it is known as scalar product.
8. $\sin \theta$ is positive in 1st and 2nd quadrant.
11. Use trigonometry

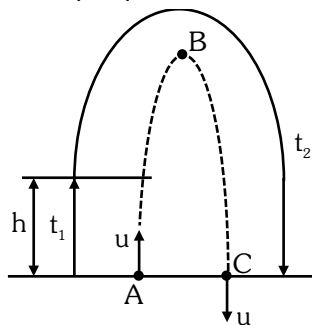
UNIT & DIMENSION

2. Reason is false because error in measurement can have any value.
 eg. $\therefore$ A student can read 23 as 25 from the device from which he is measuring a physical quantity.
3. Assertion is true because lower least count means lower error.
 Since accuracy increases when least count decreases.
 Reason is correct explanation because higher accuracy indicates error will be smaller.
5. Relative error is unitless and dimensionless. Absolute error has dimension.
6. Exact numbers have an infinite number of significant digits but they are generally not reported.
 for eg. If you count 2 person (which is exact number), then the number of person can also be written as 2.000.....
7. Shake measures time whereas light year measures distance.
8. Least count = $\frac{\text{Pitch}}{\text{No. of circular scale divisions}}$
 $\therefore$ Statement I is false.
 Statement II is true because smaller pitch means, smaller least count. $\therefore$ Greater accuracy.
9. $[\text{Stress}] = [\text{Modulus of Elasticity}] = [\text{Pressure}] = [\text{Energy density}] = [ML^{-1}T^{-2}]$
 $[\text{Strain}] = [\text{Angle}] = [M^0L^0T^0]$
 $[\text{Torque}] = [\text{Energy}] = [ML^2T^{-2}]$
10. Electrical resistance [Use $P = I^2R$]
 $R = \frac{[P]}{[I]^2} = \frac{[ML^2T^{-3}]}{[A^2]} = [ML^2T^{-3}A^{-2}]$
 Capacitance $\left[\text{use } U = \frac{1}{2} \frac{Q^2}{C} \right]$
 $[C] = \frac{[Q^2]}{[U]} = \frac{[AT]^2}{[ML^2T^{-2}]} = [M^{-1}L^{-2}T^4A^2]$
 Magnetic field [use $F = ILB$]
 $[B] = \frac{[F]}{[I][L]} = \frac{[MLT^{-2}]}{[A][L]} = [MT^{-2}A^{-1}]$
 Inductance [use $U = \frac{1}{2} LI^2$]
 $[L] = \frac{[U]}{[I]^2} = \frac{[ML^2T^{-2}]}{[A^2]} = [ML^2T^{-2}A^{-2}]$
11. $[LC] = [T^2]$
 $[LR] = [M^2L^4T^{-5}A^{-4}]$
 $[H] = [L^2T^{-2}]$
 $[S] = [L^2T^{-2}K^{-1}]$
12. Boltzmann constant (use $E = \frac{3}{2} kT$)
 $[K] = ML^2T^{-2}K^{-1}$
 Coefficient of viscosity $\left(\text{use } \eta = \frac{F}{A} \left(\frac{\Delta x}{\Delta y} \right) \right)$
 $[\eta] = [ML^{-1}T^{-1}]$
 Planck's constant (use $E = hv$)
 $[h] = [ML^2T^{-1}]$
 Thermal conductivity $\left(\text{use } K = \frac{iL}{A\Delta\theta} \right)$
 $[K] = [MLT^{-3}K^{-1}]$
13. $[\text{Energy density}] = [\text{Young's modulus}] = [ML^{-1}T^{-2}]$
14. Parsec and light year are units of distance.
15. $[h] = [\text{Angular impulse}] = [\text{Angular momentum}] = [ML^2T^{-1}]$

KINEMATICS

1. $(\vec{V}_i) = (u \cos \theta) \hat{i} + (u \sin \theta) \hat{j}$
 $(\vec{V}_f) = (u \cos \theta) \hat{i} - (u \sin \theta) \hat{j}$
 $(\vec{V}_{top}) = (u \cos \theta) \hat{i}$
 $|\vec{V}_i| = \text{Initial speed} = u$
 $|\vec{V}_f| = \text{final speed} = u$
 $|\vec{V}_{top}| = \text{speed at top} = u \cos \theta \quad (A)$
 $(B) \Delta \vec{V} = \vec{V}_f - \vec{V}_i \text{ \& } \Delta |\vec{V}| = |\vec{V}_f| - |\vec{V}_i| = 0$
 $|\Delta \vec{V}| = (2u \sin \theta) \hat{j}$
 $|\Delta \vec{V}| = 2u \sin \theta$
 $(C) |\vec{V}_i| - |\vec{V}_{top}| = u - u \cos \theta$
 $(D) |\vec{V}_i - \vec{V}_{top}| = u \sin \theta$
2. $\vec{v}_i = 3\hat{i} + 4\hat{j} \Rightarrow |\vec{v}_i| = 5$
 $\vec{v}_f = 3\hat{i} + 4\hat{j} \Rightarrow |\vec{v}_f| = 5$
 $(A) |\Delta \vec{v}| = |\vec{v}_f - \vec{v}_i| = |3\hat{i} + 4\hat{j} - 3\hat{i} + 4\hat{j}|$
 $= |8\hat{j}| = 8$
 $(B) \Delta |\vec{v}| = |\vec{v}_f| - |\vec{v}_i| = 5 - 5 = 0$
 $(C) \vec{v}_f - \vec{v}_i = 8\hat{j}$
 $(D) \vec{v}_f + \vec{v}_i = 6\hat{i}$

3.



from the figure, it is clear that time of flight is $(t_1 + t_2)$

for maximum Ht :-

time taken from B to C is $\left(\frac{t_1 + t_2}{2}\right)$

Applying 2nd eqⁿ of motion from B to C

$$H_{\max} = 0 + \frac{1}{2}g\left(\frac{t_1 + t_2}{2}\right)^2$$

for (u) $\Rightarrow$ Speed at A = Speed at C

Apply 1st eqⁿ of motion between B & C

$$u = 0 + g\left(\frac{t_1 + t_2}{2}\right)$$

$$\text{for (h)} \Rightarrow h = ut_1 - \frac{1}{2}gt_1^2$$

$$h = g\left(\frac{t_1 + t_2}{2}\right)t_1 - \frac{1}{2}gt_1^2$$

$$h = \frac{1}{2}gt_1t_2$$

5. Two projectiles having same range may have different time of flight.
6. Projectile motion is an example of motion under constant acceleration.
9. In one hour it reach to its initial position.
11. If $v^2 \propto x$
 $v^2 = kx$
 $2(vdv/dx) = k$
 $\Rightarrow a = k/2$
 $\Rightarrow a = \text{const.}$
12. When body moves with constant velocity then it moves with constant speed in same direction,
 So $|\text{Inst. velocity}| = |\text{Avg. vel.}| = \text{Inst. speed} = \text{Avg speed}$
15. Average speed for first 6 s
 $= \frac{\text{distance travelled in 6 s}}{6 \text{ s}}$
 $= \frac{\frac{1}{2} \times 15 \times 4 + \frac{1}{2} \times 2 \times 10}{6} = \frac{20}{3} \text{ ms}^{-1}$
16. Average velocity for first 12 s
 $= \frac{\text{displacement in first 12 s}}{12 \text{ s}}$
 $= \frac{-\frac{1}{2} \times 15 \times 4 + \frac{1}{2} \times 2 \times 10 + 2 \times 10 + \left(\frac{20 + 10}{2}\right)(2 - 4)}{12}$
 $= \frac{60}{12} = 5 \text{ ms}^{-1}$
17. Average acceleration from $t = 5 \text{ s}$ to $t = 15 \text{ s}$
 $= \frac{v(t = 15) - v(t = 5)}{15 - 5} = \frac{0 - 5}{10} = -\frac{1}{2} \text{ ms}^{-2}$

NEWTON'S LAWS OF MOTION & FRICTION

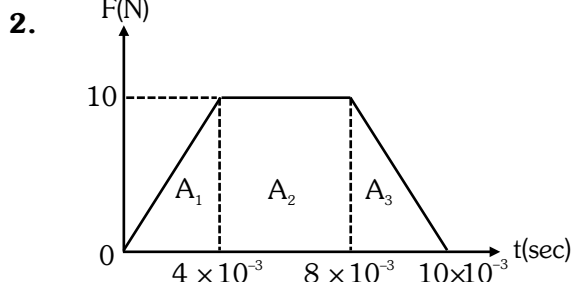
1. $W_{\text{eff}} = M(g \pm a)$

(b) $W_{\text{upward}} = M(g + a)$

$a = +ve \Rightarrow W_{\text{upward}} > Mg$

(c) $W_{\text{downward}} = M(g - a)$

$a = -ve \Rightarrow W_{\text{downward}} > Mg$



(a) $A_1 + A_2 + A_3 = m(v_f - v_i)$

$$\frac{1}{2}(4 \times 10^{-3})(10) + (4 \times 10^{-3} \times 10) + \frac{1}{2}(2 \times 10^{-3})(10)$$

$$= 70 \times 10^{-3} \left(v_f + \frac{50}{100} \right)$$

$$2 \times 10^{-2} + 4 \times 10^{-2} + 10^{-2}$$

$$= 7 \times 10^{-2} \left(v_f + \frac{50}{100} \right)$$

$$7 \times 10^{-2} = 7 \times 10^{-2} \left(v_f + \frac{50}{100} \right)$$

$$v_f + \frac{1}{2} = 1$$

$$v_f = 0.5 \text{ m/sec}$$

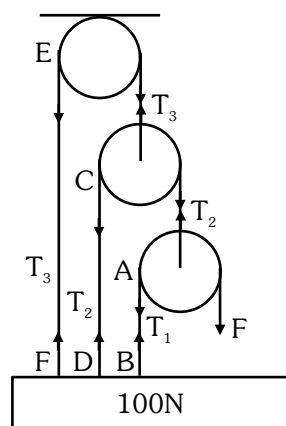
$$v_i = 50 \text{ cm/sec}$$

3. The earth revolves around the Sun. Hence, it experiences a centripetal force towards the Sun. Therefore, it is always accelerating & hence it's a non-inertial frame of reference.

Also, the Earth is always rotating about its own axis. Therefore, any point on the Earth's surface is performing circular motion about its axis.

This rotation also makes the Earth's surface a non-inertial frame of reference.

4.



$$T_1 + T_2 + T_3 = 100 \text{ N}$$

$$F + 2F + 4F = 100 \text{ N}$$

$$F = T_1$$

$$T_2 = 2T_1 = 2F$$

$$T_3 = 2T_2 = 4F$$

$$F = \frac{100}{7} \text{ N}$$

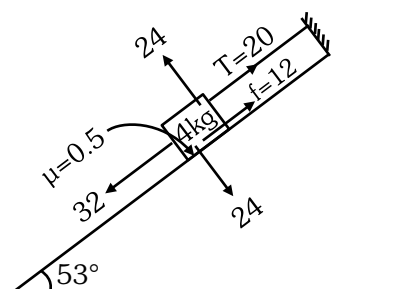
$$T_1 = F = \frac{100}{7} \text{ N}$$

$$T_2 = 2F = \frac{200}{7} \text{ N}$$

$$T_3 = 4F = \frac{400}{7} \text{ N}$$

$$T_4 = 100 \text{ N (Total force of all tension on block)}$$

5.



$$f = f_{\text{max}} = \mu N = 0.5 \times 24$$

$$\Rightarrow \boxed{f = 12 \text{ N}}$$

$$\& \quad T + f = 32$$

$$\Rightarrow \boxed{T = 20 \text{ N}}$$

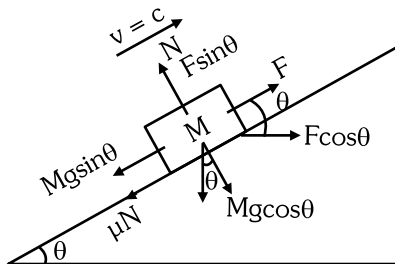
$$\text{Contact force} = \sqrt{f^2 + N^2} = \sqrt{12^2 + 24^2}$$

$$= 12\sqrt{5}$$

$$\approx 26.83 \text{ N}$$

$$\approx 27 \text{ N}$$

6. $F = Mg \sin \theta + \mu N = Mg \sin \theta + \mu Mg \cos \theta$



Net force in horizontal direction = $F \cos \theta$

$$= (Mg \sin \theta + \mu Mg \cos \theta) \cos \theta$$

$$= Mg \sin \theta \cos \theta + \mu Mg \cos^2 \theta$$

Net force in vertical direction = $F \sin \theta$

$$= (Mg \sin \theta + \mu Mg \cos \theta) \sin \theta$$

$$= Mg \sin^2 \theta + \mu Mg \cos \theta \sin \theta$$

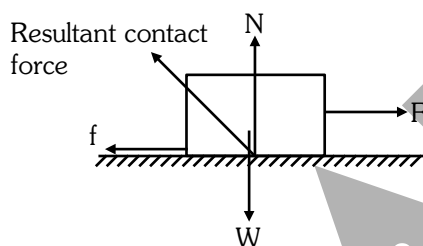
Net force along the plane

$$= F = Mg \sin \theta + \mu Mg \cos \theta$$

Net force perpendicular to plane = 0

7. $\vec{F}_{\text{net ext}} = \frac{d\vec{P}}{dt}$ (Newton's second law)

8. A body may be at rest while experiencing a static friction force.



$$f = F$$

$$\therefore \text{acc}^n = 0$$

9. Pseudo force does not actually act on a body.

It is only drawn in an F.B.D. when the frame of reference is non-inertial.

10. By definition of 'equilibrium' & 'rest'

11. Theoretical

12. The body experiences a net downward force of gravitation by Earth.

Hence, it will start moving downwards after sometime.

13. Acceleration of blocks is 2 m/s^2 .

$$\text{Displacement} = \frac{1}{2} \times 2 \times 1^2 = 1 \text{ m.}$$

One moves 1 m upwards and the other 1 m downwards. Separation = 2 m

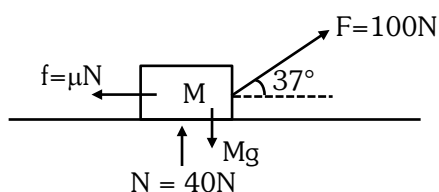
15. When a body is placed on a rough inclined plane, it is acted upon by a reactional force due to plane [electromagnetic in nature], a frictional force due to roughness of plane [electromagnetic in nature] and a gravitational force (due to its weight).

16. Statement (I) follows Newton's first law, i.e. law of inertia.

Statement (II) follows Newton's third law, i.e. action-reaction pair.

WORK, ENERGY & POWER

1. FBD of block



$$\text{So, } W_F = FS \cos 37^\circ = 800 \text{ J}$$

$$W_N = \text{zero } (\theta = 90^\circ)$$

$$W_F = -20 \times 10 = -200 \text{ J}$$

$$W_{\text{Total}} = 600 \text{ J}$$

2. Velocity at $t = 5 \text{ sec}$

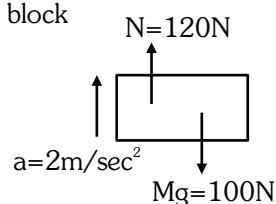
$$V = u + at$$

$$V = 10 \text{ m/s}$$

$$N - mg = ma$$

$$N = m(g + a) = 120 \text{ N}$$

FBD of block



displacement in 5 sec

$$s = ut + \frac{1}{2}at^2$$

$$s = 25 \text{ m}$$

$$\text{Power by normal at } t = 5 \text{ sec} = 120 \times 10 \\ = 1200 \text{ W}$$

Power by normal in 5 sec

$$= \frac{\text{work done by normal}}{\text{time}} = 600 \text{ W}$$

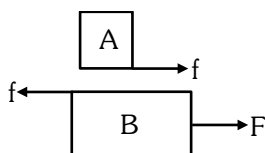
Power by gravity (Mg) at $t = 5 \text{ sec}$

$$= 100 \times 10 \times \cos 180^\circ = -1000 \text{ W}$$

Power by gravity (Mg) in 5 sec

$$\frac{\text{work done by gravity}}{\text{time}} = -500 \text{ W}$$

5. F.B.D. of block (of forces in horizontal direction)



It can be seen that W.D. by friction (f) on block A is positive and on block B is negative.

6. Statements (a) & (b) are incorrect because according to work energy theorem, work done by all forces (Internal, External, Conservative, Non conservative) is equal to change in kinetic energy.

7. Assertion is correct because according to work energy theorem work done by all forces (Internal, External, Conservative, Non conservative) is equal to change in kinetic energy.

Therefore, kinetic energy of system may increase due to internal force only.

Reason is incorrect.

8. Assertion is correct, because work done is positive if angle between $\vec{F}$ & $\vec{S}$ is acute.

Reason is false:- Work done by friction may be positive, negative or zero.

9. $\vec{OA} = \hat{i} + \hat{k}$, $\vec{OB} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{OC} = \hat{i} + 2\hat{j} + \hat{k}$
 $\Rightarrow \vec{AB} = \vec{OB} - \vec{OA} = \hat{i} + \hat{j}$, $\vec{BC} = \vec{OC} - \vec{OB} = \hat{j} - \hat{i}$,
 $\vec{AC} = \vec{OC} - \vec{OA} = 2\hat{j}$

$$W_{AB} = \vec{F} \cdot \vec{AB} = 7, W_{BC} = \vec{F} \cdot \vec{BC} = 1,$$

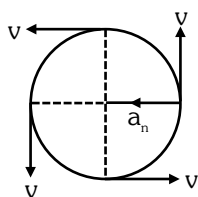
$$W_{AC} = \vec{F} \cdot \vec{AC} = 8$$

10. Since mechanical energy is conserved, balls will have same potential energy at the same height and thus will have same kinetic energy.

$$11. F = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k} \Rightarrow -(4x\hat{i} + 9y^2\hat{j} + 2\hat{k})$$

CIRCULAR MOTION

2.



a_t : tangential acceleration

a_n : Normal acceleration

$$a_t = 0$$

$$a_n \neq 0; a_n = \frac{v^2}{r}$$

4. For (A) $v_B^2 = v_A^2 - 2gR = 9gR - 2gR = 7gR$

$$\Rightarrow \text{For (B)} -2g(2R) = 5gR \Rightarrow v_B = \sqrt{7gR}$$

$$\text{For (C)} \begin{array}{c} \uparrow v_B \\ \bullet \\ \downarrow mg \end{array} \quad T_B = \frac{mv_B^2}{R} = 7mg$$

$$\text{For (D)} \begin{array}{c} \leftarrow v_C \\ \bullet \\ \downarrow T \\ \downarrow mg \end{array} \quad T_C + mg = \frac{mv_C^2}{R}$$

$$\Rightarrow T_C = 4mg$$

5. As $V_H^2 = V_L^2 - 2gh$ where $h = 2r$

$$\therefore V_H^2 = V_L^2 - 4gr$$

Now, $V_L = 2V_H$ (given)

$$\therefore V_H = \sqrt{\frac{4gr}{3}} \text{ \& } V_L = 2V_H = 2\sqrt{\frac{4gr}{3}}$$

$$T_H = T_{\min} = \frac{mv_H^2}{r} - mg = \frac{mg}{3}$$

$$T_L = T_{\max} = \frac{mv_L^2}{r} + mg = \frac{19}{3}mg$$

6. As $v = 2t$ and let radius of circular path is r then

$$a_t = \frac{dv}{dt} = 2$$

$$a_r = \frac{v^2}{r} = \frac{4t^2}{r}$$

Therefore,

$$a = \sqrt{a_t^2 + a_r^2}$$

$$a = \sqrt{4 + \frac{16t^4}{r^2}}$$

$\therefore$ (B) and (D) are correct.

7. $a_t = a$

(A) Since car has both tangential as well as centripetal acceleration.

$\therefore$ Acceleration is not towards the centre.

(B) Friction provides both centripetal and tangential acceleration.

$\therefore$ Friction force $> \frac{mv^2}{R}$

(C) Due to tangential acceleration, speed will increase, due to which, centripetal acceleration will increase.

(D) Since centripetal acceleration increases with time, $\therefore$ net acceleration will vary with time.

8. The person applies F constantly in a fixed direction. So to keep particle in uniform circular motion, some other force whose direction and magnitude varies with time should act on particle.

CENTRE OF MASS

1. Acceleration of centre of mass $\vec{a}_{cm} = \frac{\vec{F}_{ext.}}{M}$

If no external force acts on the system

$$\vec{a}_{cm} = 0 \text{ (Assertion is correct)}$$

Reason is false because; $\vec{a}_{cm} = 0$ implies velocity will not change, if C.O.M is initially at rest, then it will remain at rest in absence of external force.

2. Assertion is correct because in quick collision time duration will be less due to which higher impulse will act on a body.

Reason is false because collision may be both elastic or inelastic.

TG: @Chalnaayaaar

3. (1) False, Ring COM is outside the body.

(2) False, No mass is present physically at C.O.M of ring.

(3) True, C.O.M will lie closer to massive particle.

5. (A) Both particles will touch the ground together.
(B) C.O.M will follow same path.

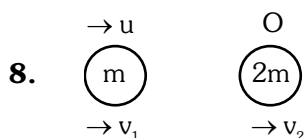
6. $I = m \cdot \Delta v$

$$= m[(-2\hat{i} + \hat{j}) - (3\hat{i} + 2\hat{j})]$$

$$= m[-5\hat{i} - \hat{j}] \rightarrow \text{Impulse on } m$$

$$\text{Impulse on } M = -(\text{Impulse on } m)$$

$$= m(5\hat{i} + \hat{j})$$



v_1 & v_2 are velocities after collision

Conservation of linear momentum

$$mu = mv_1 + 2mv_2$$

$$u = v_1 + v_2 \quad \dots\dots\dots(1)$$

$$e = 1 = \frac{v_2 - v_1}{u}$$

$$u = v_2 - v_1 \quad \dots\dots\dots(2)$$

From (1) & (2)

$$v_2 = \frac{2u}{3}; v_1 = -\frac{u}{3}$$

$$\text{Momentum of 1st particle} = -\frac{mu}{3} = -\frac{P}{3}$$

$$\text{Momentum of 2nd particle} = +2m\left(\frac{2u}{3}\right) = \frac{4P}{3}$$

$$\text{K.E. of 1st particle} = \frac{1}{2}m\left(\frac{u}{3}\right)^2 = \frac{K}{9}$$

$$\text{K.E. of 2nd particle} = \frac{1}{2} \times 2m\left(\frac{2u}{3}\right)^2 = \frac{8K}{9}$$

ROTATIONAL MOTION

3. In conical pendulum, net torque about the axis of rotation due to gravity and the tension is zero. Therefore, the angular momentum is conserved.

Assertion is true but Reason is false.

4. **Axis-1:**

$$(M.I.)_{\text{Rod Hz.}} = \frac{m\ell^2}{3}; (M.I.)_{\text{Rod Vert.}} = 0$$

- Axis-2:**

$$(M.I.)_{\text{Rod Hz.}} = 0; (M.I.)_{\text{Rod Vert.}} = \frac{m\ell^2}{3}$$

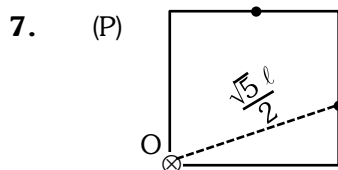
- Axis-3:**

$$(M.I.)_{\text{Rod Hz.}} = \frac{m\ell^2}{3}; (M.I.)_{\text{Rod Vert.}} = \frac{m\ell^2}{3}$$

- Axis-4:**

$$(M.I.)_{\text{Rod Hz.}} = \frac{m\ell^2}{3}; (M.I.)_{\text{Rod Vert.}} = mL^2$$

5. about (A) $10(2)\hat{k} + 6(2)\hat{k} = 32\hat{k}$
about (B) $10 \times 2\hat{k} - 10 \times 2\hat{k} + 6 \times 2\hat{k} - 12 \times 2\hat{k} = -12\hat{k}$
about (C) $-(10)(2)\hat{k} = -20\hat{k}$
about (D) $-12(2)\hat{k} - 10(2)\hat{k} - 10(2)\hat{k} = -64\hat{k}$



M.O.I. about z-axis passing through 'O'

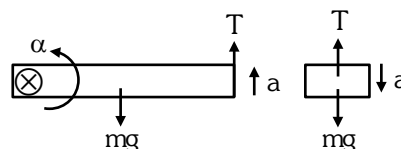
$$= 2\left[\frac{m\ell^2}{3}\right] + 2\left[\frac{m\ell^2}{12} + \frac{5}{4}m\ell^2\right] = \frac{10}{2}m\ell^2$$

$$(Q) \text{ about side } I = 2\left(\frac{m\ell^2}{3}\right) + m\ell^2 = \frac{5m\ell^2}{3}$$

$$(R) \text{ about x-axis } I = 2\left(\frac{m\ell^2}{3}\right) + 2(m)\left(\frac{\ell}{2}\right)^2 = \frac{2}{3}m\ell^2$$

$$(S) \text{ about z-axis } I = 4\left[\frac{m\ell^2}{12} + m\left(\frac{\ell}{2}\right)^2\right] \\ = \frac{m\ell^2}{3} + m\ell^2 = \frac{4m\ell^2}{3}$$

- 10.



$$a = \ell\alpha \quad \dots(1)$$

$$mg - T = ma \quad \dots(2)$$

$$T\ell - mg\frac{\ell}{2} = I\alpha \quad \dots(3)$$

on solving (1), (2), (3)

$$a = 3g/8$$

$$a = \frac{3g}{3\ell}$$

$$T = \frac{5mg}{8}$$

GRAVITATION

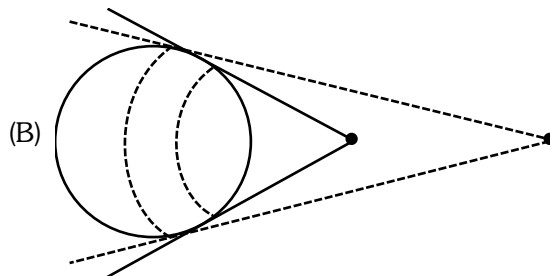
1. After the force disappears, it continues to move in a straight line due to inertia of direction and orbital speed $V_0 = \sqrt{\frac{GM}{r}}$.

So, $V_0 \propto \frac{1}{\sqrt{r}} \Rightarrow r \uparrow \Rightarrow V_0 \downarrow$
2. Work has to be done on a satellite by external agency to shift it in higher orbit. So total energy will not remain conserved.
4. (A) The magnitude of force between 1 Kg bodies is $6.67 \times 10^{-11} \text{ N}$
(B) is correct, if escape velocity for a planet is high, the thermal speeds of lighter gases are less than the escape velocity.
 $\therefore$ Lighter gases are not able to escape.
(C) is wrong; friction force is electromagnetic in nature.
(D) is correct.
5. Satellite in circular orbit has constant speed, but not constant velocity and it will have centripetal acceleration.
6. **I.** Gravitational force on satellite due to earth acts always towards the centre of earth.
II. Since distance of the satellite from the earth changes, then the total mechanical energy will be change.
III. Since v changes in magnitude & direction.
 $\therefore$ Linear momentum is not conserved.

7. $r \rightarrow 2r$

(A) Angular Momentum : $L = mvr = m\sqrt{\frac{GM}{r}} \cdot r$

$L \propto \sqrt{r} \quad r \uparrow \Rightarrow L \uparrow$



Area covered will $\uparrow$

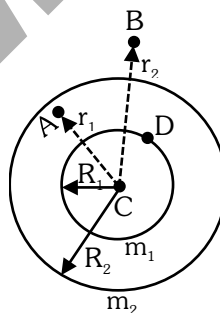
(C) $U = -\frac{GMm}{r} = -U_0$

$U' = -\frac{GMm}{2r} = -\left(\frac{U_0}{2}\right)$

P.E. will $\uparrow$

(D) $KE = \frac{GMm}{2r} \Rightarrow KE \text{ will } \downarrow$

8.



(A) $U_A = -\frac{Gm_1}{r_1} - \frac{Gm_2}{R_2}$; $U_B = -\frac{Gm_1}{r_2} - \frac{Gm_2}{r_2}$

$U_A < U_B$

(B) $I_A = \frac{Gm_1}{r_1^2}$; $I_B = \frac{G(m_1 + m_2)}{r_2^2}$

Can't compare as data is insufficient.

(C) $U_C = -\frac{Gm_1}{R_1} - \frac{Gm_2}{R_2}$; $U_D = -\frac{Gm_1}{R_1} - \frac{Gm_2}{R_2}$

remains same

(D) $I_A = \frac{Gm_1}{r_1^2}$; $I_D = \frac{Gm_1}{R_1^2}$

$I_D > I_A$

PROPERTIES OF MATTER AND FLUID MECHANICS

1. I. In water adhesive force is greater than cohesive force. $\therefore$ It wets the capillary, whereas in mercury, cohesive force is greater.

$$\text{II. } \Delta P = \frac{4T}{R} \Rightarrow \Delta P \propto \frac{1}{R}$$

3. In case of ice and water, the volume of water displaced by ice is equal to the volume of water formed by melting the ice, therefore level remains unchanged.

5. Variation of pressure in vertical direction

$$\Delta P = \rho(a_v) \cdot (\text{Vertical distance between points}).$$

Variation of pressure in horizontal direction.

$$\Delta P = \rho(a_h) \cdot (\text{Horizontal distance between points})$$

6. Velocity of efflux is given by

$$V = \sqrt{\frac{2gh}{1 - \frac{a^2}{A^2}}}$$

$a \rightarrow$ area of orifice (hole)

$A \rightarrow$ Area of tank

8.
$$\Delta \ell = \frac{F\ell}{Ay}$$

for copper wire :

$$(\Delta \ell)_1 = \frac{500 \times 8}{0.5 \times 10^{-4} \times 10^{11}}$$

$$(\Delta \ell)_1 = 0.8 \text{ mm}$$

for steel wire :

$$(\Delta \ell)_2 = \frac{500 \times 4}{0.5 \times 10^{-4} \times 2 \times 10^{11}}$$

$$(\Delta \ell)_2 = 0.2 \text{ mm}$$

Total elongation :

$$\Delta \ell = (\Delta \ell)_1 + (\Delta \ell)_2$$

$$= 1 \text{ mm}$$

9.
$$\therefore P_{e_x} = \frac{4T}{r}$$

$$\therefore P_{e_x} \propto \frac{1}{r}$$

The air pressure is greater inside the smaller bubble.

Hence air flows from the smaller to the larger bubble.

10. In given condition

$$\therefore \text{Tension in a rod} = F$$

$$\therefore \text{Stress} = F/A$$

11. When a needle is placed carefully on the surface of water, it floats on the surface of water due to the surface tension of water, which does not allow the needle to sink. In case of a steel ball, the surface tension of the water is not sufficient to keep it floating, so it sinks down.

12. On spraying, surface area increases and hence surface energy; this increase in surface energy is on the expense of decrease in internal energy and hence temperature decreases.

13. Viscosity of liquids is greater than gases.

Insoluble impurities decreases the cohesive force acting between the molecule of liquid. Therefore surface tension decreases.

14. For a substance:

$$B = \frac{1}{C}$$

$$BC = 1$$

for ideal rigid body

$$\text{strain} = 0$$

$$\therefore \eta = \frac{\text{shear stress}}{\text{shear strain}}$$

$$\eta = \infty$$

15. For concave meniscus

$$F_A > \frac{F_C}{\sqrt{2}} \Rightarrow \theta_c < 90^\circ$$

For convex meniscus

$$F_A > \frac{F_C}{\sqrt{2}} \Rightarrow \theta_c > 90^\circ$$

For plane meniscus

$$F_A = \frac{F_C}{\sqrt{2}} \Rightarrow \theta_c = 90^\circ$$

THERMAL PHYSICS TG: @Chalnaayaaar

1. Celsius scale ranges from 0°C to 100°C
 Fahrenheit scale ranges from 32°F to 212°F
 Kelvin scale ranges from 273.15 K to 373.15 K
 unknown scale ranges from x to y .
2. When a bimetallic strip is heated it bends in the form of an arc with material having greater temperature expansion coefficient going on outer side. It is used in thermostat.
3. Since the coefficient of linear expansion for brass is greater than steel so system should be cooled to take out the disc from the plate.
4. When heat is supplied to the solid its temperature may increase (phase heating) or it may remain constant (state change).
5. A good cooking utensil should have low specific heat, low thermal capacity and high coefficient of thermal conductivity.
6. In BC temperature remains constant
 ice $\rightarrow$ water
 As density of ice is less than water so after melting volume decreases.
7. During melting temperature of substance remains constant.
8. Heat $\rightarrow$ Energy in transit
 Internal energy $\rightarrow$ It is sum of energy due to attractive force and due to random motion.
 Temperature $\rightarrow$ Degree of hotness or coldness of a substance.
 Specific heat $\rightarrow$ Amount of heat required to raise temperature of unit mass by 1°C .
9. If $t = 0$
 $\Rightarrow a(\uparrow), r(\downarrow), e(\uparrow)$ {Here, $a + r = 1$ and $a \propto e$ }
 $\Rightarrow a(\downarrow), r(\uparrow), e(\downarrow)$
11. $P = \frac{dQ}{dt} = \sigma AT^4 \propto r^2$
 $R = \frac{d\theta}{dt} = \frac{\sigma AT^4}{ms} \propto \frac{A}{v} \propto \frac{r^2}{r^3} \propto \frac{1}{r}$
13. $\lambda_m \propto \frac{1}{T}$
 (As temperature increases λ_m decreases)
14. $V_{\text{RMS}} \propto \sqrt{\frac{T}{M_w}}$
 V_{RMS} also depends on nature of gas.

15. $V \propto \sqrt{\frac{T}{M_w}}$
 $\Rightarrow T \rightarrow \text{const.} \Rightarrow V \propto \frac{1}{\sqrt{M_w}}$
 $\Rightarrow (M_w)_{\text{N}_2} > (M_w)_{\text{He}}$
 $\Rightarrow V_{\text{He}} > V_{\text{N}_2}$
 and $\text{KE} = \frac{f}{2} kT$; $\text{KE} \propto f$
 for N_2 gas; $f \uparrow \text{KE} \uparrow$
16. As $\frac{p}{\rho} = \frac{RT}{M_w} \Rightarrow p = \left(\frac{M}{V}\right)\left(\frac{RT}{M_w}\right)$
 at constant temperature $\Rightarrow p \propto \frac{1}{V}$
 $V \uparrow$ and $P \downarrow$
18. Remember
 $V_s = \sqrt{\frac{\gamma RT}{M_w}}$; $V_{\text{Avg}} = \sqrt{\frac{8 RT}{\pi M_w}}$
 $V_{\text{RMS}} = \sqrt{\frac{3RT}{M_w}}$; $V_{\text{mp}} = \sqrt{\frac{2RT}{M_w}}$
19. (A) Adiabatic bulk modulus
 $= -\left(\frac{dP}{dV}\right) \times V = -\left(-\gamma \frac{P}{V}\right) \times V = +\gamma P$
 (B) Slope of isothermal curve $PV = K$
 $PdV + VdP = 0$
 $(\text{slope})_{\text{is}} = \frac{dP}{dV} = -\frac{P}{V}$
 (C) $C_v = \frac{f}{2}R$, $C_p = \left(\frac{f}{2} + 1\right)r$, $\gamma = \frac{C_p}{C_v}$
 Therefore $\gamma = 1 + \frac{2}{f}$
 So, $f = \frac{2}{\gamma - 1}$
 (D) $C_p = \frac{\gamma R}{\gamma - 1}$

$$\frac{C_p}{R} = \frac{\gamma}{\gamma - 1}$$

20. $Q = W + \Delta U$

$$Q = 0$$

$$\Delta U = -W$$

21. Process ab $\rightarrow$ (Isothermal process)

$$T = \text{constant} \Rightarrow U = \text{constant} \Rightarrow \Delta U = 0$$

and p increasing

Process bc $\Rightarrow p = \text{constant}$

then isochoric process

Process cd $\Rightarrow$ (Isothermal process)

$$T = \text{constant} \Rightarrow U = \text{constant} \Rightarrow \Delta U = 0$$

and p decreasing.

Process da $\Rightarrow$ (Isothermal process)

$p = \text{constant}$ then isochoric process

22. (i) $dU = nC_v dT$ is valid for all types of process.

(ii) Temperature of the system can change in adiabatic process, polytropic process.

(iii) $dQ = dU + d\omega$ (First law) is valid for all types of process.

(iv) The process in which heat exchange does not occurs is adiabatic process and free expansion.

23. $C_p - C_v = R$

$$C_p \Rightarrow Q = W + \Delta U$$

$$C_v \Rightarrow Q = \Delta U$$

(in isobaric process)

24. In a cyclic process $dU = 0 \Rightarrow U \rightarrow \text{constant}$

In an isothermal process $dU = 0$

$$\Rightarrow U \rightarrow \text{constant}$$

If $\Delta Q = \Delta\omega$ means $dU = 0 \Rightarrow U \rightarrow \text{constant}$

25. $Q = 0$ for adiabatic

(i) $nCdT = 0 \Rightarrow C = 0$

(ii) $C = \frac{\Delta Q}{mdT}$ for IT ; $dT = 0$

$$C = \infty$$

(iii) $C_v = \frac{f}{2}R$; $C_p = \left(\frac{f+2}{2}\right)R$

f depends on nature of gas

(iv) $Q = mS\Delta T$... (1)

$Q = nC\Delta T$... (2)

$$n = \frac{m}{M_w}$$

$$mS = nC$$

$$mS = \frac{m}{M_w} C \Rightarrow C = M_w S$$

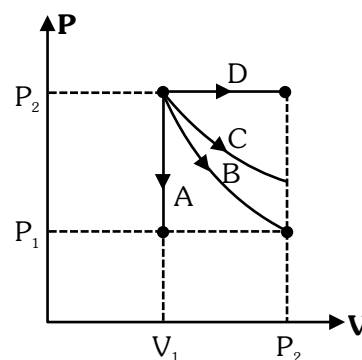
26. (A) Adiabatic expansion – Temperature decrease
so decrease in internal energy

(B) Isobaric expansion – Volume increases and hence
temperature increases since $P \rightarrow \text{constant}$ ($V \propto T$)
So internal energy increases

(C) Isothermal expansion – $T \rightarrow \text{constant}$
So, constant internal energy

(D) Isochoric process – $V \rightarrow \text{constant}$
So, No work done
As $\omega = P\Delta V = 0$

27.



For (i) : $W = 0$ & $\Delta T < 0 \Rightarrow Q < 0$ & $\Delta U < 0$

For (ii) : $Q = 0$, $W > 0 \Rightarrow \Delta U < 0$

For (iii) : $\Delta T = 0$, $\Rightarrow \Delta U = 0 \Rightarrow Q = W > 0$

For (iv) : $V \propto T \Rightarrow W > 0$ & $\Delta U > 0 \Rightarrow Q > 0$

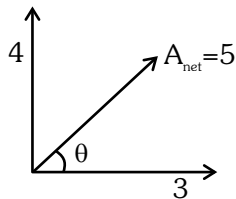
OSCILLATIONS (S.H.M.)

1. $x = 3 \sin 4\pi t + 4 \cos 4\pi t$

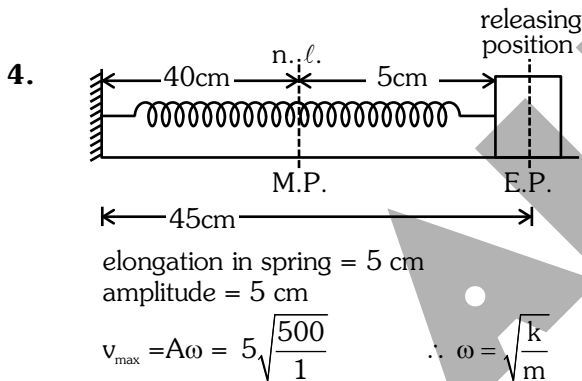
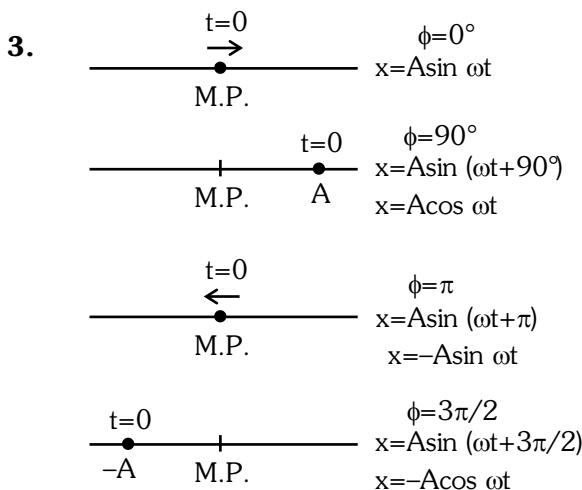
$\tan \theta = \frac{4}{3}$

$\theta = 53^\circ$

$x = 5 \sin (4\pi t + 53^\circ)$
(SHM)



2. Theory



5. (a) Lift is at rest $a = 0$
(b) lift is moving downward with constant speed
 $\Rightarrow$ constant velocity $\Rightarrow a = 0$
in both cases $a = 0$ so $g_{\text{eff}} = g$
 $T = 2\pi\sqrt{\frac{\ell}{g}}$
(c) In this case lift is moving with increasing speed so acceleration is downward.
 $g_{\text{eff}} = g - a.$

6. Energy $E = \frac{1}{2}KA^2 = \frac{1}{2}m\omega^2A^2$

$v_{\max} = A\omega = A\sqrt{\frac{K}{m}}$

$T = 2\pi\sqrt{\frac{m}{K}}$

$(P.E.)_{\max} = \frac{1}{2}KA^2$

(i) Mass is doubled

$\Rightarrow$ Time period become $\sqrt{2}$ times.

(ii) Amplitude $\Rightarrow$ double

Energy $\Rightarrow$ 4 times

$v_{\max} \Rightarrow$ 2 times

$(P.E.)_{\max} \Rightarrow$ 4 times

(iii) $K \Rightarrow$ double

Energy = double

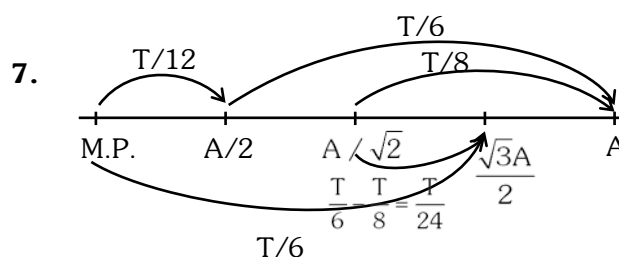
$v_{\max} = \sqrt{2}$ times

$T \Rightarrow \frac{1}{\sqrt{2}}$ times

$(P.E.)_{\max} = 2$ times

(iv) Angular frequency $\left(\omega = \sqrt{\frac{k}{m}}\right)$ is halved

means k becomes $1/4$ times so
maximum speed is halved.

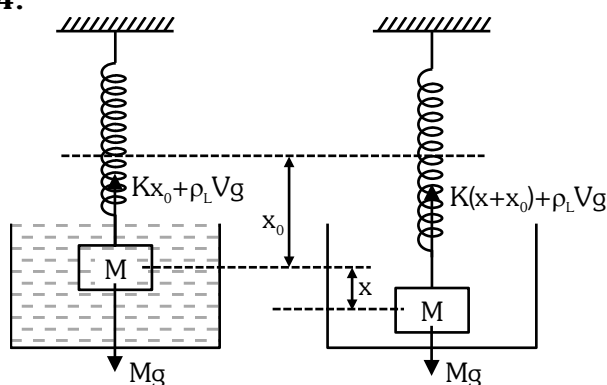


11. $a = -\omega^2x$

13. $A = A_0 e^{-\frac{bt}{2m}}$

$A = \frac{F_0}{m\sqrt{(\omega_d^2 - \omega^2)^2 + \left(\frac{b\omega_d}{m}\right)^2}}$

14.



- (1) $\rho_L Vg + Kx_0 = Mg$
- (2) $F_R = -[K(x + x_0) + \rho_L Vg - Mg]$
 $= -[Kx + Kx_0 + \rho_L Vg - Mg]$
 $= -Kx$
- (3) $a = \frac{F_R}{M} = -\frac{Kx}{M} \Rightarrow \omega = \sqrt{\frac{K}{M}} \Rightarrow T = 2\pi\sqrt{\frac{M}{K}}$

15. It is a periodic oscillatory and since released from small height it is SHM

$$16. x = 2 + 3 \frac{1 - \cos 2\omega t}{2} = 2 + \frac{3}{2} - \frac{3}{2} \cos 2\omega t$$

$$= \frac{7}{2} - \frac{3}{2} \cos 2\omega t \Rightarrow \left(x - \frac{7}{2}\right) = \frac{3}{2} \cos 2\omega t$$

$$17. \frac{xy}{z} = \frac{WA.f}{W^2 A} = \frac{f}{W} = \frac{f}{2af} = \frac{1}{2\pi}$$

a constant and dimension less.

18. At the mean position if $U_0 < 0$, then $K_{\max} > E_{\text{total}}$

$$19. T = 2\pi\sqrt{\frac{m}{k}}$$

$$22. \text{ Here } T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow \frac{\ell}{g} = \frac{1}{\pi^2} \text{ But}$$

$g = \pi^2$ therefore $\ell = 1 \text{ m}$

23. It is due to resonance i.e. bridge may collapse.

WAVE MOTION

1. Same length of organ pipe means wavelength in both the pipe are same but velocity of sound is different in both the pipes due to different gasses inside it.

$$3. v = \sqrt{\frac{\gamma RT}{M_0}} = \sqrt{\frac{\gamma \rho}{\rho}}$$

$$(M_0)_{\text{dry air}} > (M_0)_{\text{moist air}}$$

$$(\rho)_{\text{dry air}} > (\rho)_{\text{moist air}}$$

4. $y = 0.02 \sin 2\pi (10t - 5x)$
 compare the about equation with
 $y = A \sin (\omega t - kx)$

$$A = 0.02 \text{ m}$$

$$\omega = 20\pi \text{ rad/s}$$

$$k = 10\pi \text{ m}^{-1}$$

$$v = \frac{\omega}{k} = \frac{20\pi}{10\pi} = 2$$

$$\omega = 20\pi = 2\pi f$$

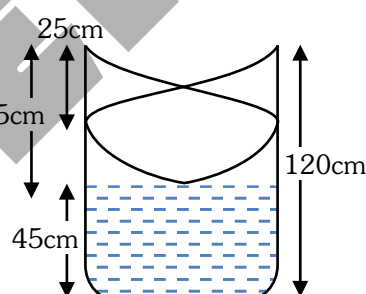
$$f = 10 \text{ Hz}$$

$$k = 10\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = 0.2 \text{ m}$$

$$v_{\max} = A\omega = 0.02 \times 20\pi = 0.4\pi \text{ m/s}$$

5. As $V = v\lambda$

$$\lambda = \frac{V}{v} = \frac{340}{340} = 1 \text{ m}$$



first Resonance length

$$R_1 = \frac{\lambda}{4} = \frac{1}{4} \text{ m} = 25 \text{ cm}$$

$$\therefore R_2 = \frac{3\lambda}{4} = \frac{3}{4} \text{ m} = 75 \text{ cm}$$

$$\therefore R_3 = \frac{5\lambda}{4} = \frac{5}{4} \text{ m} = 125 \text{ cm}$$

i.e. third resonance does not establish

$\therefore$ minimum length of H_2O column to have the resonance = 45 cm

$$\therefore \text{Distance between two successive nodes} = \frac{\lambda}{2}$$

$$= \frac{1}{2} \text{ m} = 50 \text{ cm}$$

and maximum length of H_2O column to create resonance i.e. $120 - 25 = 95 \text{ cm}$

OR

$$h = 120 \text{ cm}$$

$$f = 340 \text{ Hz}$$

$$v = 340 \text{ m/s}$$

$$\lambda = \frac{v}{f} = \frac{340}{340} = 100 \text{ cm}$$

$$(1) \frac{\lambda}{4} = \frac{100}{4} = 25 \text{ cm (air)} \Rightarrow \text{water level} = 95 \text{ cm}$$

$$(2) \frac{3\lambda}{4} = 75 \text{ cm (air)} \Rightarrow \text{water level} = 45 \text{ cm}$$

$$(3) \frac{5\lambda}{4} = 125 \text{ cm (not possible)}$$

So maximum water level = 95 cm

Minimum water level = 45 cm

Distance between two successive nodes

$$= \frac{\lambda}{2} = \frac{100}{2} = 50 \text{ cm}$$

$$6. \text{ From } \sqrt{\frac{\gamma RT}{M\omega}} = \sqrt{\frac{\gamma P}{\rho}}$$

(i) Temperature 2 times $\Rightarrow$ speed double

(ii) Temperature constant there

$$P \uparrow \quad \rho \uparrow \quad v_{\text{sound}} = \text{const}$$

(iv) $M\omega$ is 4 times $\Rightarrow$ speed half.

7. For closed organ pipe fundamental frequency

$$f = \frac{v}{4\ell}$$

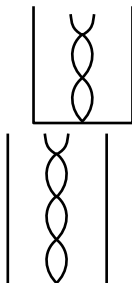
Third overtone $f = \frac{7v}{4\ell}$ (7 time of fundamental frequency)

$$\text{Second overtone } f = \frac{5v}{4\ell}$$

No. of nodes = 3

$$\text{Third overtone } f = \frac{7v}{4\ell}$$

No. of A.N. = 4



8. Theory

$$9. \quad v = \sqrt{\frac{\gamma RT}{M\omega}} = \sqrt{\frac{\gamma P}{\rho}}$$

$$(M_{\omega})_{\text{dry air}} > (M_{\omega})_{\text{moist air}}$$

$$(\rho)_{\text{dry air}} > (\rho)_{\text{moist air}}$$

$$10. \text{ First resonating length } \ell_1 = \frac{\lambda}{4}$$

$$\text{Second resonating length } \ell_2 = \frac{3\lambda}{4}$$

11. Thoery

13. Wavelength depends on length which is fixed. Therefore wavelength does not change.

$$\text{Further } v = \sqrt{T/m} \quad \text{or } v \propto T^{1/2}$$

$$\therefore \% \text{ change in } v = \frac{1}{2} \quad (\% \text{ change in } T)$$

$$= \frac{1}{2}(2) = 1\%$$

i.e., Speed and hence frequency will change by 1%.

Change in frequency is 15 Hz which is 1% of 1500 Hz.

Therefore, original frequency should be 1500 Hz.

OR

$$f = \frac{v}{2\ell} = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$\frac{\Delta f}{f} = \frac{1}{2} \frac{\Delta T}{T}$$

$$\frac{15}{f} = \frac{1}{2} \frac{2}{100}$$

$$f = 1500 \text{ Hz (original frequency)}$$

$$v = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta T}{T}$$

$$\frac{\Delta v}{v} \times 100 = \frac{1}{2} \times 2\% = 1\%$$

$\therefore$ Length of string fixed that mens wavelength the remain same.

14. For closed pipe,

$$f = n \left(\frac{v}{4\ell} \right) m = 1, 3, 5, \dots$$

$$\text{For } n = 1, f_1 = \left(\frac{v}{4\ell} \right) n = 1, 3, 5, \dots$$

$$\text{For } n = 1, f_1 = \frac{v}{4\ell} = \frac{320}{4 \times 1} = 80 \text{ Hz}$$

$$\text{For } n = 3, f_3 = 3f_1 = 240 \text{ Hz}$$

$$\text{For } n = 5, f_5 = 5f_1 = 400 \text{ Hz}$$

OR

$$L = 1\text{m} = 100\text{ cm}$$

$$v = 320\text{ m/s}$$

$$\text{fundament freq. } f = \frac{v}{4\ell} = \frac{320}{4 \times 1} = 80\text{ Hz}$$

$$3^{\text{rd}} \text{ harmonic} = 3 \times 80 = 240\text{ Hz}$$

$$5^{\text{th}} \text{ harmonic} = 5 \times 80 = 400\text{ Hz}$$

$$\begin{aligned} 15. \quad y &= 8 \sin^2 \pi \left[\frac{t}{0.02} - \frac{x}{100} \right] \\ &= 4 \left[1 - \cos 2\pi \left(\frac{t}{0.02} - \frac{x}{100} \right) \right] \\ \Rightarrow y - 4 &= -4 \cos \left(\frac{2\pi t}{0.02} - \frac{2\pi x}{100} \right) \end{aligned}$$

For (A) : Amplitude = 4 cm

For (B) :

$$\text{Frequency } f = \frac{\omega}{2\pi} = \left(\frac{2\pi}{0.02} \right) \frac{1}{2\pi} = 50\text{ Hz}$$

$$\begin{aligned} \text{For (C) : Velocity of wave} &= \frac{\omega}{k} = 5000\text{ cm/s} \\ &= 50\text{ ms}^{-1} \end{aligned}$$

For (D) : Maximum particle velocity

$$\begin{aligned} &= A\omega = (4 \times 10^{-2}) \left(\frac{2\pi}{0.02} \right) = 4\pi\text{ ms}^{-1} \\ &= 400\pi\text{ cm-s}^{-1} \end{aligned}$$

16. The wave speed depends on properties of the medium, not on how you generate the wave. For a string, $v = \sqrt{T_s / \mu}$. Increasing the tension or decreasing the linear density (lighter string) will increase the wave speed.

$$17. \quad v_1 = \frac{\omega}{k} = \frac{4}{2} = 2\text{ m/s},$$

$$v_2 = \frac{10}{4} = 2.5\text{ m/s},$$

$$v_3 = \frac{12}{6} = 2\text{ m/s},$$

$$v_4 = \frac{16}{8} = 2\text{ m/s},$$

$$v_5 = \frac{20}{10} = 2\text{ m/s}$$

as $v = \sqrt{\frac{T}{\mu}}$ so T_2 is different.

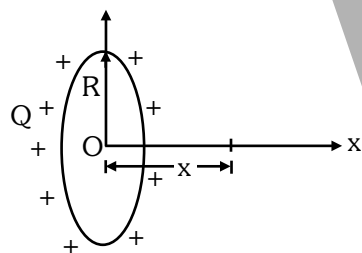
18. At node $\cos(10\pi x) = 0$ &

at antinode $\cos(10\pi x) = 1$

$$\& \omega = 50\pi, k = 10\pi \& v = \frac{\omega}{k}, k = \frac{2\pi}{\lambda}$$

ELECTROSTATICS

1.



$$E = \frac{KQx}{(R^2 + x^2)^{3/2}}, \text{ for } (x, 0, 0)$$

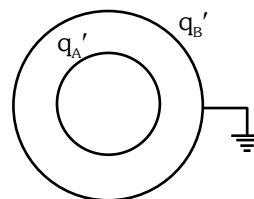
$$E = 0 \& V = \frac{KQ}{R} \text{ at centre/origin.}$$

2. $|q| \propto \text{No. of ELF}$

Neutral point of system of two unlike charges is outside the line joining the two charges near to & near the charge of smaller magnitude.

3. $\therefore$ Both are unlike so neutral point exist outside & near the smaller charge.

4.



$$2V = \frac{kq_A'}{R} + \frac{kq_B'}{2R}$$

$$\frac{3V}{2} = \frac{k(q_A' + q_B')}{2R}$$

on solving

$$\frac{q_A'}{q_B'} = \frac{1}{2}$$

$$V_B = 0$$

$$\frac{kq_A'}{b} + \frac{kq_B'}{b} = 0$$

$$q_B' = -q_A'$$

$$\frac{|q_A'|}{|q_B'|} = 1$$

5. Charge particle is in stable equilibrium if it is displaced along y-axis.
6. Think according to options.

7. For infinite non-conducting sheet ; $E = \frac{\sigma}{2\epsilon_0}$

For infinite conducting sheet ; $E = \frac{\sigma}{\epsilon_0}$

For non conducting sphere (at surface) ; $E = \frac{\rho R}{3\epsilon_0}$

For non conducting sphere at centre ; $E = 0$

8. Charge can't exist without mass, but mass can exist without charge. Charge is conserved but mass is not conserved.
9. $[F] = [MLT^{-2}]$

$\therefore F = \frac{E}{q}$

$\therefore E = \frac{MLT^{-2}}{AT} \Rightarrow E = [MLT^{-3}A^{-1}]$

ϵ_r is dimensionless

$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \Rightarrow \epsilon_0 = \frac{[A^2 T^2]}{[MLT^{-2}][L^2]}$
 $= [M^{-1}L^{-3}A^2T^4]$

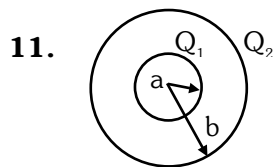
10. $V_{R/2} = \frac{kQ}{2R^3} \left[3R^2 - \left(\frac{R}{2} \right)^2 \right] = \frac{kQ}{2R^3} \left(\frac{11R^2}{4} \right)$

$V_{R/2} = \frac{11kQ}{8R} = \frac{11V}{8}$; $V_s = \frac{kQ}{R}$

$V_{2R} = \frac{kQ}{2R} = \frac{V}{2}$

$E_{R/2} = \frac{kQ}{R^3} \cdot \frac{R}{2} = \frac{kQ}{2R^2} = \frac{V}{2}$ ($R = 1m$)

$E_{2R} = \frac{kQ}{(2R)^2} = \frac{V}{4}$

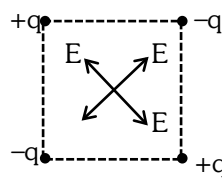


$\Delta V = KQ_1 \left(\frac{1}{a} - \frac{1}{b} \right)$ does not depends on outer charge.

$E_{in} = \frac{KQ_1}{a^2}$ does not depends on outer charge.

12. Direction of force depends on nature of charge.
13. Potential depends on frame of reference while potential difference is a absolute.

14.

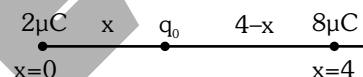


$E = 0, V = \frac{Kq}{x} + \frac{Kq}{x} - \frac{Kq}{x} - \frac{Kq}{x} = 0$

It is not necessary that if potential is zero then electric field is also zero.

15. Extra charge on aircraft body due to air friction is transferred to ground.
16. Electric field can not enter in conductor.
17. Charge of isolated body can not be changed.
18. Net flux is due to inside charges but electric field is due to all charges.
19. There is loss / gain of electrons so mass may decrease or increase.

21.



22.

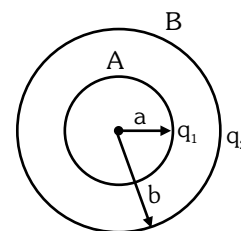
$\frac{k2q_0}{x^2} = \frac{k8q_0}{(4-x)^2}$; $\frac{1}{x} = \frac{2}{4-x}$

$4-x = 2x$; $3x=4$

$x = \frac{4}{3} = \frac{\alpha}{\beta} \Rightarrow \alpha + \beta = 4 + 3 = 7$

23. Refer page 33 ; article 1.14 (NCERT Class XII Part 1)

24.



$V_A = \frac{q_2}{b} + \frac{q_1}{a}$

$V_B = \frac{q_1}{b} + \frac{q_2}{b}$

$V_A - V_B = q_1 \left[\frac{1}{a} - \frac{1}{b} \right]$

25. $V = \frac{kQ}{r}$ & $E = \frac{kQ}{r^2}$

27. Centre of mass is taken but there is no concept of centre of charge.

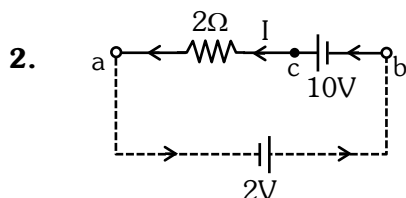
CURRENT ELECTRICITY

1. $\therefore V_1$ and V_2 are in series

$$\therefore V \propto R$$

$$\therefore V_1 \neq V_2 \text{ (as } R_1 \neq R_2 \text{)}$$

$$V_1 + V_2 = V_3 \text{ (as } V \text{ is same in parallel)}$$



$$I = \frac{12}{2} = 6A$$

Apply KVL from a to c

$$V_a + 6 \times 2 = V_c$$

$$V_a - V_c = -12 \text{ Volt}$$

3. $E = 10V, I_{\max} = \frac{E}{r} = 2 \text{ Amp.}$

$$r = \frac{E}{2} \Rightarrow r = \frac{10}{2} = 5\Omega$$

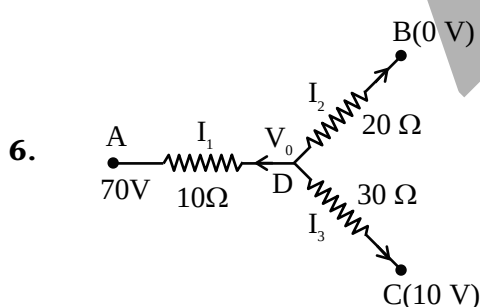
4. Sensitivity $\propto \frac{1}{P.G.} \propto \frac{L_{pm}}{V_{pm}}$

5. $\frac{V^2}{R} t = \text{constant}$

$$\therefore t \propto R$$

In series $t_s = t_1 + t_2$

In parallel $t_p = \frac{t_1 t_2}{t_1 + t_2}$



$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_0 - 70}{10} + \frac{V_0 - 0}{20} + \frac{V_0 - 10}{30} = 0$$

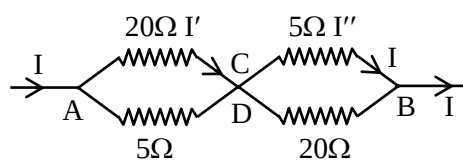
$$\Rightarrow V_0 = 40 \text{ volt}$$

$$I_1 = 3A$$

$$I_2 = 2A$$

$$I_3 = 1A$$

- 7.



$$I' = \frac{5}{25} I = \frac{I}{5}$$

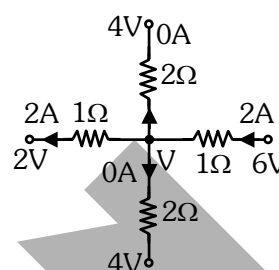
$$R_{eq} = 4 + 4 = 8\Omega$$

$$I'' = \frac{20}{25} I = \frac{4I}{5}$$

$$\text{Also } = V_c = V_d$$

$$\text{Current in DC} = \frac{4I}{5} - \frac{I}{5} = 3I/5$$

- 8.



$$\frac{V-2}{1} + \frac{V-6}{1} + \frac{V-4}{2} + \frac{V-4}{2} = 0$$

$$2V - 4 + 2V - 12 + V - 4 + V - 4 = 0$$

$$6V = 24$$

$$V = 4$$

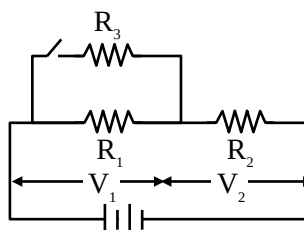
$$2A, -2A, 0A, 0A$$

- 9.

$$i = \text{constant}$$

$$\text{as area } \uparrow \Rightarrow R/\ell \downarrow \text{ \& } E \downarrow \Rightarrow \frac{V}{L} \downarrow \text{ \& } J = \frac{i}{A} \downarrow$$

- 10.



After closing the switch $V_1 \downarrow$ and $V_2 \uparrow$

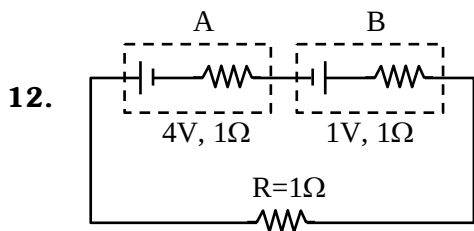
$\Rightarrow i$ through $R_1 \downarrow$ and i through $R_2 \uparrow$.

11. $V = \frac{\text{Work}}{q} = \frac{[ML^2T^{-2}]}{[AT]} = [ML^2T^{-3}A^{-1}]$

$$R = \frac{V}{i} = [ML^2T^{-3}A^{-2}]$$

$$\rho = \frac{RA}{\ell} = [ML^3T^{-3}A^{-2}]$$

$$\sigma = \frac{1}{\rho} = [M^{-1}L^{-3}T^3A^2]$$



$$i = \frac{4-1}{1+1+1} = 1\text{A}$$

Now TPD of A = $4 - 1 \times 1 = 3\text{V}$

TPD of B = $1 + 1 \times 1 = 2\text{V}$

Power consumed by R = $1^2 \times 1 = 1\text{W}$

P.D. of R = $1 \times 1 = 1\text{V}$

13. Increase in temperature causes atoms to vibrate more, which increases the number of collisions electrons suffer and decrease the drift velocity.

OR

$$\vec{v}_d = \mu E = \left(\frac{e\tau}{m} \right) \vec{E} \quad \& \quad \sigma = \frac{ne^2\tau}{m}$$

$v_d \propto \tau$, $\sigma \propto \tau$ (temp. $\uparrow$; $\tau \downarrow$)

14. $S(i - i_0) = i_0 G$

$$\Rightarrow i = i_0 \left(1 + \frac{G}{S} \right)$$

$\therefore$ As $S \uparrow$ $i \downarrow$

$\therefore$ Value of shunt is more in mili ammeter.

$$R_{\text{net}} = \frac{GS}{S+G} = \frac{G}{1+\frac{G}{S}}$$

$\therefore S \uparrow \Rightarrow R_{\text{net}} \uparrow$

$\Rightarrow R_{\text{mili ammeter}} > R_{\text{ammeter}}$

15. A current carrying wire is not charged.
16. In balanced condition, current through galvanometer is zero and does not affect main current.
17. For rating, $P = \frac{V^2}{R}$ $\therefore P \uparrow \Rightarrow R \downarrow$
18. * For maximum power $R = r$
* $P = I^2 R \propto I^2$ (if $R = \text{constant}$)
19. At higher currents, joule's heating will be larger, increasing the temperature sharply. Since, $R = R_0(1 + \alpha \Delta T)$, resistance varies and $V \propto I$ doesn't hold anymore.

20. For meter bridge experiment, unknown resistance $S = \frac{100 - \ell}{\ell} R$

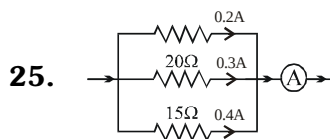
21. $\Delta KE = q(\Delta V)$

22. Unknown emf should not be more than voltage across potentiometer wire.

23. As i changes (in same resistor); n remain constant

by $i = JA$; J become double

by $i = nAeV_d$; V_d become double



$$0.2R_1 = 20 \times 0.3 = 15 \times 0.4$$

27. $\ell_{\text{min}} = \left(\frac{100}{16} \times 10 \right)$ when $R_{\text{rheostat}} = 0$

30. For (A) :

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k} = \frac{1}{x^2} \hat{i} + \frac{1}{y^2} \hat{j} + \frac{1}{z^2} \hat{k}$$

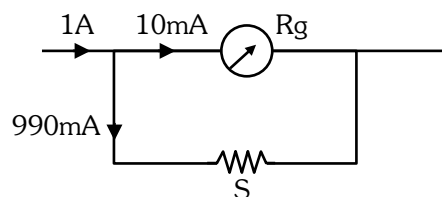
$$\Rightarrow |\vec{E}| = |\hat{i} + \hat{j} + \hat{k}| = \sqrt{3}$$

For (B) :

$$I = \frac{dQ}{dt} = 4 - 4t, \text{ current will be zero at } t=1\text{s}$$

Total heat produced

$$\int_0^1 I^2 R dt = \int_0^1 (4 - 4t)^2 (6) dt = (16 \times 6) \left[\frac{(1-t)^2}{-3} \right]_0^1 = 32$$



For(C) $(10 \text{ mA}) (R_g) = (990 \text{ mA}) (s)$

$$\Rightarrow s = \frac{0.99}{99} = 0.01\Omega$$

For(D) $U = -\vec{p} \cdot \vec{E}$ Here $\vec{p} = \pm qd\hat{i}$

$$= \pm (10 \times 10^{-3}) (8 \times 10^{-2}) = \pm (8 \times 10^{-4}) \hat{i}$$

CAPACITANCE

1. $\therefore$ battery is connected

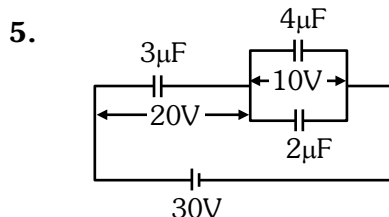
$$\therefore V = \text{constant}$$

$$U = \frac{1}{2} CV^2, Q = CV$$

$$U \propto C, Q \propto C$$

4. $U = \frac{1}{2} \epsilon_0 E^2 (Ad)$

$$U \propto E^2$$



$$\therefore V \propto \frac{1}{C}$$

$$Q_{\text{Battery}} = C_{\text{eff}} V$$

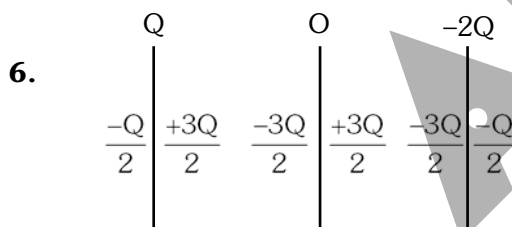
$$= 2 \times 20 \mu C$$

$$Q_{\text{battery}} = 60 \mu C$$

$$Q_{3\mu F} = 60 \mu C$$

$$Q_{4\mu F} = 40 \mu C$$

$$Q_{2\mu F} = 20 \mu C$$



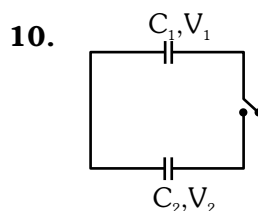
7. $U_{\text{self}} = \frac{KQ^2}{2R}$ Energy density = $\frac{1}{2} \epsilon_0 E^2$

$$F = \frac{Q^2}{2A\epsilon_0} \quad W_{\text{field}} = U_i - U_f = \frac{kQ^2}{R} - 0$$

$$W_{\text{field}} = \frac{kQ^2}{R}$$

8. Force between two point charges is independent of the presence of other charges.

9. $C = 4\pi\epsilon_0 R \Rightarrow 1 = \frac{1}{9 \times 10^9} R \Rightarrow R = 9 \times 10^9 \text{ m}$



$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

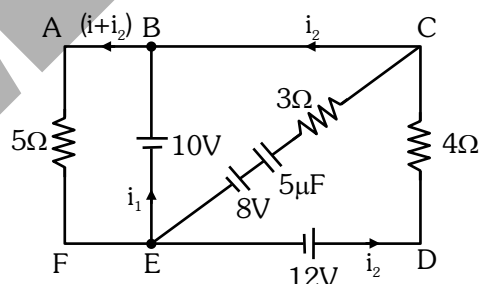
When two capacitors are connected then energy of system decreases. (but Not always)

11. A high capacity capacitor charged to high voltage has very high charges, that may discharge to ground through human body, when touched with bare hands.

12. $U = \frac{Q^2}{2C}$ and $C = \frac{\epsilon_0 A}{d}$ $d \uparrow \quad C \downarrow \quad U \uparrow$

13. At facing surface same magnitude and opposite nature charges are there.

- 14.



When a steady state is reached, no current passes through the capacitor or the branch CE.

Considering the loop ABEFA,
 $5 \times (i_1 + i_2) = 10$ or $i_1 + i_2 = 2A \dots (i)$

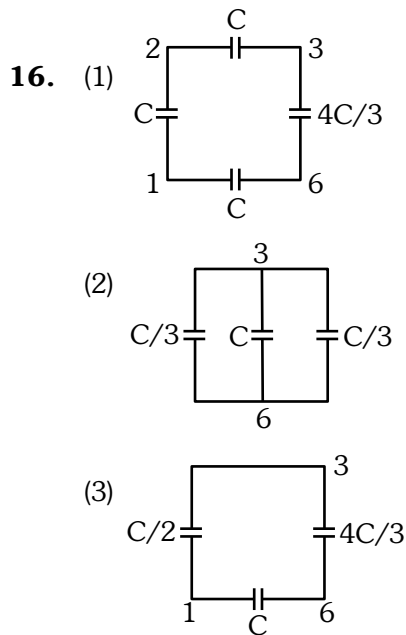
Considering the loop BCDEB
 $4i_2 = 12 - 10 = 2 \Rightarrow i_2 = 0.5 A$
So $i_1 = 2 - 0.5 = 1.5 A$

To find the charge on capacitors, we must know potential difference across the plates.

Consider the loop CEDC :

$$-12 + 4i_2 + 3 \times 0 - V_C + 8 = 0 \Rightarrow V_C = -2V.$$

So charge on capacitor $Q = CV = 10 \mu C$

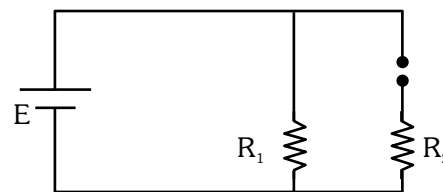


18. In steady state current through battery

$$I = \frac{50}{23 + 3 + 17 + 7} = 1A$$

$$\Rightarrow Q_{10\mu F} = CV = 10 \times 20 = 200\mu C$$

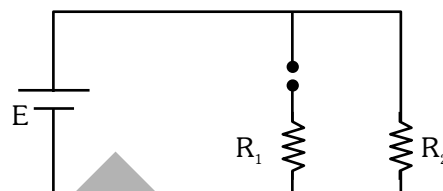
19.



At $t = 0$

$$I_1 = \frac{E}{R_1} = \frac{25}{5} = 5A$$

$$I_2 = 0$$



At $t = \infty$

$$I_1 = 0$$

$$I_2 = \frac{E}{R_2} = \frac{25}{10} = 2.5 A$$

MAGNETIC EFFECT OF CURRENT AND MAGNETISM

1. Magnetic field due to coil of radius 5 cm

$$B_1 = \frac{N\mu_0 I}{2R} = \frac{50 \times 4\pi \times 10^{-7} \times 2}{2 \times (5 \times 10^{-2})}$$

$$= 4\pi \times 10^{-4} T$$

Magnetic field due to coil of radius 10 cm

$$B_2 = \frac{N\mu_0 I}{2R} = \frac{100 \times 4\pi \times 10^{-7} \times 2}{2 \times (10 \times 10^{-2})}$$

$$= 4\pi \times 10^{-4} T$$

$\therefore$ Current in same sense ; $B_{Net} = 8\pi \times 10^{-4} T$

$\therefore$ Current in opposite sense ; $B_{Net} = 0$

2. Angle between $\vec{M}$ and $\vec{B}$ is 180° .
3. Current carrying wire is electrically neutral.
4. K.E. = $qV \Rightarrow$ If V doubles, K.E. doubles

$$R = \frac{1}{B} \sqrt{\frac{2mV}{q}} \Rightarrow \text{If } V \text{ doubles,}$$

R becomes $\sqrt{2}$ times

$$\text{Time period } T = \frac{2\pi m}{qB} \Rightarrow \text{Independent of speed}$$

$\Rightarrow$ Angular velocity will not change.

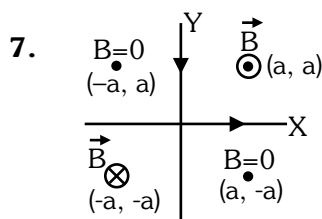
5. $\vec{B}$ at a is inwards

$\vec{B}$ at f is zero (point on axis of wire)

$\vec{B}$ at c is outwards

$\vec{B}$ at g is outwards

6. (a) $\theta = 180^\circ$ ($Q \rightarrow$ angle between $\vec{M}$ & $\vec{B}$)
- (b) $\theta = 90^\circ$
- (c) $\theta = 0^\circ$
- (d) θ is obtuse



8. Wires carrying current in same direction attract each other.
Wires carrying current in opposite direction repel each other.

9. Use ACL and solve
for loop (1) $\oint \vec{B} \cdot d\vec{\ell} = \mu_0(-i)$
for loop (2) $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$
for loop (3) $\oint \vec{B} \cdot d\vec{\ell} = 0$
for loop (4) $\oint \vec{B} \cdot d\vec{\ell} = 0$
10. $F_M = Bi\ell \sin \theta$
 $F_{AB} = Bia$ $F_{CH} = Bia$
 $F_{BC} = 0$ $F_{CE} = Bi\sqrt{2}a$
11. $m_{\text{eff.}} = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$
12. Net force on a current carrying loop in a non uniform magnetic field may or may not be zero.
e.g. when a circular current loop is placed in a circular MF at a line of force then $F = 0$.
14. Magnetic field lines always form close curve.
15. There is moving charge in a current carrying element.
16. Current carrying wires (current = constant) produces magnetic field but not electric field.
Charge particle beam produces both electric and magnetic field.

17. Ampere's law is valid for infinite wire of symmetrical current distribution.
18. At geomagnetic poles magnetic field of earth is vertical.

OR

At poles, Dip angle is 90°

In vertical plane perpendicular to magnetic meridian apparent dip 90° .

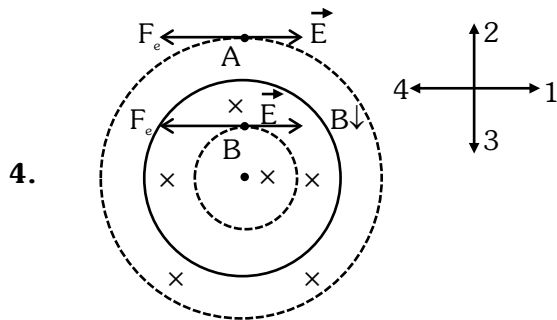
19. At curie point ferro start behaving as para.
20. Solenoid is used to produce uniform magnetic field.
22. (A) When $V=0$ or $\vec{V} \parallel \vec{E} \rightarrow$ path will be straight
When $\vec{V} \neq \vec{E} \rightarrow$ path will be parabolic
(B) When $\vec{V} \parallel \vec{B} \rightarrow$ path will be straight
When $\vec{V} \perp \vec{B} \rightarrow$ path will be circular

ELECTROMAGNETIC INDUCTION (EMI)

1. $e = (vB \ell_{\text{eff}})$
 $\ell_{\text{eff}} = L/2$
 $\Rightarrow e = \frac{1}{2} BLv$ [P is at higher potential]
2. (a) $\phi = LI = \text{Henry} - \text{ampere}$
(b) Energy = $\frac{1}{2} LI^2$
 $L \rightarrow \left(\frac{J}{A^2} \right)$
(c) $\text{emf} = -L \frac{dI}{dt}$
 $L \rightarrow \frac{\text{volt-second}}{\text{ampere}}$
(d) S.I. unit of magnetic induction is Tesla.

$$3. \left. \begin{aligned} q &= 2t^2 \\ I &= \frac{dq}{dt} = 4t \\ \frac{dI}{dt} &= 4 \end{aligned} \right\} \begin{aligned} \text{At } t=1 \text{ sec} \\ q &= 2C \\ I &= 4A \\ \frac{dI}{dt} &= 4A/s \end{aligned}$$

- (i) $V_a - L \frac{dI}{dt} = V_b \Rightarrow V_a - V_b = L \frac{dI}{dt} = 4V$
(ii) $V_b - \frac{q}{C} = V_c \Rightarrow V_b - V_c = \frac{q}{C} = 1V$
(iii) $V_c - IR = V_d \Rightarrow V_c - V_d = IR = 16V$
(iv) $V_a - L \frac{dI}{dt} - \frac{q}{C} - 4I = V_d$
 $\Rightarrow V_a - V_d = 21V$



5. $B = \frac{F}{i\ell} = \frac{MLT^{-2}}{AL}$

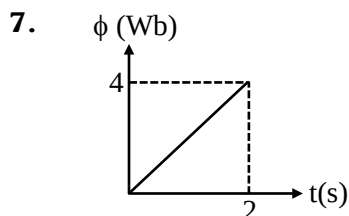
$$L = \frac{\text{Energy}}{i^2} = [ML^2T^{-2}A^{-2}]$$

$$T = 2\pi\sqrt{LC} \Rightarrow LC = [T^2]$$

$$\phi = LI = [M^1L^2T^{-2}A^{-1}]$$

6. At $t = 0$ $V_L = V_0 = 10V$

$$\begin{aligned} \text{At } t = 1 \quad & \begin{cases} V_R = V_0 = 0V \\ V_L = V_0 e^{-t/\tau} \\ V_L = 10 e^{-1} \end{cases} \quad \begin{cases} V_R = V_0 [1 - e^{-t/\tau}] \\ V_R = 10 \left[1 - \frac{1}{e}\right] \end{cases} \\ & V_L = \frac{10}{e} \end{aligned}$$



$$R = 2\Omega$$

$$\Delta\phi = 4$$

$$e_{in} = \frac{\Delta\phi}{\Delta t} = \frac{4}{2} = 2 \text{ volt}$$

$$i_{in} = \frac{e_{in}}{R} = \frac{2}{2} = 1 \text{ amp}$$

$$\Delta q = \frac{\Delta\phi}{R} = \frac{4}{2} = 2C$$

$$H = I^2 R t$$

$$H = (1)^2 \times 2 \times 2 = 4 \text{ J}$$

$$a \rightarrow s, b \rightarrow q, c \rightarrow s, d \rightarrow p$$

8. D.C. voltage is steady with time.

9. According to Lenz's law induced current always opposes the change in flux.

10. Inductor tends to keep flux constant.

11. $M_{1,2} = M_{2,1}$

Independent of current.

12. $\oint \vec{E} \cdot d\vec{\ell} = -A \frac{dB}{dt}$

13. $\phi = \vec{B} \cdot \vec{A}$

14. When flux is zero, at that instant rate of flux change is maximum

ALTERNATING CURRENT

1. At frequency lower than resonant frequency since $X_C > X_L$

$\therefore$ Current leads voltage

$\rightarrow$ At resonance, impedance is minimum.

$\rightarrow$ At resonance, phase diff. between voltage

across L and current in C is $\frac{\pi}{2}$

$$\rightarrow f = \frac{1}{2\pi\sqrt{LC}}$$

2. $\tau = RC \Rightarrow R = \frac{\tau}{C}$

$$[R] = \left[\frac{1}{fC} \right]$$

$$\tau = \frac{L}{R} \Rightarrow R = \frac{L}{\tau} = (fL)$$

$\therefore$ b & c are correct.

$$3. \quad V_1 = \sqrt{V_R^2 + V_L^2} = 100 \quad \dots(1)$$

$$V_C - V_L = 120 \quad \dots(2)$$

$$130 = \sqrt{V_R^2 + (V_L - V_C)^2} \Rightarrow V_R = 50V$$

$$V_L = 50\sqrt{3}V = 86.6V \text{ \& } V_C = 206.6V$$

$$\cos\phi = \frac{V_R}{130} = \frac{5}{13}$$

$$4. \quad \begin{array}{c} i_0 \\ 0 \end{array} \quad \begin{array}{c} i_{AV} = 0, i_{rms} = \frac{i_0}{\sqrt{2}} \\ t \end{array}$$

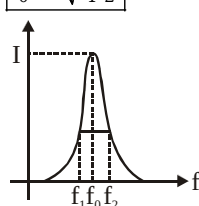
$$\begin{array}{c} i_0 \\ 0 \end{array} \quad \begin{array}{c} i_{av} = 0, i_{rms} = \frac{i_0}{\sqrt{2}} \\ t \end{array}$$

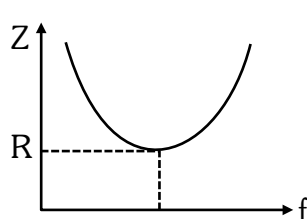
$$\begin{array}{c} i_0 \\ 0 \end{array} \quad \begin{array}{c} i_{AV} = i_{rms} = i_0 \\ t \end{array}$$

$$\begin{array}{c} i_0 \\ -i_0 \end{array} \quad \begin{array}{c} i_{AV} = 0, i_{rms} = i_0 \\ t \end{array}$$

$$5. \quad Q_{\text{factor}} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$6. \quad \begin{array}{c} V_L \\ V_R \\ V_C \end{array} \quad \begin{array}{c} i \\ V_R \end{array}$$

$$7. \quad \boxed{f_0 = \sqrt{f_1 f_2}}$$


$$8. \quad z = \sqrt{R^2 + (x_L - x_C)^2}$$


Impedance can be greater than or equal to resistance, but it can't be less than resistance of circuit.

$$9. \quad \begin{array}{c} V_L \\ V_C \end{array} \quad \begin{array}{c} V_R \\ I \end{array}$$

At resonance

$$V_L = V_C \text{ \& } Z = R$$

there V_L & V_C are out of phase

$$10. \quad \text{For capacitor and inductor}$$

$$\phi = 90^\circ \Rightarrow \cos\phi = 0 \Rightarrow P_{av} = 0$$

$$11. \quad \text{Some part of energy is lost in form of heat and radiation.}$$

$$12. \quad \text{For (A) : Current and voltage are in phase}$$

$$\Rightarrow \text{only R or RLC in resonance.}$$

$$\text{For (B) : Current leads voltage by } \frac{\pi}{2} \Rightarrow \text{only C}$$

$$\Rightarrow \text{power factor} = 0$$

$$\text{For (C) : Current leads voltage} \Rightarrow \text{RC}$$

$$\text{or RLC with } X_C > X_L$$

$$\text{For (D) : Current leads voltage}$$

$$\Rightarrow \text{RL or RLC with } X_L > X_C$$

ELECTROMAGNETIC WAVES (EMW)

$$2. \quad \vec{E} = 30 \sin(6 \times 10^8 t + kz) \hat{i}$$

$$E = E_0 \sin(\omega t + kz)$$

Direction of propagation = -z axis

$$\text{wave length } \lambda = \pi = 3.14 \text{ m}$$

Direction of M. field direction of $\vec{v}$ = direction of $\vec{E} \times \vec{B}$

$$-\hat{k} = \hat{i} \times \hat{B}$$

$$\hat{B} = -\hat{j}$$

$$a \rightarrow s, b \rightarrow r, c \rightarrow p, d \rightarrow q$$

$$3. \quad \text{Energy of x ray is higher than radio waves.}$$

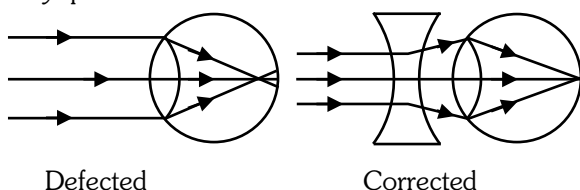
TG: @Chalnaayaaar

RAY OPTICS

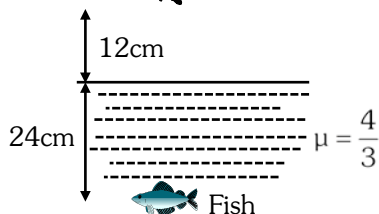
2. Simple microscope forms erect image while compound microscope form inverted image.

3. $M.P \propto \frac{1}{f}$ (for normal adjustment $M.P = \frac{D}{f}$)

4. Myopia



5. Bird



(A) $(d_{\text{to surface}}) = \frac{24}{4/3} \times 1 = 18 \text{ cm}$

(B) $(d_{\text{to surface}}) = \frac{12}{1} \times \frac{4}{3} = 16 \text{ cm}$

(C) $d_{\text{to}} = 12 + \frac{24}{4/3} \times 1 = 12 + 18 = 30 \text{ cm}$

(D) $d_{\text{to}} = 24 + \frac{12}{1} \times \frac{4}{3} = 24 + 16 = 40 \text{ cm}$

7. When mirror moves parallel to its plane, then image will not move.

8. $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ Both are same for thin lens.

9. On covering upper half of the lens its aperture decreases which results in decreasing of intensity but nature, position, shape and size of image remain same.

10. $M = \frac{D}{f}$ and D may different for different persons.

11. $v = \frac{uf}{u-f}$

(A) $u = -f, f = +f$

(B) $u = -f, f = -f$

(C) $u = -2f, f = -f$

(D) $u = -2f, f = +f$

$v = \frac{-f \times f}{-f - f} = \frac{+f}{2}$

$v = \frac{-f \times -f}{-f - (-f)} = \infty$

$v = \frac{-2f \times -f}{-2f - (-f)} = -2f$

$v = \frac{-2f \times f}{-2f - f} = \frac{+2}{3}f$

12. For $A > 2\theta_c$; the light ray never crosses the prism for any angle of incidence.

16. $e = (A + \delta) - i = (30 + 30) - 60 = 0$

WAVE OPTICS

2. $\beta = \frac{\lambda D}{d}, \quad \alpha = \frac{\lambda}{d}$

$(y_{n^{\text{th}}})_{\text{Bright}} = \frac{n\lambda D}{d}; \quad (y_{n^{\text{th}}})_{\text{Dark}} = \frac{(2n-1)\lambda D}{2d}$

4. $I \propto w$ and $I \propto A^2$; So intensity of source are I & 4I

$I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2 = (\sqrt{I} + 2\sqrt{I})^2 = 9I$

$I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2 = (\sqrt{I} - 2\sqrt{I})^2 = I$

$I_{\text{max}}/I_{\text{min}} = 9 : 1$

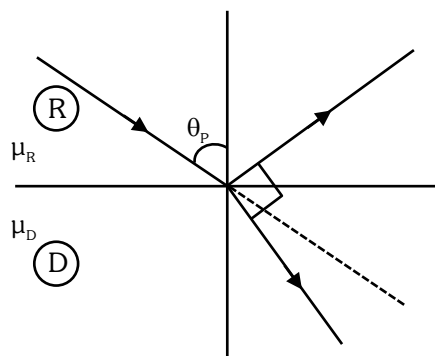
7. For higher order maxima in diffraction, the light waves from major part of aperture meet destructively and light from small part remain present. This remaining effective part of aperture decreases with increase in order of maximum.

10. $Z_f = \frac{a^2}{\lambda}$

11. For first maxima; $a \sin \theta = (2n+1) \frac{\lambda}{2}$

$\Delta x = \frac{3}{2}\lambda \quad \& \quad \phi = 3\pi \quad \& \quad [n=1]$

12.



$$\theta_p = \tan^{-1} \left(\frac{\mu_D}{\mu_R} \right)$$

$$i_c = \sin^{-1} \left(\frac{\mu_R}{\mu_D} \right)$$

$$\tan \theta_p = \frac{\mu_D}{\mu_R} = \frac{1}{\sin i_c}$$

$$\theta_p = \tan^{-1}(\operatorname{cosec} i_c)$$

$$i_c = \sin^{-1}(\cot \theta_p)$$

for D → R

$$\mu_D \sin \theta_p = \mu_R \sin r$$

$$\mu_D \sin \theta_p = \mu_R \sin(90^\circ - \theta_p)$$

$$\tan \theta_p = \frac{\mu_R}{\mu_D}$$

$$\tan \theta_p = \sin i_c$$

$$\theta_p = \tan^{-1}(\sin i_c)$$

$$i_c = \operatorname{cosec}^{-1}(\cot \theta_p)$$

13. Reflected ray has vibrations in the plane perpendicular to incident plane but incident light has no vibrations in that plane.

14. In the nicol prism, the e-ray and o-ray are produced by double refraction and the o-ray is furnished in another direction by TIR.

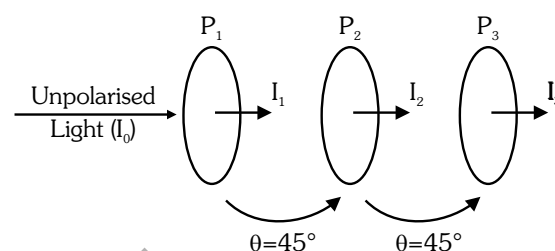
15. Reflected ray is completely polarised. Hence, polarising angle is 53° .

$$\theta_p = \tan^{-1}(\mu)$$

$$\mu = \tan \theta_p = \tan 53^\circ = \frac{4}{3}$$

$$r = 90^\circ - \theta_p = 90^\circ - 53^\circ = 37^\circ$$

16.



$$I_1 = \frac{I_0}{2}, I_2 = I_1 \cos^2 \theta$$

$$= \frac{I_0}{2} \cos^2 45^\circ = \frac{I_0}{4}$$

$$I_3 = I_2 \cos^2 45^\circ = \frac{I_0}{4} \cos^2 45^\circ = \frac{I_0}{8}$$

$$I_0 \xrightarrow{P_1} \frac{I_0}{2} \xrightarrow{P_2} \frac{I_0}{4} \xrightarrow{P_3} \frac{I_0}{8}$$

Hence, intensity becomes 50% each time, by Malus law.

MODERN PHYSICS

Radioactivity

1. Assertion is true but reason is false.

Matter Wave

5. Both assertion and reaction are correct and reason in the correct explanation of assertion.
6. Statement-I is true, statement-II is false.

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda \propto \frac{1}{\sqrt{m}} (\because KE = \text{same})$$

7. (A) $\lambda = \frac{h}{mv}$ or $\lambda \propto \frac{1}{m}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{m_\alpha}{m_p} = \frac{4}{1} (\because m_\alpha \approx 4m_p)$$

$$(B) \lambda = \frac{h}{\sqrt{2mqV}} \text{ or } \lambda \propto \frac{1}{\sqrt{m \cdot q}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\left(\frac{m_\alpha}{m_p}\right) \left(\frac{q_\alpha}{q_p}\right)} = \sqrt{\frac{4}{1} \times \frac{2}{1}} = \frac{2\sqrt{2}}{1}$$

$$(\because q_\alpha = 2q_p)$$

$$(C) \lambda = \frac{h}{\sqrt{2mE_k}} \text{ or } \lambda \propto \frac{1}{\sqrt{m}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_p}} = \sqrt{\frac{4}{1}} = \frac{2}{1}$$

$$(D) \lambda = \frac{h}{p}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{1}{1}$$

8. Give $\lambda_p = 2\lambda_e$

$$\text{So } \frac{p_p}{p_e} = \frac{\lambda_e}{\lambda_p} = \frac{1}{2} \left(\because \lambda = \frac{h}{p} \right)$$

$$\frac{E_e}{E_p} = \frac{\frac{1}{2} m_e v^2}{\frac{hc}{\lambda}} = \frac{\frac{1}{2} (mv) v}{\left(\frac{h}{\lambda}\right) c}$$

$$= \frac{1}{2} \frac{p_e \cdot v}{p_p c} = \frac{1}{2} \left(\frac{p_e}{p_p} \right) \frac{v}{c}$$

$$\frac{E_e}{E_p} = \frac{1}{2} \left(\frac{2}{1} \right) \left(\frac{1}{100} \right) = 10^{-2}$$

$$\text{and } \frac{p_e}{m_e c} = \frac{m_e v_e}{m_e c} = \frac{v_e}{c} = \frac{1}{100} = 10^{-2}$$

Nuclear Physics

9. Informative facts. At distance less than 0.7 fm, the nuclear force becomes repulsive.

10. (i) $R = R_0 A^{1/3}$

$$\frac{R_1}{R_2} = \left(\frac{2}{16} \right)^{1/3} = \frac{1}{2}$$

(ii) $\rho = \text{constant}$ $\frac{\rho_1}{\rho_2} = \frac{1}{1}$

(iii) $v \propto A$, $\frac{v_1}{v_2} = \frac{2}{16} = \frac{1}{8}$

(iv) $\frac{m_1}{m_2} = \frac{v_1}{v_2} = \frac{2}{16} = \frac{1}{8}$

12. larger is the binding energy per nucleon more is the stability of nucleus.

Photoelectric Effect

13. Photo electric equation

$$KE_{\max} = E - \phi = h\nu - \phi$$

$$(4.14 \times 10^{-15} \text{ eV-s}) (6 \times 10^{14} \text{ Hz}) - 2.0 \text{ eV}$$

$$= 2.48 - 2.0 = 0.48 \text{ eV} \approx 0.5 \text{ eV}$$

$$= 0.5 \times 1.6 \times 10^{-19} = 8 \times 10^{-20} \text{ J}$$

Stopping potential

$$eV_s = KE_{\max} = 0.48 \text{ eV}$$

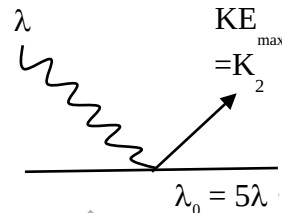
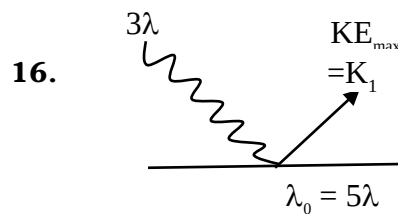
$$V_s = 0.48 \text{ volt}$$

$$n = \frac{I\lambda}{hc} \times A = \frac{I}{hv} \times A$$

$$= \frac{60 \times 10^3 \times 6.6 \times 10^{-4}}{6.6 \times 10^{-34} \times 6 \times 10^{14}}$$

$$n = 10^{20} \text{ per second}$$

$$i = ne = 10^{20} \times 1.6 \times 10^{-19} = 16 \text{ A}$$



$$K_1 = \frac{hc}{3\lambda} - \frac{hc}{5\lambda} = \frac{hc}{\lambda} \left[\frac{1}{3} - \frac{1}{5} \right] = \frac{hc}{\lambda} \left[\frac{2}{15} \right]$$

$$K_2 = \frac{hc}{\lambda} - \frac{hc}{5\lambda} = \frac{hc}{\lambda} \left[1 - \frac{1}{5} \right] = \frac{hc}{\lambda} \left[\frac{4}{5} \right]$$

(a) Ratio of KE

$$\frac{K_1}{K_2} = \frac{\frac{hc}{\lambda} \left[\frac{2}{15} \right]}{\frac{hc}{\lambda} \left[\frac{4}{5} \right]} = \frac{1}{6} = 1:6$$

(b) Ratio of speeds of incident photons

$$\frac{c_1}{c_2} = \frac{1}{1} = 1:1$$

Ratio of momentum of incident photons

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{\lambda}{3\lambda} = 1:3$$

(c) Ratio of fastest speed of photo electrons

$$\frac{v_1}{v_2} = \sqrt{\frac{K_1}{K_2}} = \sqrt{\frac{1}{6}} = 1:\sqrt{6}$$

TG: @Chalnaayaaar SEMICONDUCTOR & DIGITAL ELECTRONICS

1. Indium is a trivalent impurity.

3. In first region, transistor is in cut off mode and works as OFF switch.

In second region, transistor is in active mode and works as amplifier.

In third region, transistor is in saturation mode and works as ON switch.

6. When temp. increases in semiconductor then number of charge carriers increase.

9. In an unbiased P-N junction, majority charge carriers diffuse due to concentration difference.

IMPORTANT NOTE

ALLEN